Heart-Care Machine Learning / Deep Learning powered Cardiovascular Disease Prediction Application

Dhanushka Gunasinghe

Student ID : 23052728

MSc in Data Science

Project Supervisor : Mahmoud Elbattah

University of West of England (UWE)

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Abstract

Cardiovascular diseases are the world number one silent killer that effects all parts of the world, without any economical, geographical, sociological and cultural boundaries. CVD (cardiovascular diseases) including heart attacks, strokers and heart related medical consequences claim significant number of deaths per year. 90% of CVD causes human lifestyle behaviors like smoking, alcohol and physical inactivity. The economic impact behind the CV diseases is significantly considerable and social consequences can't be underestimated.

However, the ability to identify and make predictions of CV diseases based on human behavioral features promising new hopes for global health. With the exponential growth of AI and Data Science more predictive approaches open to the world. In this study focuses to develop highly accurate, user-friendly application to predict CV diseases with simple very straightforward steps to protect yourself and your loved ones.

This study uses a significantly larger dataset and eleven human behavioral features. To effective, highly accurate and trustworthy results initial preprocessing of removing outliers, feature selection and standardization carried out. These steps are crucial in refining the data, ensuring that the model is both effective and reliable in its predictions.

Throughout this study explorer with wide range of machine learning classifiers specially, ensemble model with random forest, extreme gradient boost and light gradient boost. Linear base classification model like logistic regression, tree base classifiers like decision tree and distance base classifiers like k-nearest neighbors and support vector machine. Further this analysis focuses on deep learning classifiers like artificial neural networks(ANN) and Google's deep learning model for tabular data, TabNet. All classifiers train a using wide range of hyper parameter space and analyze using different matrices such as train, test scores, accuracy, precision, recall and F1-Scores.

TabNet, deep learning classifier achieved nearly 80% train, test scores with decent levels of related measures. As outcome of this research developed Heart Care responsive application using react java script framework with modern tailwind user interface to effective prediction of CVD. This application is deployed for public use.

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1. Introductions and Objectives

1.1 Introduction

1.1.1 Cardiovascular Disease

The World Health Report(WHR)(2023) states, over 500 million humans are affected by cardiovascular diseases(CVD),which claims 20.5 million deaths in 2021.

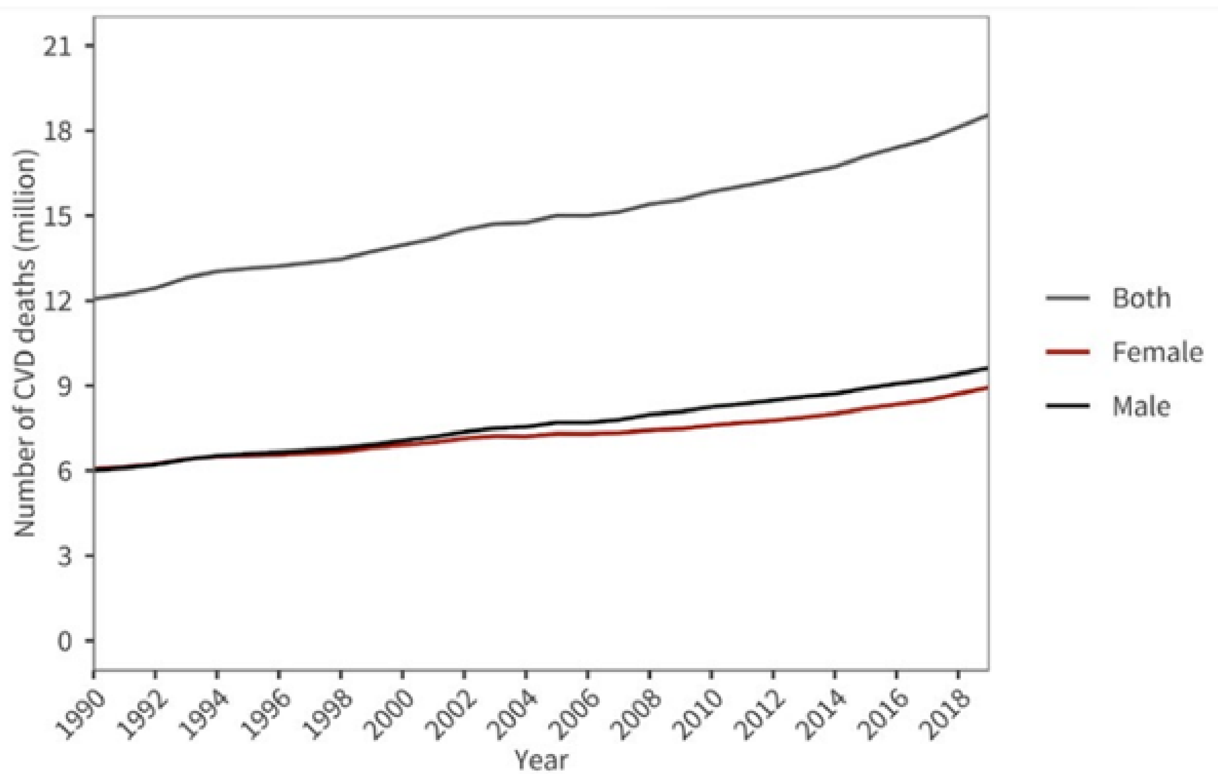


Figure 1- Global trends in number of deaths due to cardiovascular diseases, 1990-2019.

Source: Institute for Health Metrics and Evaluation (IHME)

Over past 30 years, CVD deaths have steadily increased, which contributes 33% of global deaths in 2021 and marking CVD the leading cause of death. Strokes is the main CVD which causes long-term disability, affect human brain related activities such as motor skills, speech, and cognition. Prevention and early detection are crucial, can avoid or reduce risks. WHR(2023) indicates 80% of CVD deaths occurred in low or middle-income countries, while high-income countries in declined trend.

1.1.2 Current Medical Practice on CVD detection

Existing medical practices for detecting CVD are reactive, relying on conventional methods like blood tests, ECG, MRI,X-rays, CT scans, and ultrasounds. These methods often detect CVD after the symptom appears, which delays reacting and potentially creates severe outcomes. Furthermore, these medical processes take longer time, high cost on access and less availability. Long wait times on limited resources create social health consequences.

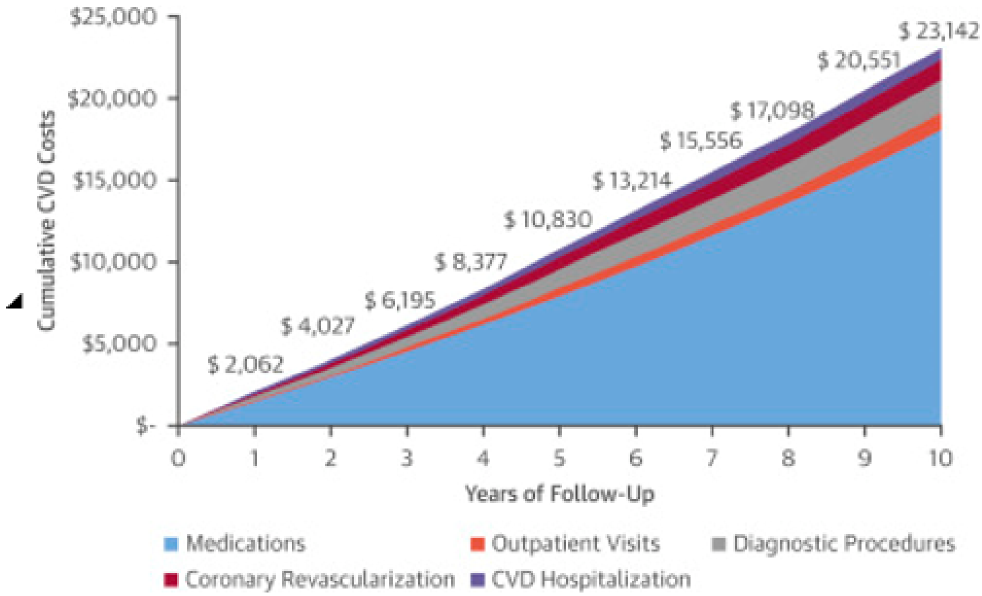


Figure 2- Leslee J., Abhinav Goyal MD (2018) [5] discuss CVD cost over the years.

According to figure 2, The costs associated with cardiovascular diseases (CVD) have increased consecutively and continuously. Implementing effective pre-detection and identification mechanisms will provide economic and sociological relief to the world.

1.1.3 Current Data Driven Practices in Cardiovascular disease (CVD) detection

Recent decades data-driven decision-marking, make revolutionary chargers in healthcare specially in CVD identification. However, limitation of enough true-valid human biological data for training and testing of algorithms will cause rapid development.

Under GDPR classification medical data are highly sensitive and vulnerable. Sharma and Yadav(2020), describe the privacy and security issues in healthcare data management and related complications. Char, Shah, and Magnus (2018), addressing ethical challenges associated with machine learning artifacts on healthcare.

Stefan Harrer(2019) discusses ethical implications of deploying effective machine learning applications in healthcare. Especially discussed bias and overfitting can cause significant enforcement for the quality of the result. Confidentiality restrictions significantly limit access to realistic and real-world data limits the effective development of machine learning algorithms.

Few applications provide free-open access to the world regarding cardiovascular disease predictions. Framingham Risk Score for Hard Coronary Heart Disease, QRISK Calculator, American College of Cardiology (ACC) ASCVD Risk Estimator, all are using traditional point scoring mechanisms for risk calculation.

1.1.4 Identified Gaps

Many studies and research on CVD predictions focused on relatively low datasets with limited number of features leading to bias and unfair results.

Secondly, most of the research limited conventional machine learning algorithms (Base models) restricting potential growth and accuracy results. Advanced machine learning algorithms and deep learning mechanisms could increase accuracy and trustworthiness.

As a third point both free and commercial products basically focus on point and weight base hard calculations methods which are relatively outdated and lack trust. There is a significant gap in machine learning or deep learning solutions, both free and commercial in cardiovascular prediction domain.

1.2 Research Aim, Scope and Objectives

1.2.1 Research Aim

The aim of this project is identifying the suitable diversified dataset with minimal bias and applying various machine learning and deep learning algorithms and techniques. The project will involve evaluating the above models, improving their accuracy and finally deploying the most suitable model as a fully responsive application to the public using cloud infrastructure.

1.2.2 Research Scope

This research project will focus on building an effective machine learning or deep learning model that is capable of accurately predicting cardiovascular diseases. Sametime project will focus to create user- friendly responsive application that communicates with the deployed model using web-API framework to provide valuable predictions as pro-active measure to the public.

1.2.3 Research Hypotheses

The initial hypothesis of the research is dataset in this study is high quality with low bias, and good variance without low noise. Particularly in health-related data, the high level of privacy concerns presents a significant challenge.

1.2.4 Research Limitation

In this research, faced several challenges. Collecting live clinical data in the field of health is a major and significant challenge due to privacy and sensitivity. This challenge addresses using publicly available clinical dataset which contains large number of records and vast number of features.

1.2.5 Research Objectives

There are key objectives discussed and outlined.

1. Identify a highly quality, diversified and balanced data set with minimal bias and low noise.
2. Feature analysis and determination of significant features and identity features that highly co-related with the target.

3. Training different machine learning and deep learning models using cleaned dataset, make hyper parameter tunning. Evaluate them with various evaluation matrixes and scores.
4. Identify best model with less bias and less overfitted incidents
5. Deploy model using web-API framework like fast API, Django or flask and exposure features as REST-API.
6. Develop front end application that allows users to make requests, validate and send them for the model prediction by consuming web services. Handling response and display friendly and understandable manner.

1.3 Ethical and Legal Considerations

The Dataset used for this research project is an open-source publicly available in Kaggle. Although data collection mechanisms are anonymized for each individual scenario, it does not contain any personal privacy data or personal identification trackers. Thus, this data collection and usage does not create any legal, professional or any privacy issues.

1.4 Code and Resource

Resources including codes, dataset and supporting elements can be accessed on GitHub.
https://gitlab.uwe.ac.uk/d2-gunasinghem/heartcare

1.5 Structure and Overview

This dissertation is organized into eight chapters.Chapter1 introduces CVD, discusses their implications and traditional treatments after occurred. Furthermore, Chapter outlines research aim, existing gaps, objectives, scope and limitations.Chapter2 review existing research and literature related to the topic. Third chapter dedicated to data description, data preparation techniques, methods and models.4th chapter reveals the results, performance scores and further comparison and best model selection.Chapter5 describes the learning outcomes, novelty discussion and personal reflection.Chapter6 dedicated to concluding the study and future directions, while 7,8 for reference and appendix.

2. Literature Review

This chapter overviews the research on early CVD detection using machine learning and deep learning.Chapter begins with discussing about machine learning approach for analysis of large complex clinical data to determine the patterns, behaviors and uncover the relationships of predictive variables. Secondly discuss previous studies and research related to CVD prediction using machine learning and deep learning. Third focus on discussing the research works on evaluation, scoring and comparison.

2.1 Machine Learning and Deep Learning in Healthcare

Recently machine learning and related concepts have been used in various medical fields to detect, predict and analyze several diseases. For example, cancer detection, heart disease prediction and risk assessments, detecting abnormalities, detect consequences of medical images. Etc.
According to Mohammed and Nagy(2023), discuss the integration of machine learning and deep learning into healthcare and the significant advantages over use of traditional data management concepts. The paper discusses machine learning and deep learning concepts, algorithms and tools that enable predictive analysis effectively and efficiently.

Anandhavalli and Sehrish(2019), reveals the deep learning concepts on predictive analysis in healthcare. This paper specifically discusses convolutional neural networks(CNNs) and generative adversarial networks(GANs).CNNs are used to predict and make decisions on medical imaging, identify patterns and structures to effective classifications such as cancer detections, neurological disorders identification and various complex image analysis.

2.2 Machine Learning and Deep Learning for Cardiovascular Predictive Analysis

Move over several studies focused on machine learning and deep learning concepts and algorithms for predictive analysis of cardiovascular diseases.

Ramalingam v(2018), discuss effective use of supervises learning techniques such as Decision tree(DT), K- nearest neighbor (KNN), Random Forest(RF), Support vector machine(SVM) and Naïve Bayes. The paper detail explains the underlying mathematical concepts on statistical enforcement of above models and algorithms. Machine learning concepts have a vast array of algorithms and models, each tailored to address specific problem domains with varying levels of accuracy. That gives great opportunity to and freedom to evaluate better solutions. On the other hand, this study discussed accuracy and efficiency of ensemble models by use of best base model combinations.

Nabaouia and Samira D(2021) proposed machine learning model for pre-alerting cardiovascular diseases. They used 303 records with 13 feature attributes to perform analysis on SVM, Extreme gradian boost(XG-boost) and Adaptive boosting(Ada boost) with voting classifier. Furthermore, paper discuss about missing data handling mechanisms like MICE(Multiple Imputation by Chained Equations), KNN technique and RF handling missing values.

Rahul and Sunit(2020) study about risk factors of cardiovascular diseases discussed machine learning techniques and algorithm that can be used for successful predictive analysis. They used SVM, LR, KNN, Naïve Bayes, RF along with the Artificial Neural Network(ANN) and Deep Neural Network(DNN).They manage to get 95% highest accuracy on LR,93.4% in SVM and 92% in ANN. They recommended further research on deep learning for better success.
Xiao-Yan(2015)proposed an ensemble method to enhance the accuracy of predicting heart diseases. Their research focused on combining multiple machine learning algorithms to create a more robust and reliable model. They use ensemble techniques for better accuracy enhancement with the KNN, SVM, DT, RF, NB. They go with the principal component analysis(PCA) and linear discriminant analysis(LDA) as features selection algorithms. In this research they achieved 98.4% accuracy with the DT ensemble technique.
Research uses dataset of 1025 instances with 13 independent attributes collected from Kaggle which is relatively low, recommended to expanding dataset.
Geeta and Surya(2021) proposed cardiovascular disease prediction model using UCI dataset medical records of 303 patients. They continued their study using KNN algorithm and achieved 87% accuracy. In this analysis feature selection was done manually without any statistical approach. Same as earlier research dataset is limited and larger and more diverse datasets, and to explore other machine learning algorithms for potential improvements.

Anna Karen(2020) conduct research on cardiovascular disease predictions using machine learning classifiers including DT, Gradient boost, LR, NB and RF. Interesting finding on this research is, dataset contain 70+ features and they used different mechanism to identify the highly co-related features. Principal Component Analysis PCA) used to find principal components and Chi-Square used for identifying effective features among PCA identifications. PCA specially for reduced multi-dimensional features suitable for large feature sets.
Amin and Jian(2018) proposed hybrid intelligent machine-learning based predictive system. Their approach integrates multiple machine learning techniques and advanced feature selection methods to improve classification accuracy and reliability. Hybrid system works with K-NN, ANN, SVM, NB, DT, and RF classifies. To enchase the performance, they use LSSO(Least Absolute Shrinkage and Selection Operator), mRMR-(Minimum Redundancy Maximum Relevance) to feature selection and K-fold cross validation for validation purpose

3. Research Methodology

This chapter describes the various approaches employed in this research project including describing dataset, data preparation and cleaning, model employed and complete description of architecture, training, testing as well as detailed implementation.

3.1 About Dataset

The Dataset used by this study is the CVD dataset, which is publicly available under open license in Kaggle repository. This data was gathered by Svetlana Ulianova in2018.Data collected from medical examinations and information provided by trustable and reputable sources in different geographical locations.

3.1.1 Data Description

The dataset contains 70000 clinical records with a total of 11 independent features and 1 target variable. These features are very familiar and easy to understand.

Variable	Definition	Data Type
Age	Age of the patient	Int(dates)
Gender	Patient Gender	Int (categorical code; (2= female,1 =male)
Height	Height of the Patient	Int(centimeters)
Weight	Weight of the Patient	Float(kilogram)
Ap_hi	High pressure of the patient	Int(mmHg)
Ap_lo	Low pressure of the patient	Int(mmHg)
cholesterol	Patient cholesterol level	Int (categorical 1 = normal, 2 = above normal, 3 = well above normal)
gluc	Patient glucose level	Int (categorical (1 = normal, 2 = above normal, 3 = well above normal)
Smoke	smoking or not	Int (categorical Smoke-binary; (1 = smoker, 0 = non-smoker)
Alco	Alcohol use or not	Int (categorical Alco-binary; (1 = yes, 0 = no)
Active	Patient physical active or not	Int (categorical (active = 1, inactive = 0)

Target Variable

Cardio	Patient is in risk or not	Int(Categorical (1 = Presence = 1, 0 = absence of cardiovascular disease)

Table-1: Description of the Dataset

Variable	Count	mean	std	min	25%	50%	75%	max
Age	70000	19468	2467	10798	17664	19703	21327	23713
Gender (categorical)	70000	-	-	1	-	-	-	2
Height	70000	164.36	8.2	55	159	165	170	250
Weight	70000	74.2	14.39	10	65	72	82	200
Ap_hi	70000	128	154	-150	120	120	140	16020
Ap_lo	70000	96.63	188.475	-70	80	80	90	11000
cholesterol (categorical)	70000	-	-	1		2		3
gluc(categorical)	70000	-	-	1				3
Smoke(categorical)	70000	-	-	0				1
Alco(categorical)	70000	-	-	0				1
Active(categorical)	70000	-	-	0				1

Table-2: Descriptive Statistics of the dataset

3.2 Data Pre- Processing.

Data often contains noise, errors and misleads which can harm successful to analysis. Avoid outliers, noises and errors, preprocessing techniques play crucial and critical role in data science and machine learning pipeline. Kotsiantis and Kanellopoulos(2006) discuss about Importance of handling inadequate, extraneous and irrelevant information, it generates less accurate and less understandable results. Furthermore, research describes data pre-processing steps addressing problems like noise, redundancy and missing data implications.

3.2.1 Fill empty columns

Checking for null or empty values is the first step in data pre-processing. Figure 3. Provide code snippet and result set for null check.

```
# check null values
print(df_heart.isnull().sum())
```

```
id             0
age            0
gender         0
height         0
weight         0
ap_hi          0
ap_lo          0
cholesterol    0
gluc           0
smoke          0
alco           0
active         0
cardio         0
dtype: int64
```

Figure 3- Result set for null check. No null observed

3.2.2 Check Duplicate Entries

Removing duplicates increase the data quality and integrity and improving model performance and accuracy.Figure-4 Provide code snippet and result for the duplicate check.

```
print(df_heart.duplicated().sum())
```

```
0
```

Figure 4- Result set for duplicates check. No duplicates observed

3.2.3 Feature Engineering

In this phase, new feature introduced as BMI (Body Mass Index) by using height and weight attributes. BMI is the most familiar attribute, especially in healthcare. 'Age' converted from days to years for easier interpretation.Figure5-demonstrate the code snippet.

Introduced BMI = weight/height* height

```
df_heart=df_heart.drop(['id'], axis=1)
```

```
df_heart['BMI'] =df_heart['weight']/(df_heart['height']/100 * df_heart['height']/100)
df_heart
```

convert age into years

```
df_heart['age']=np.round(df_heart['age']/365).astype(int)
```

Figure 5- Feature engineering on BMI and Age

3.3 Exploratory Data Analysis (EDA)

EDA involves examining the data, discovering patterns, understanding relationships and determining anomalies. EDA enables data analysis to deep drive into the features, by visualizing, tableting and simplifying using various statistical approaches.

3.3.1 Detecting and Removing Outliers

Outliers are the data points significantly deviate from the regular pattern; outliers make considerable statistical influence on machine learning model. Initially, outliers identified visually followed by the domain knowledge and statistical methodologies .Figures 6 to 13

Shows data distribution of various attributes by plotting the data. For effective identification the clusters, employed density base cluster identification algorithms(DBSCAN) over each attribute.

Martin and Peter(2018) discuss advantages of DBSCAN algorithm on outlier detection, paper further revealed applications on DBSCAN. One unique use case of DBSCAN for outlier detection is in fraud detection and anomaly detection. This approach identifies data points outside the selected cluster to be as the outlier.

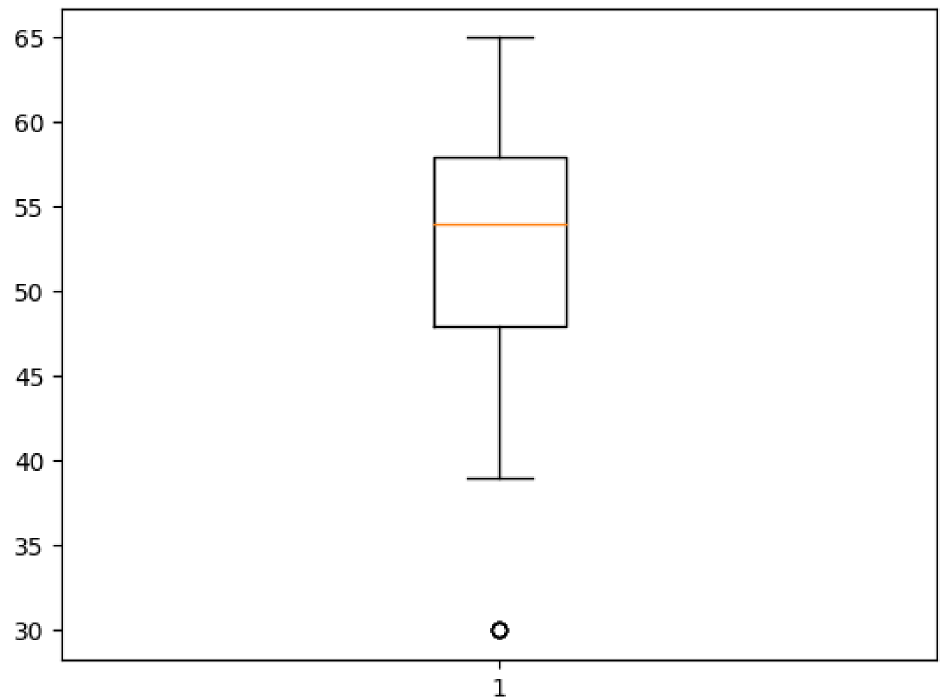


Figure 6- Boxplot [Age]

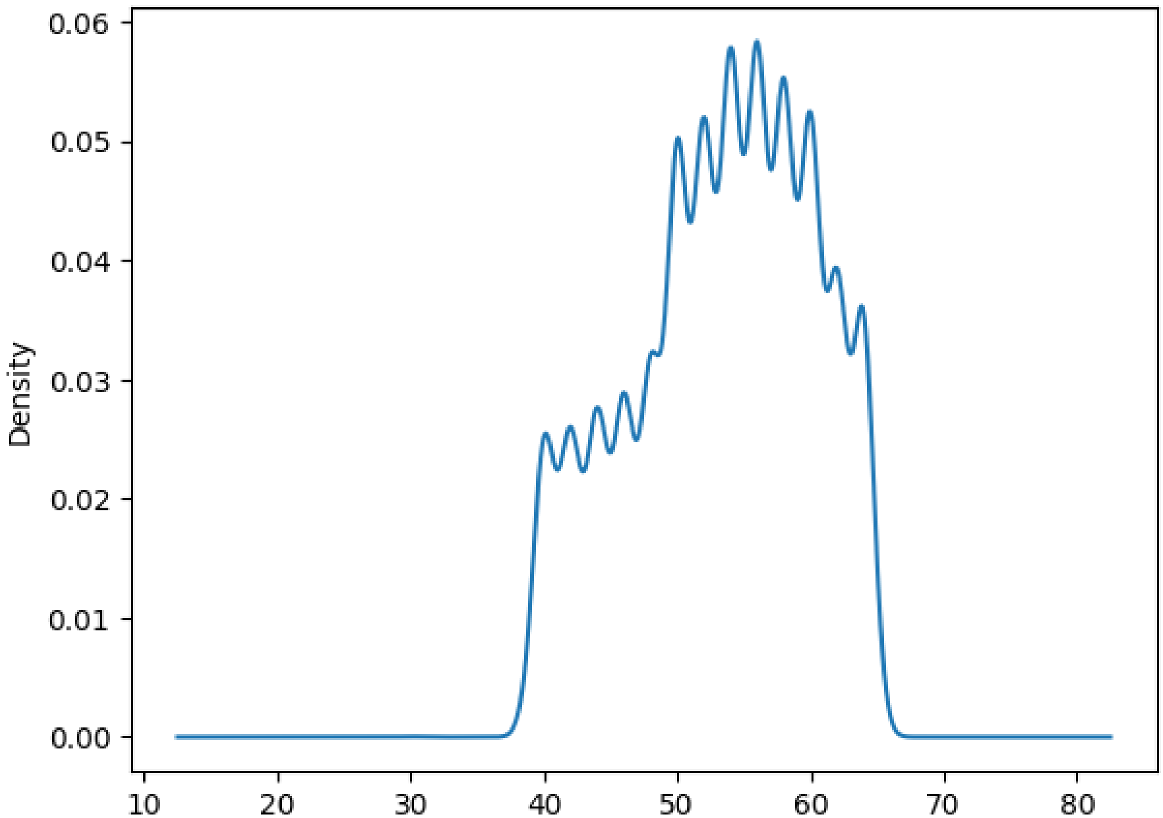


Figure 6- Density [Age]

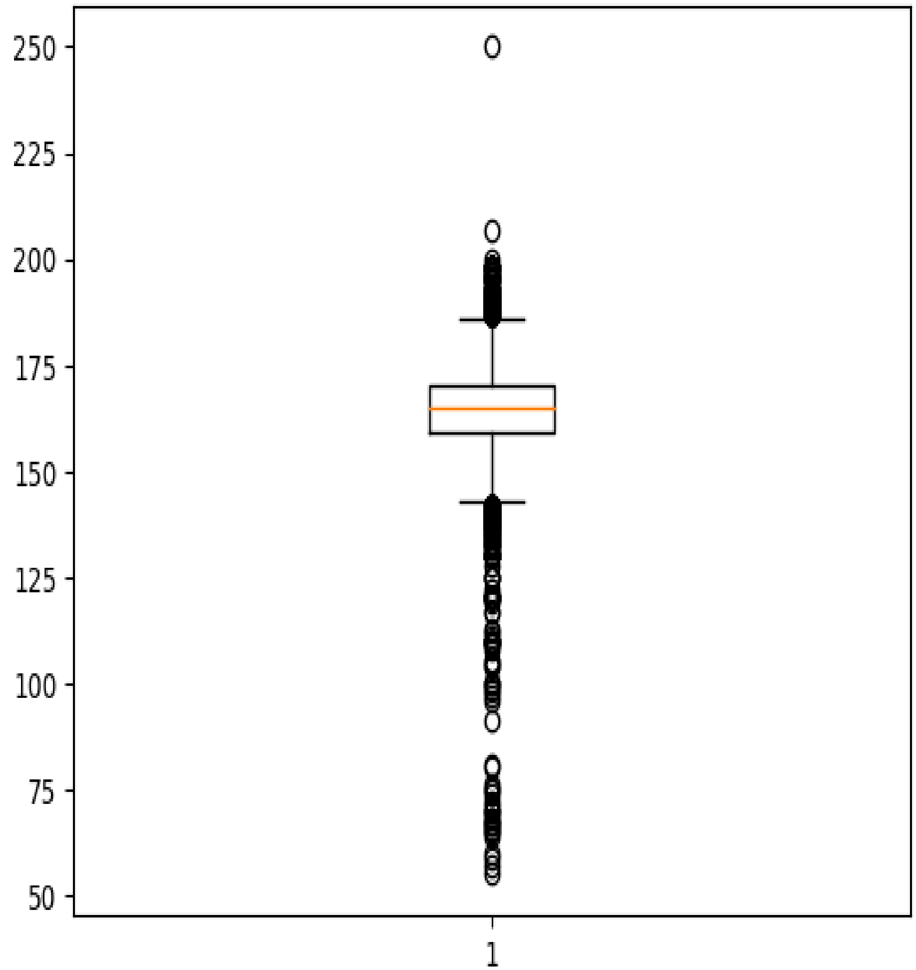


Figure 7- Boxplot [Height]

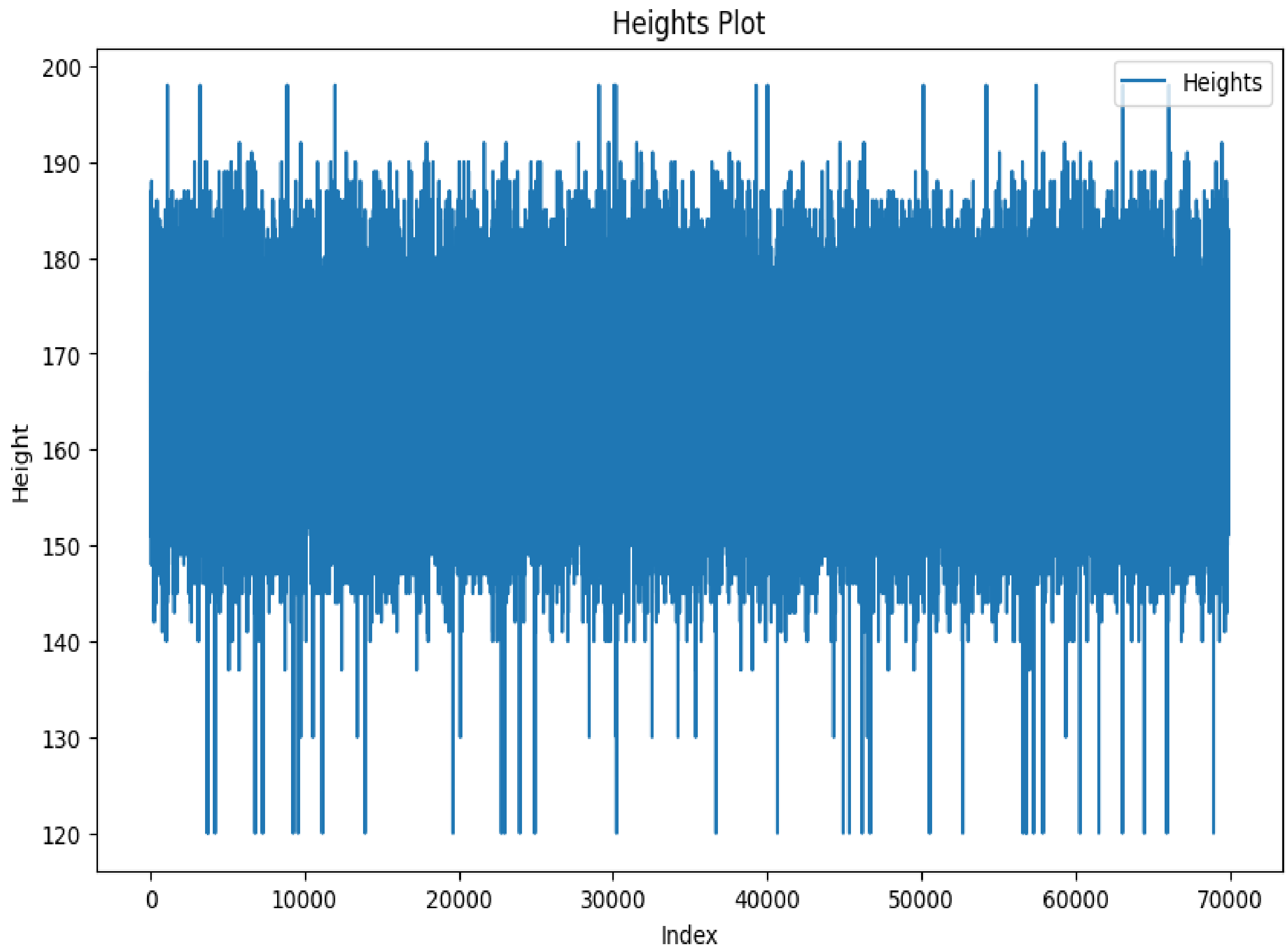


Figure 8- plot [Height and Record Index]

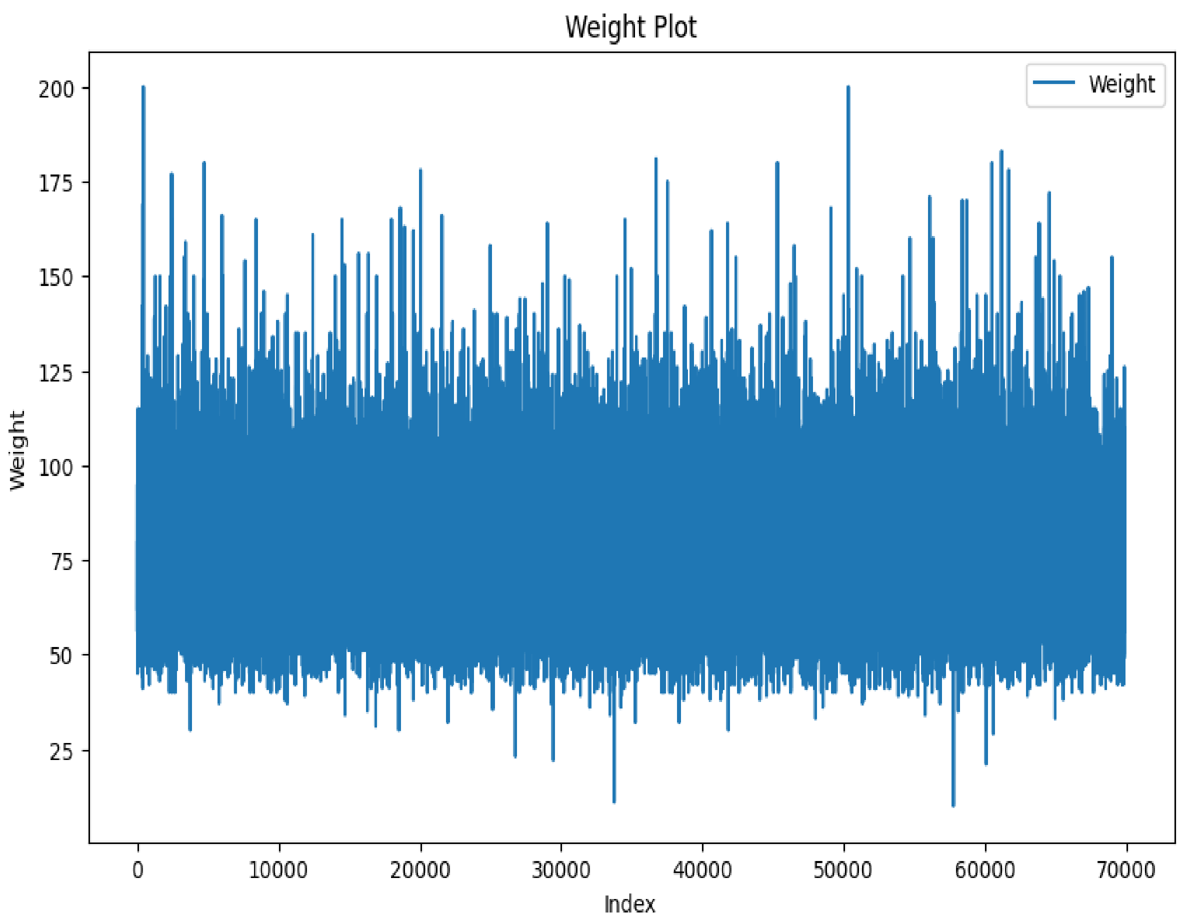


Figure 9- plot [Weight and Record Index]

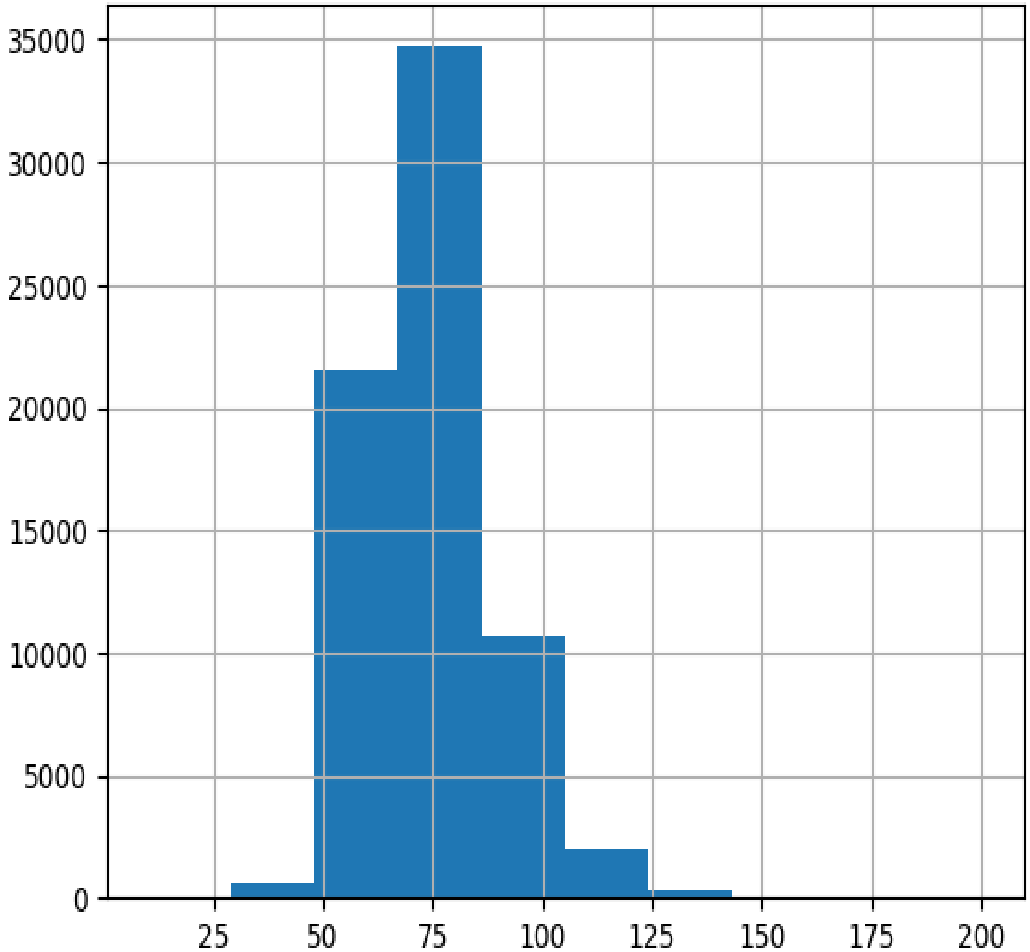


Figure9a-Histogram [Weight]

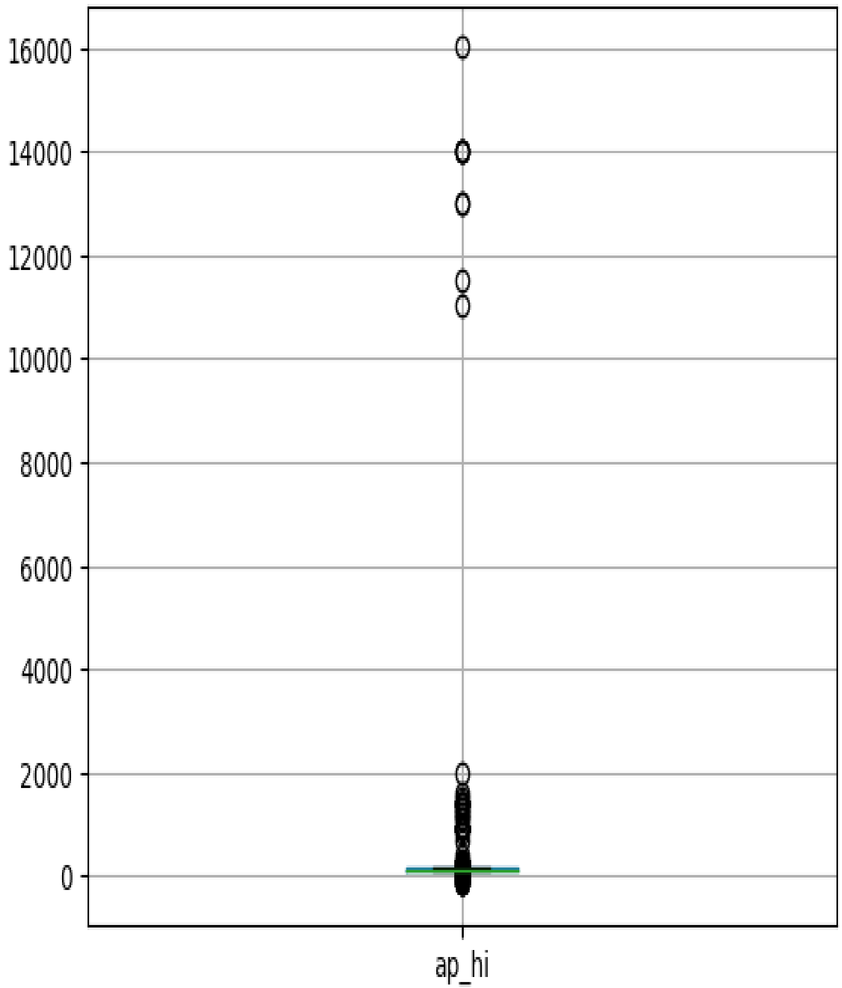


Figure 10- Boxplot [High Pressure]

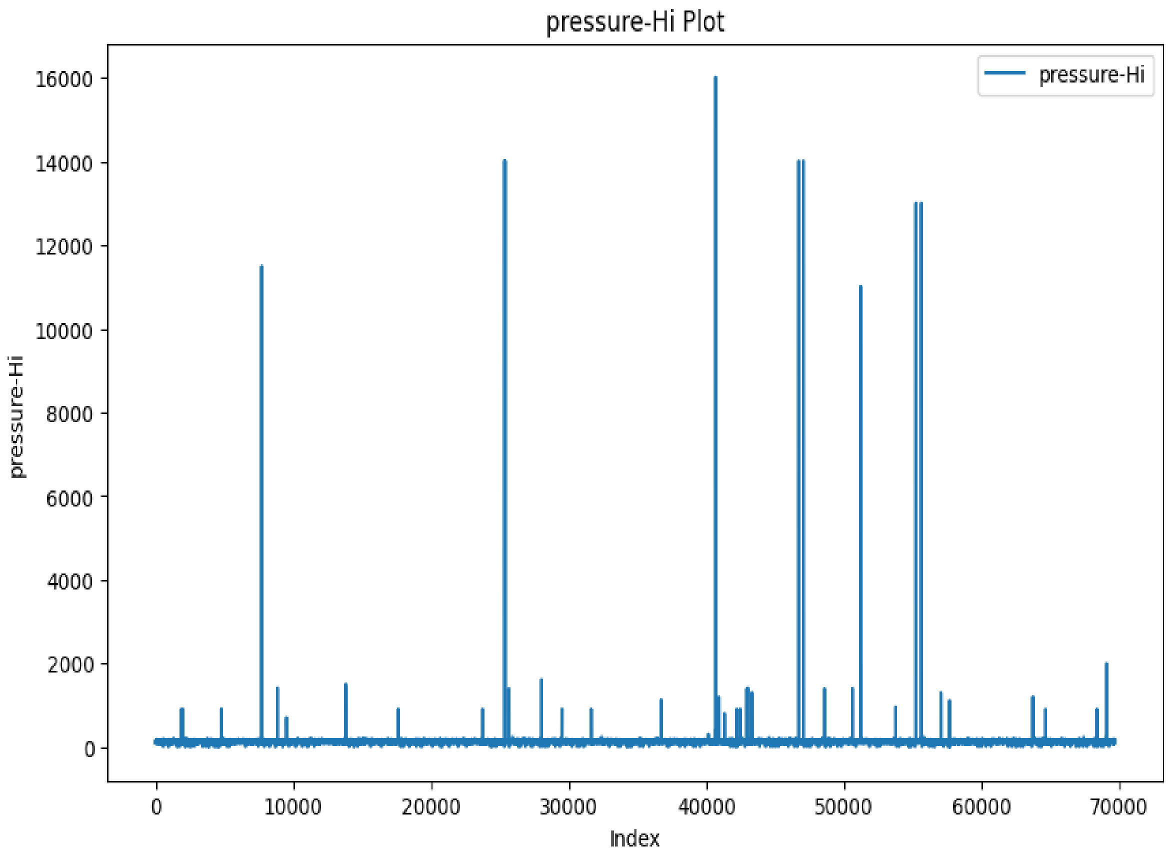


Figure 11- plot [Hi-Pressure and Record Index]

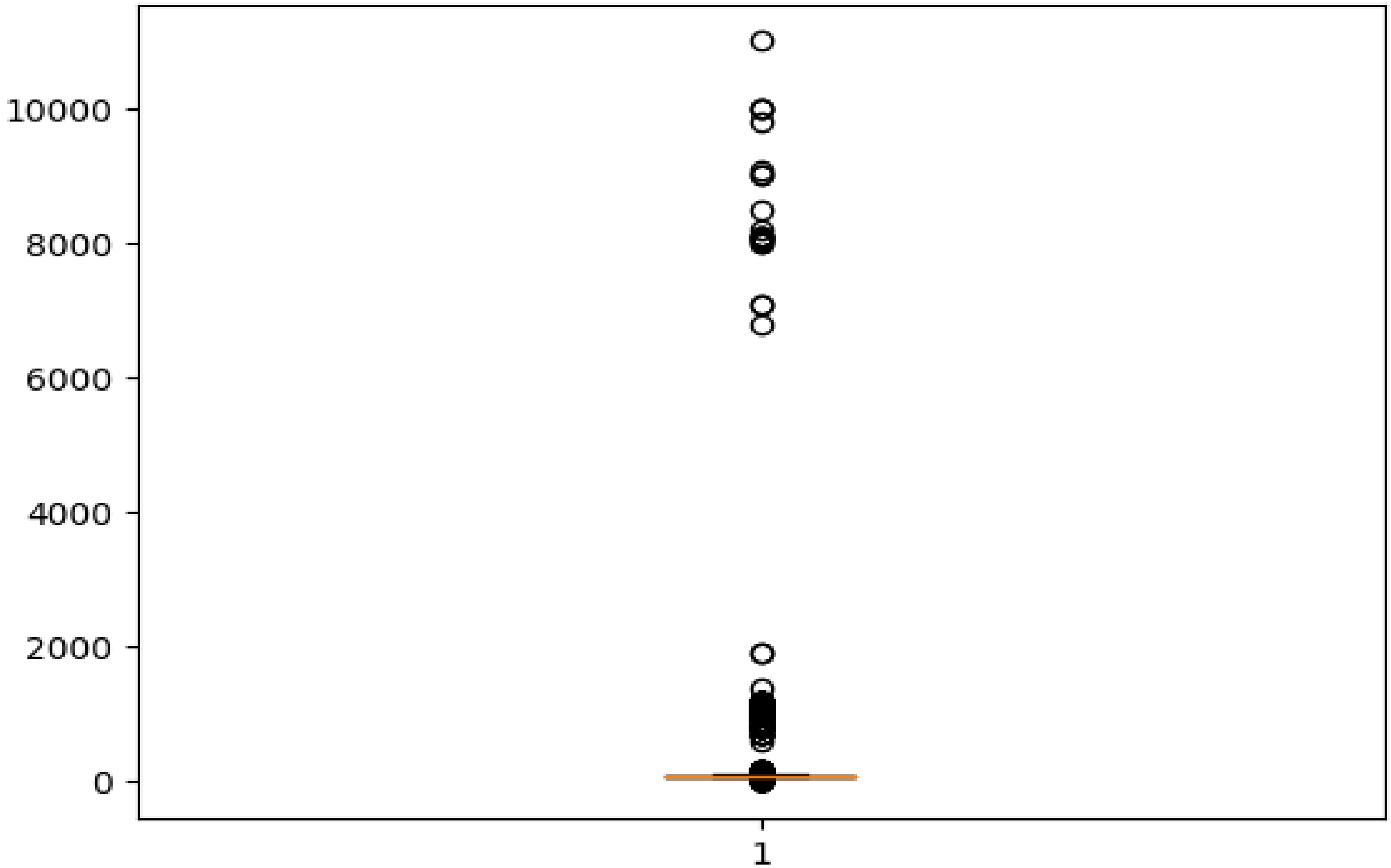


Figure 12- Boxplot [Low Pressure]

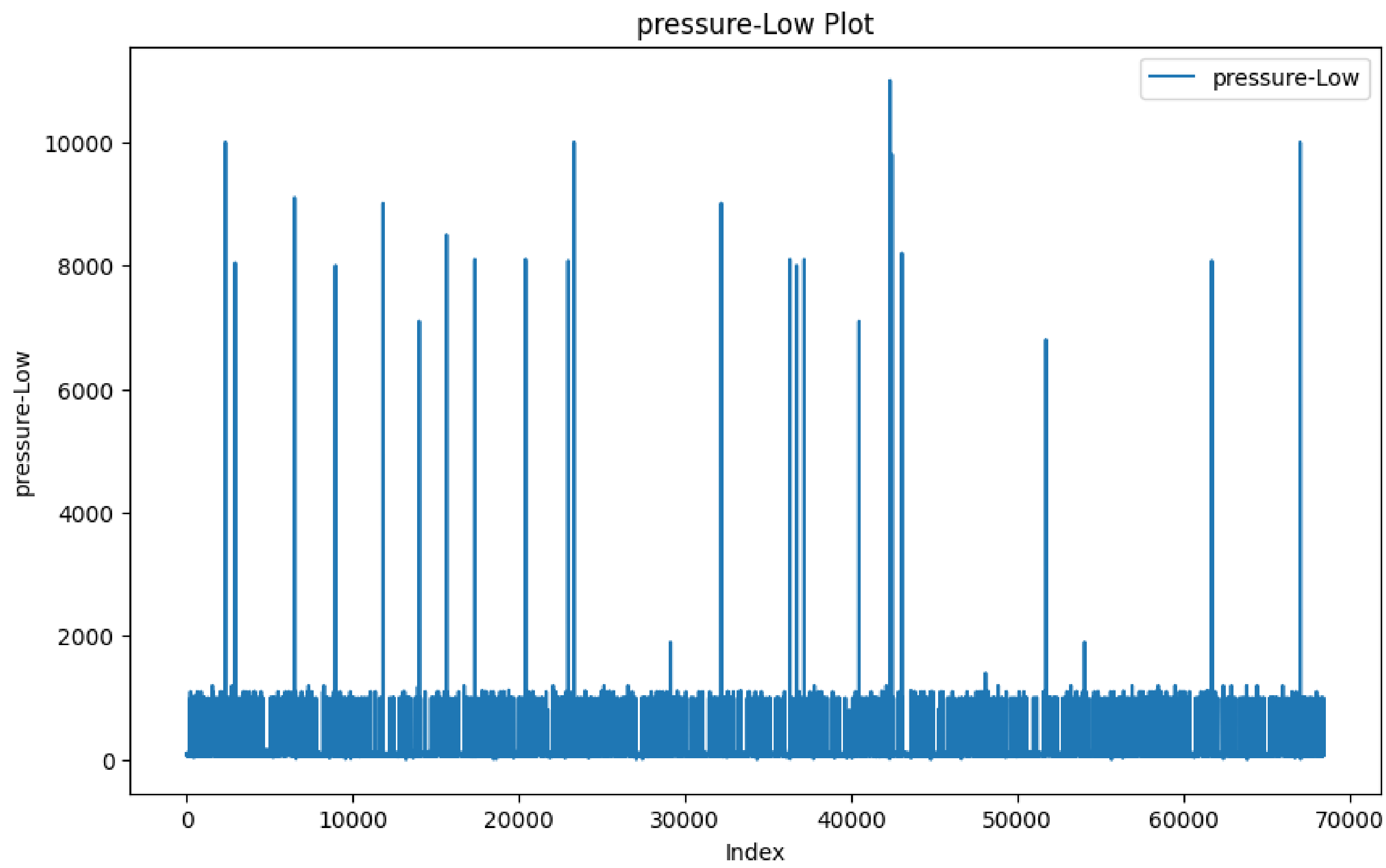


Figure 13- plot [Low-Pressure and Record Index]

Density-Based Spatial Clustering Of Applications With Noise (DBSCAN), is a clustering and noise detection algorithm that defines EPS and MinPts as key parameters for defining a cluster.

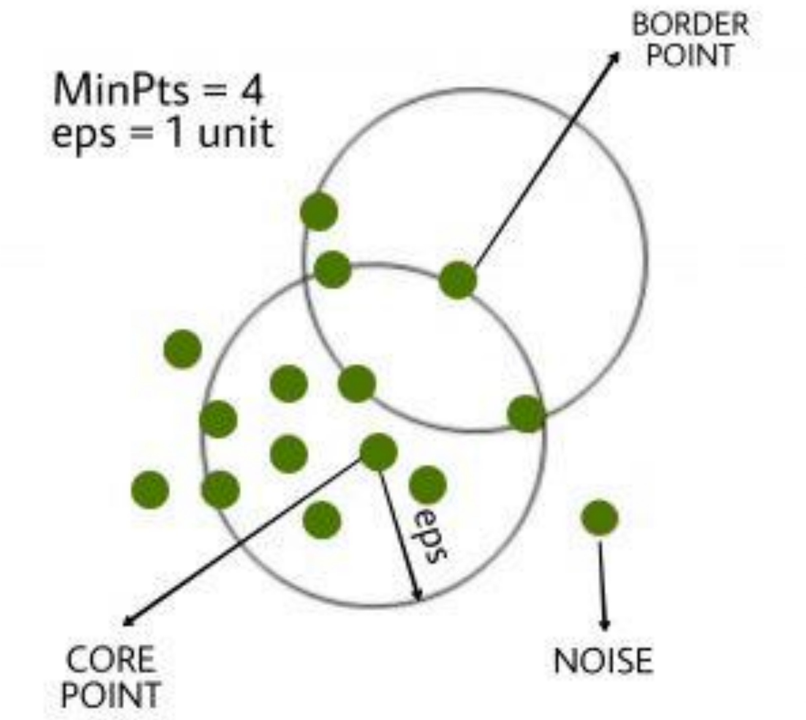


Figure 13a.
MinPts - Minimum number of neighbors
Eps – Minimum distance to qualify as a neighbor
Fig. 13a. visualize the key concepts of the DBSCAN which are useful for discovering clusters and effectively handling noise in datasets.

Figure 14 to Figure 19 visualize cluster identification after applying sklearn.cluster.DBSCAN for each attribute and remove non cluster data points which are detected as outliers.

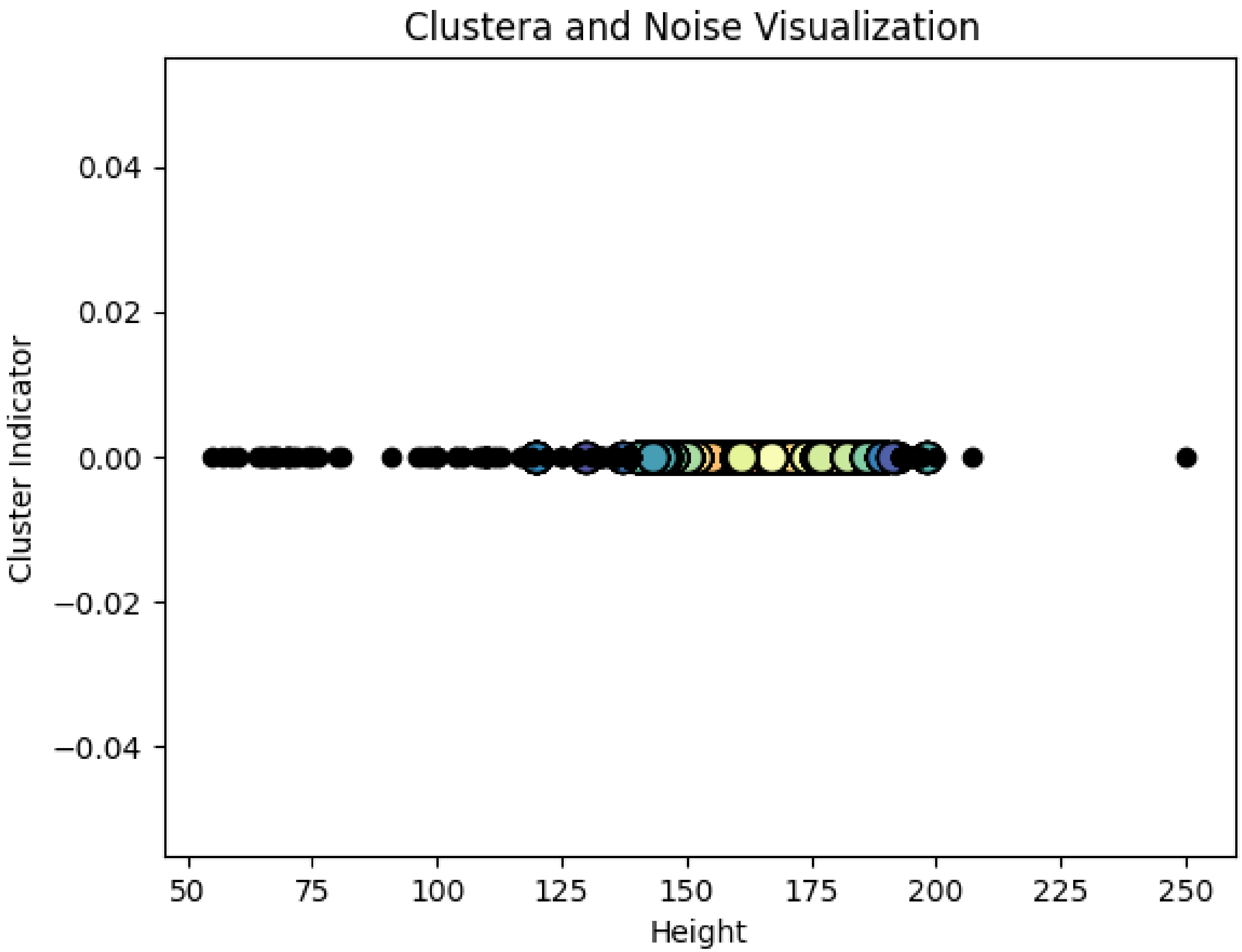


Figure 14- Cluster Distribution [Height]

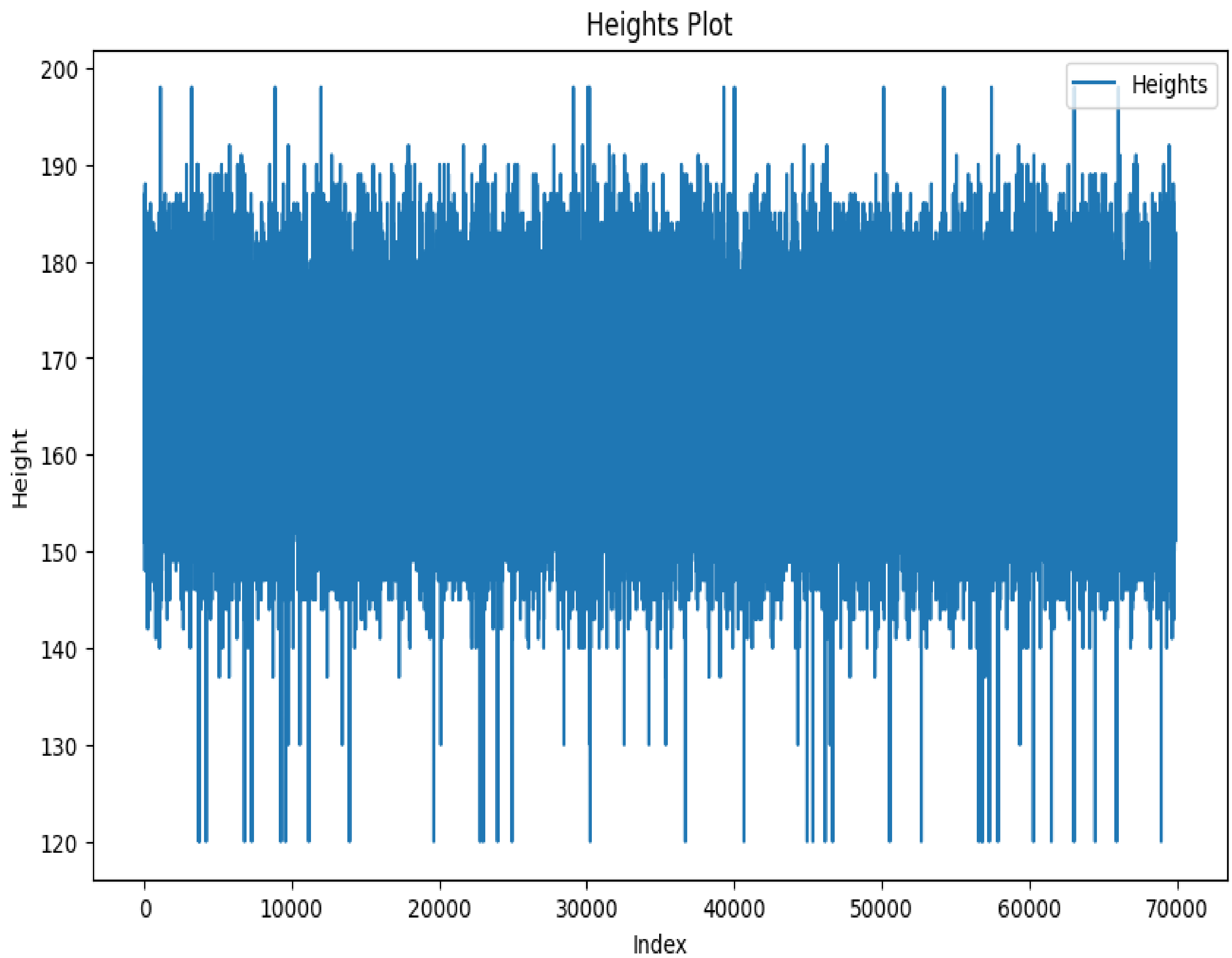


Figure 15-After DBSCAN Plot Distribution - [Height]

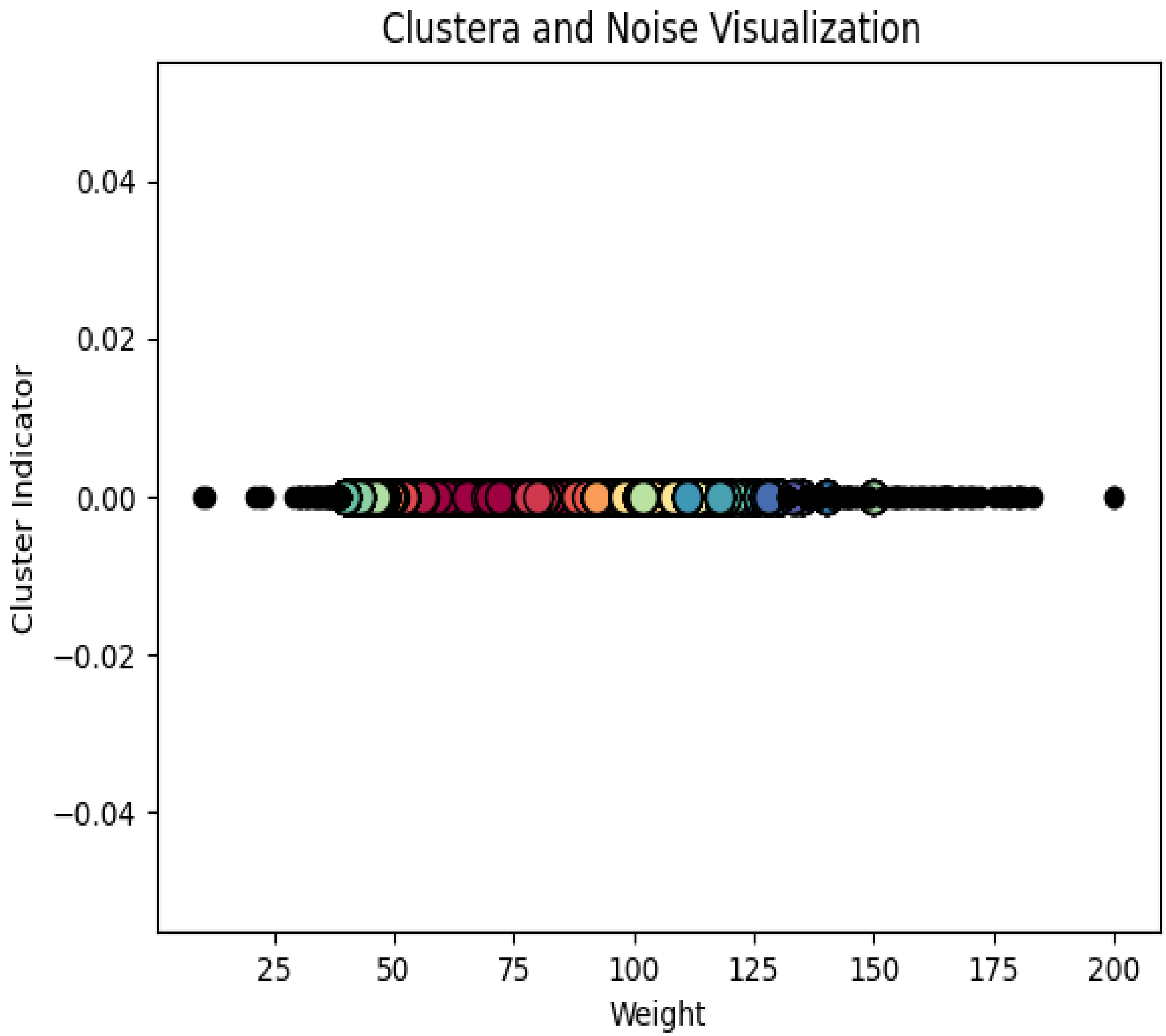


Figure 16- Cluster Distribution - [Weight]

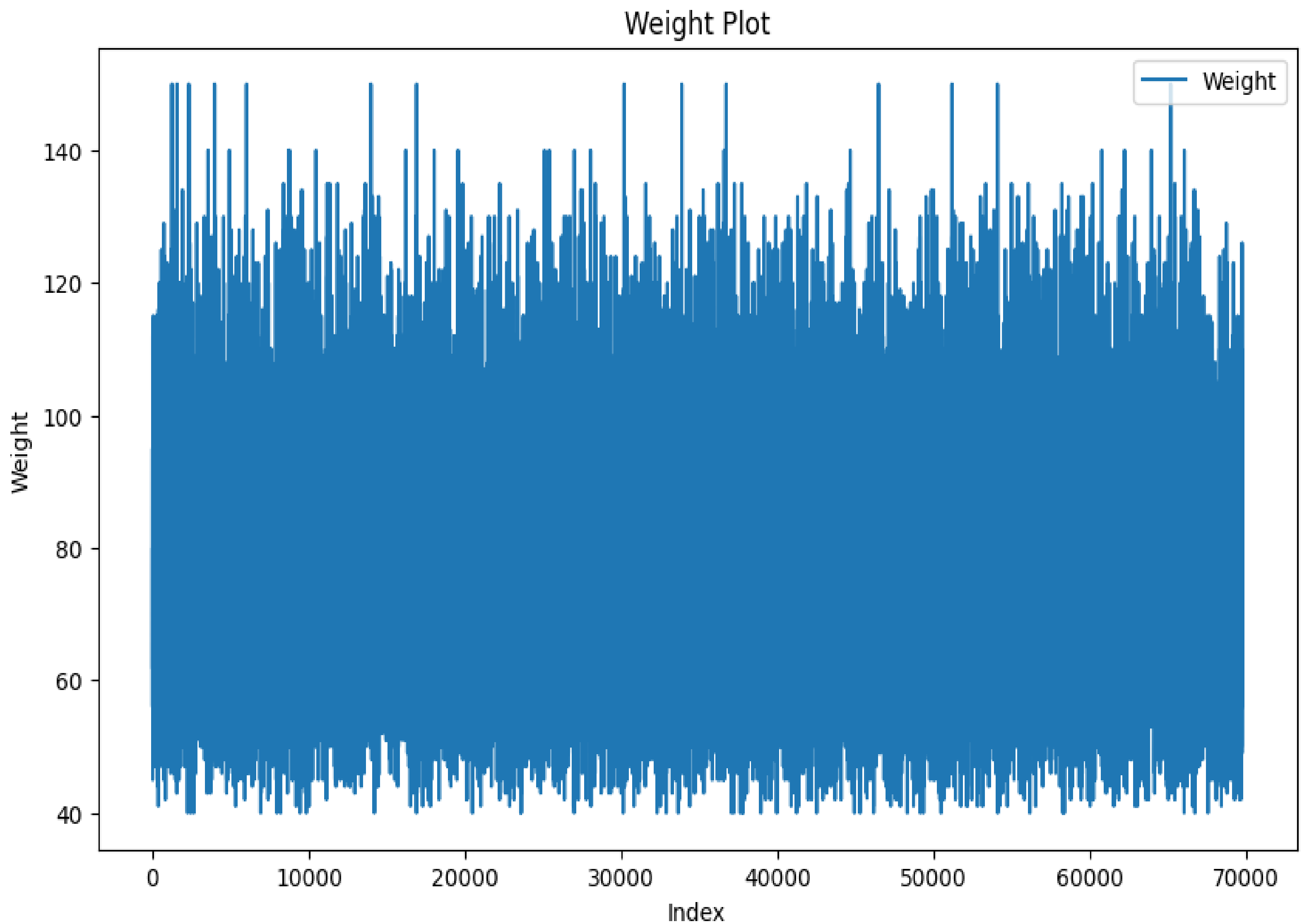


Figure 17- After DBSCAN Plot Distribution - [Weight]

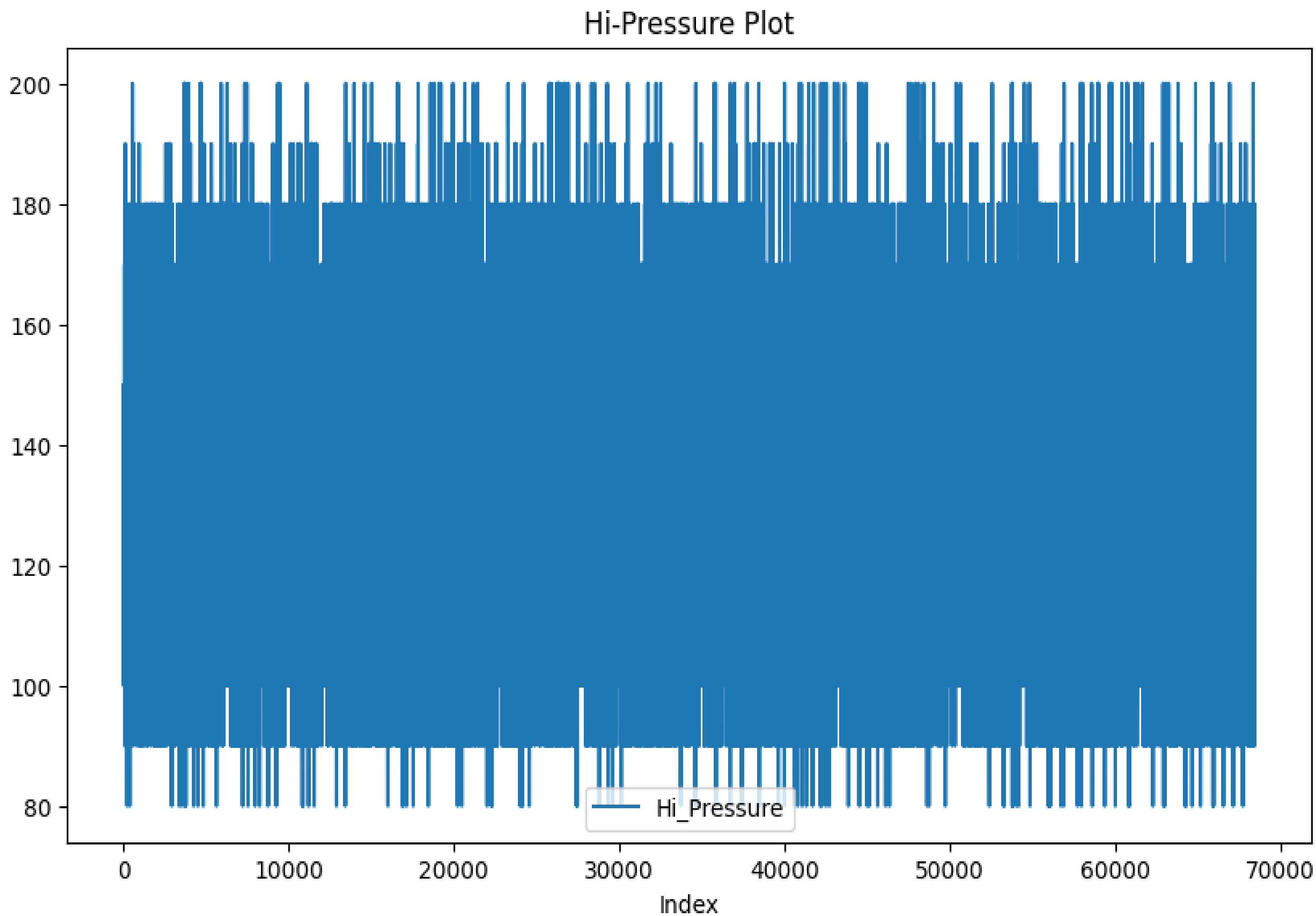


Figure 18- After DBSCAN Plot Distribution - [Hi-Pressure]

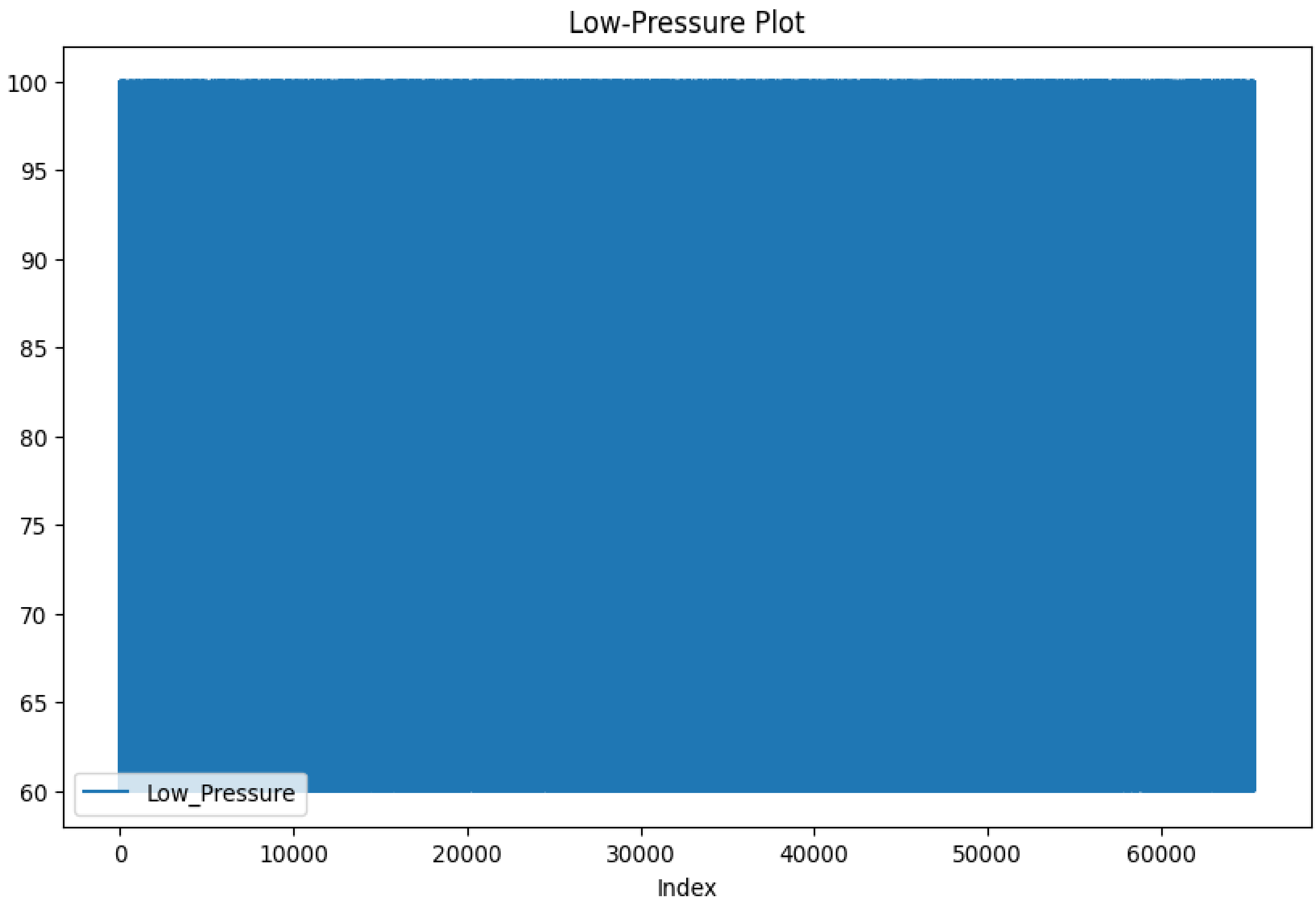


Figure 19- After DBSCAN Plot Distribution - [Low-Pressure]

3.3.2 Categorical Attributes

Dataset included five categorical attributes includes the cholesterol, gluc, smoke, alco, active and the target attribute cardio describe the patient risk level.Figure20 to Figure24 demonstrate the distribution of each categorical variable.

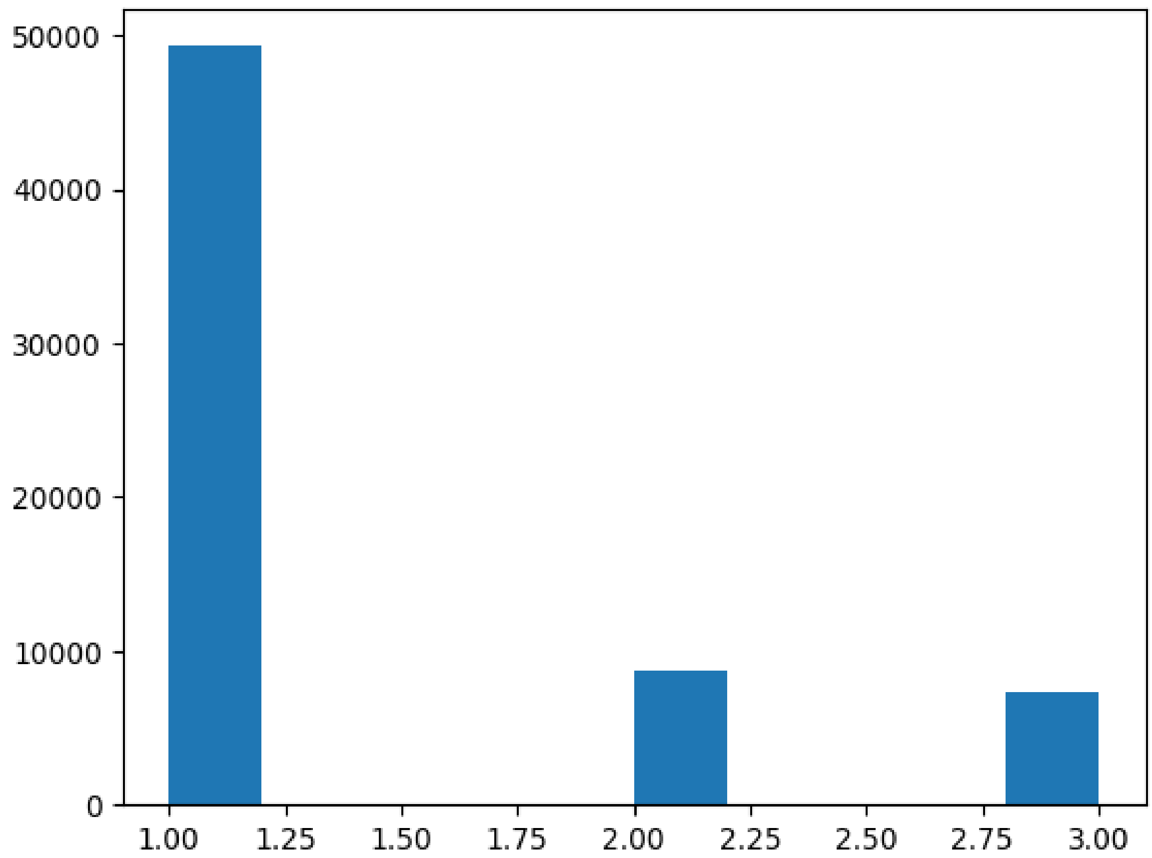


Figure 20- Histogram - [Cholesterol]

Fig 20. Shows histogram of the cholesterol distribution of the dataset.
1- Normal, 2- Above Normal and 3-Well Above Normal

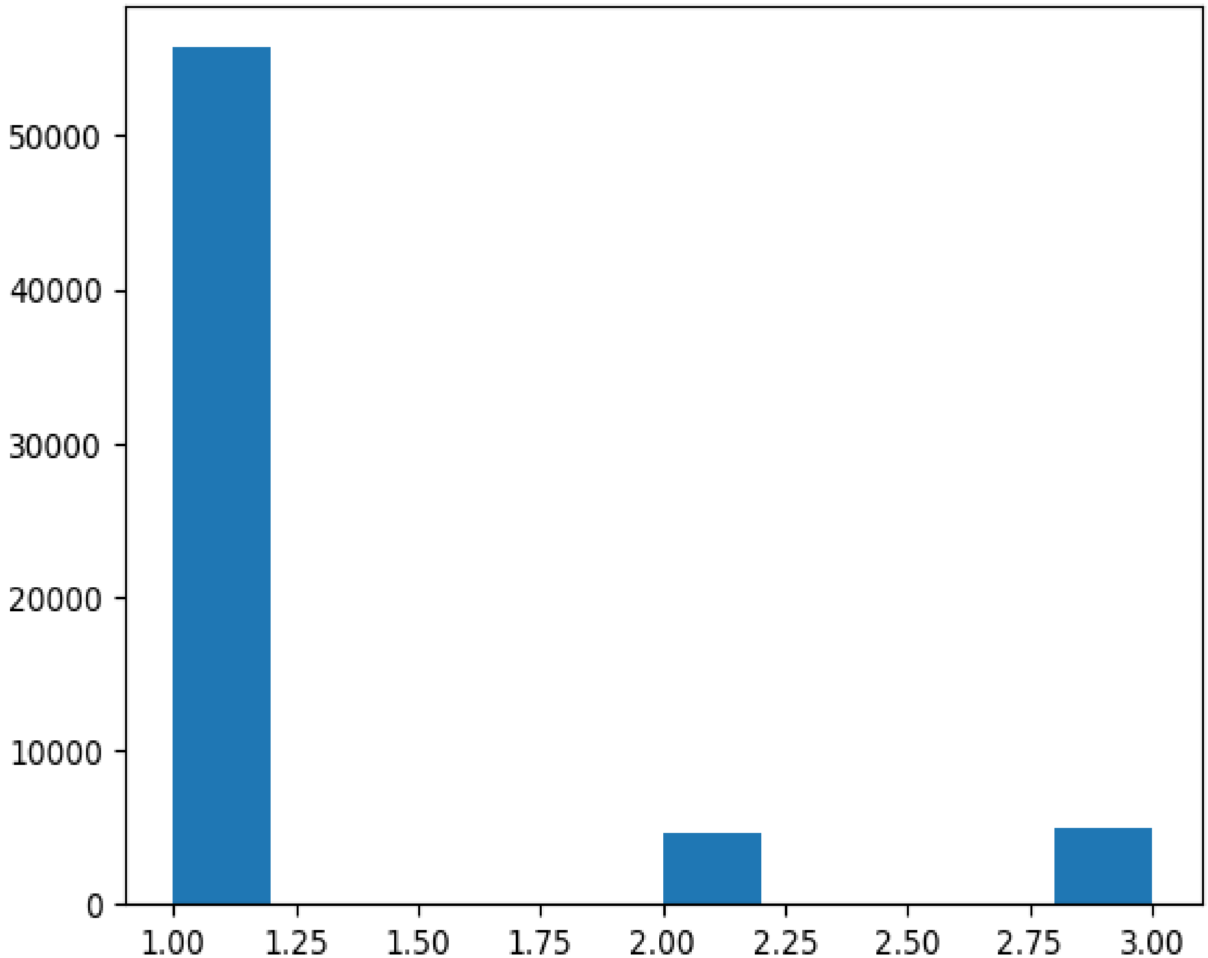


Figure 21- Histogram - [Glucose]

Fig 21. Shows histogram of the Glucose (blood sugar) distribution of the dataset.
1- Normal, 2- Above Normal and 3-Well Above Normal

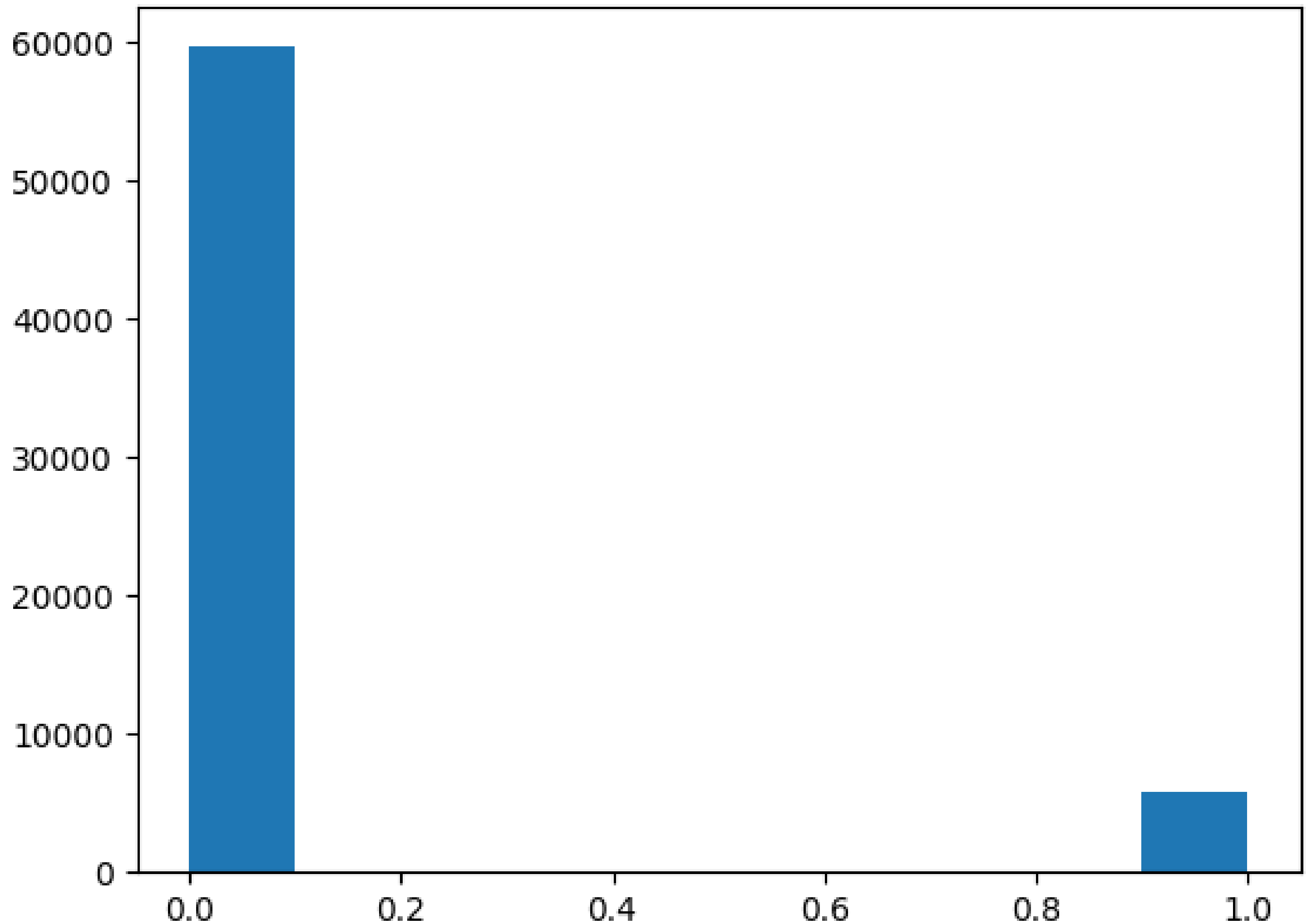


Figure 22- Histogram - [smoke]

Fig 22. Shows histogram of the Smoker and Non-Smoker distribution of the dataset.
0- Non-Smoker, 1- Smoker

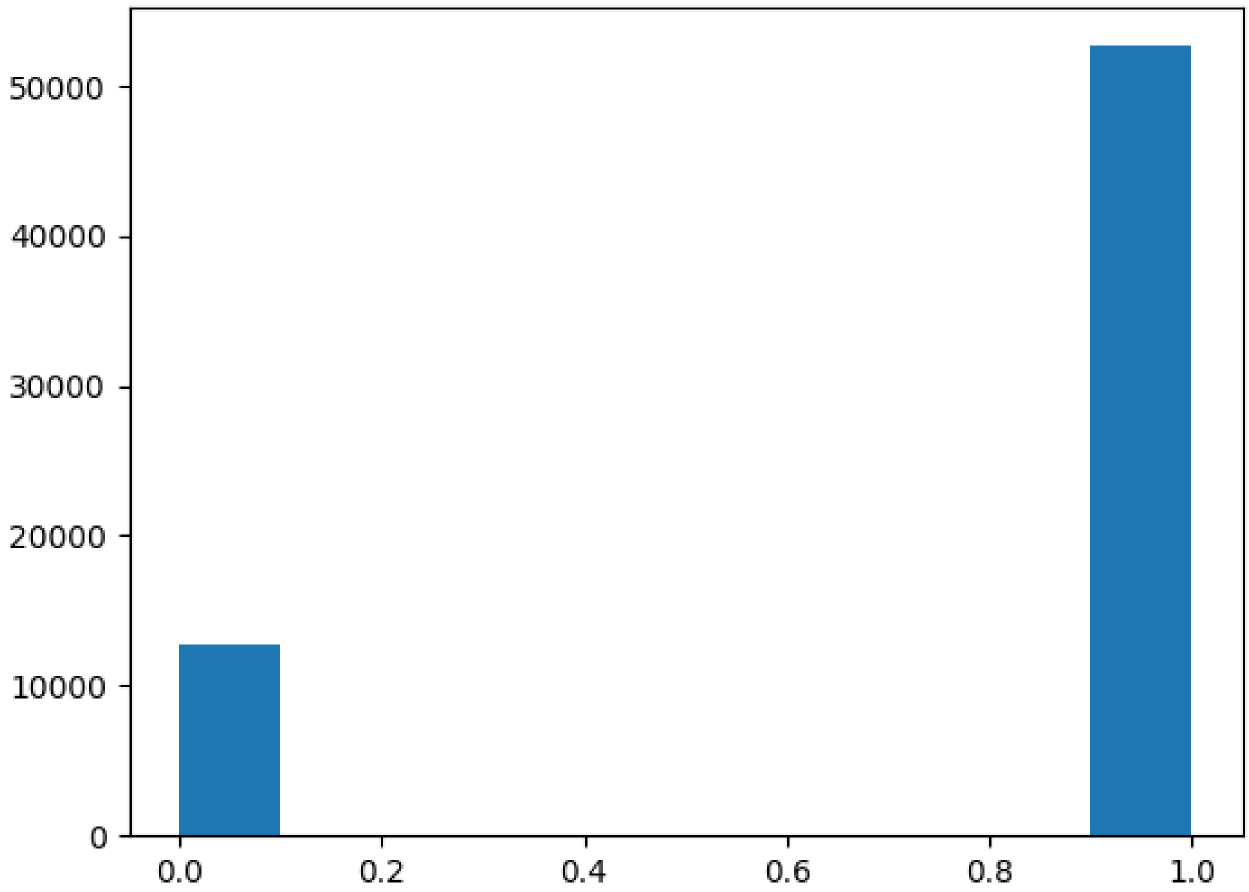


Figure 23- Histogram - [Alcohol]

Fig 23. Histogram of the Alcohol user and non-alcohol user distribution of the dataset.
0- Non-Alcohol User, 1- Alcohol User

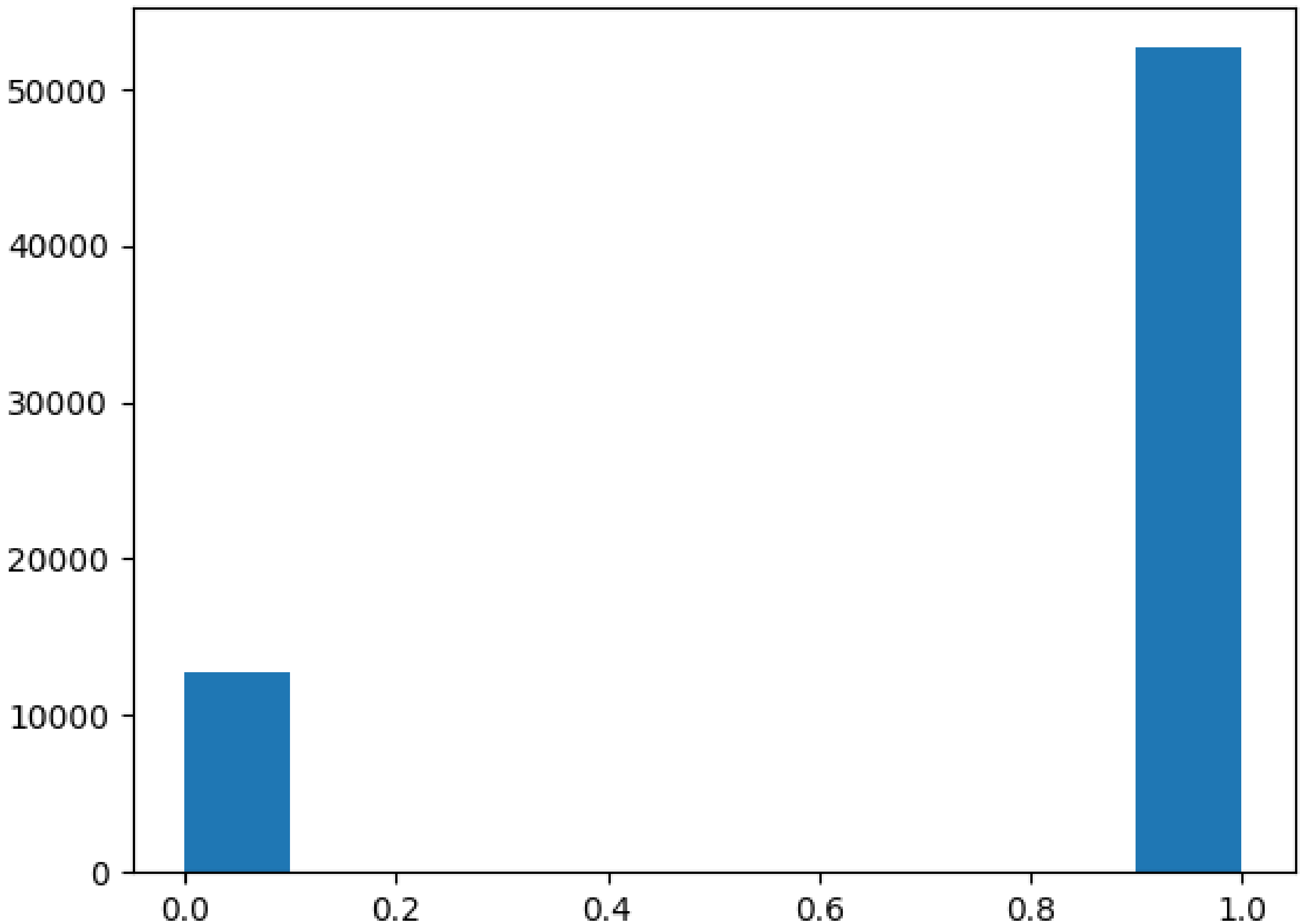


Figure 24- Histogram - [Active]

Fig 24. Visualize patient physical active [1] or not active [2]

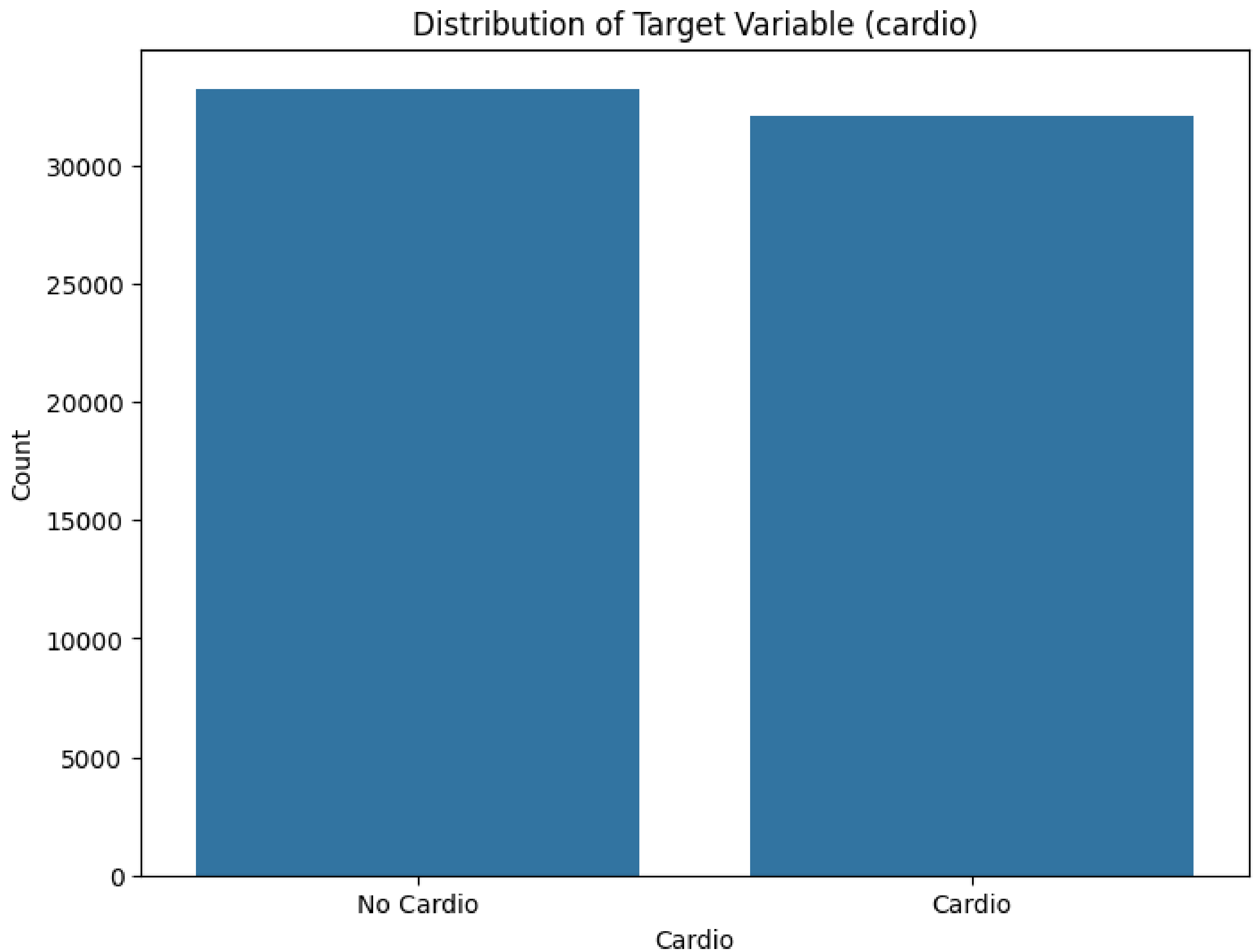


Figure 25- Histogram - [Active]

Fig 25. visualizes the target variable distribution between non-risk [0] and risk [1]. The dataset is well balanced between the target variables.

3.3.3 Numerical Attributes Identification and Distribution

Data distribution of numerical attributes after removing outliers visualized.

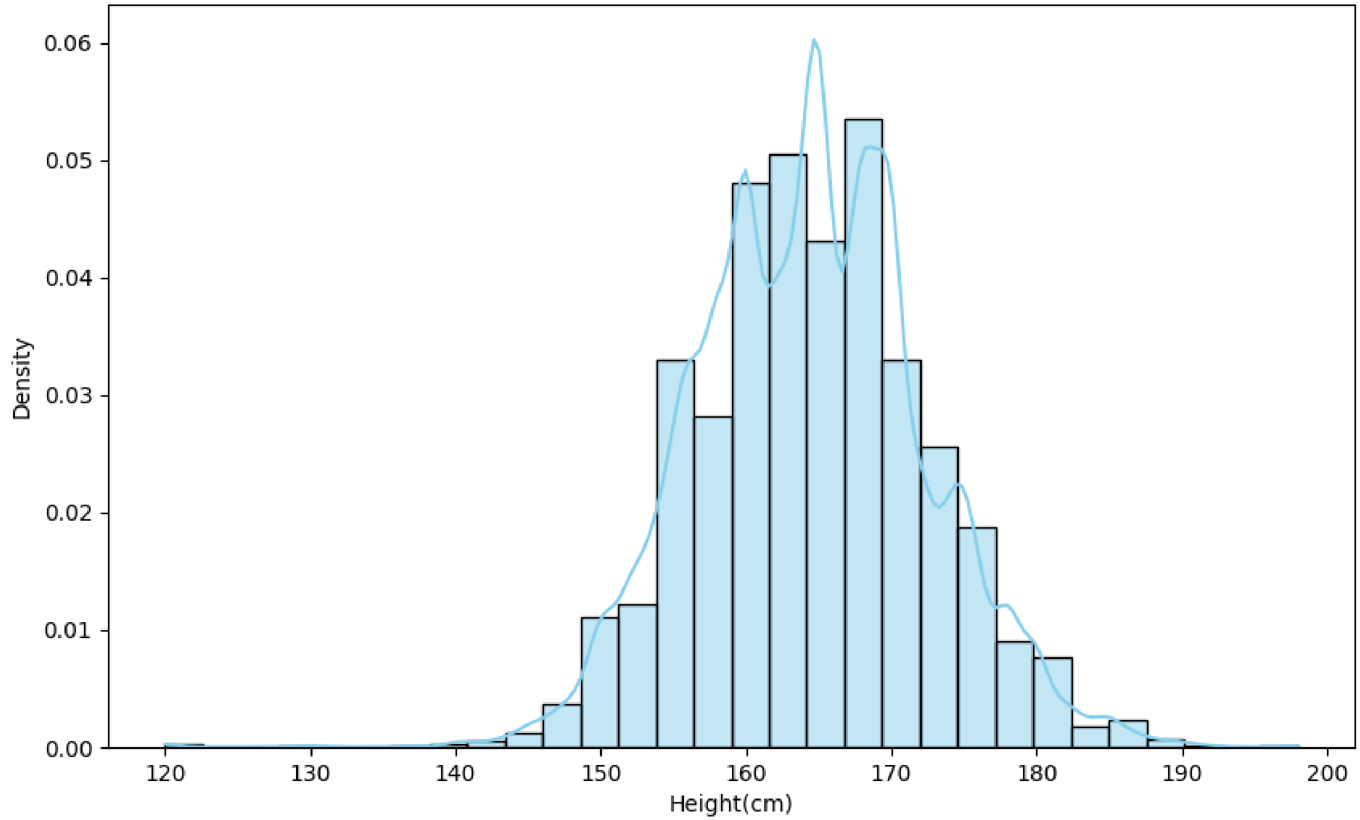


Figure 26- Histogram with KDE (kernel estimate density) - [Hight (cm)]

Fig.26 shows that most of the patients[candidates] height falls into the 155 to 175 range. Most individuals have heights concentrated around the central peak.

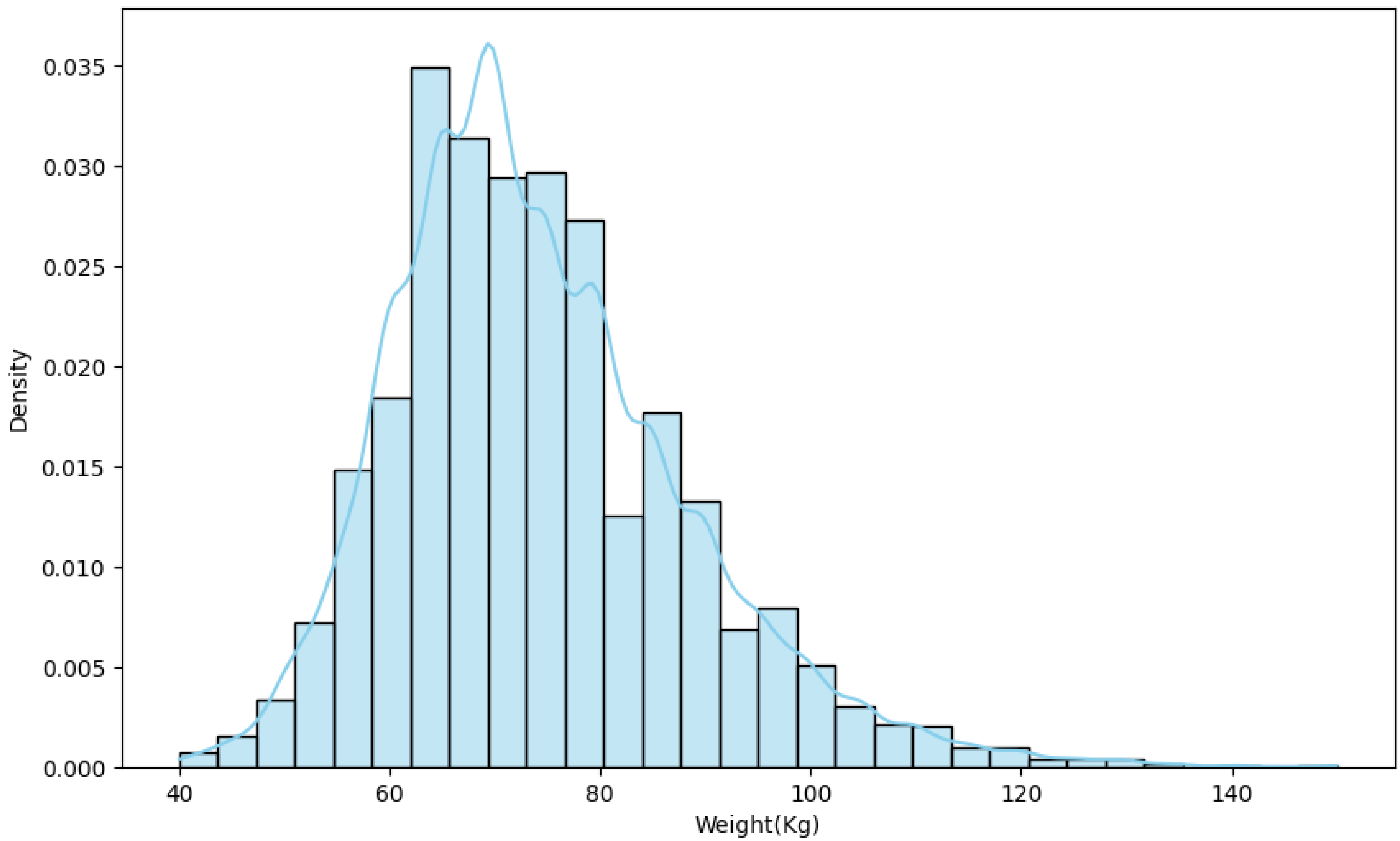


Figure 27- Histogram with KDE (kernel estimate density) - [Weight (kg)]

Fig 27 illustrates the weight distribution over the dataset. The distribution appears to be unimodal with peak around 60-70 kg. Most patient[candidate] around this range.

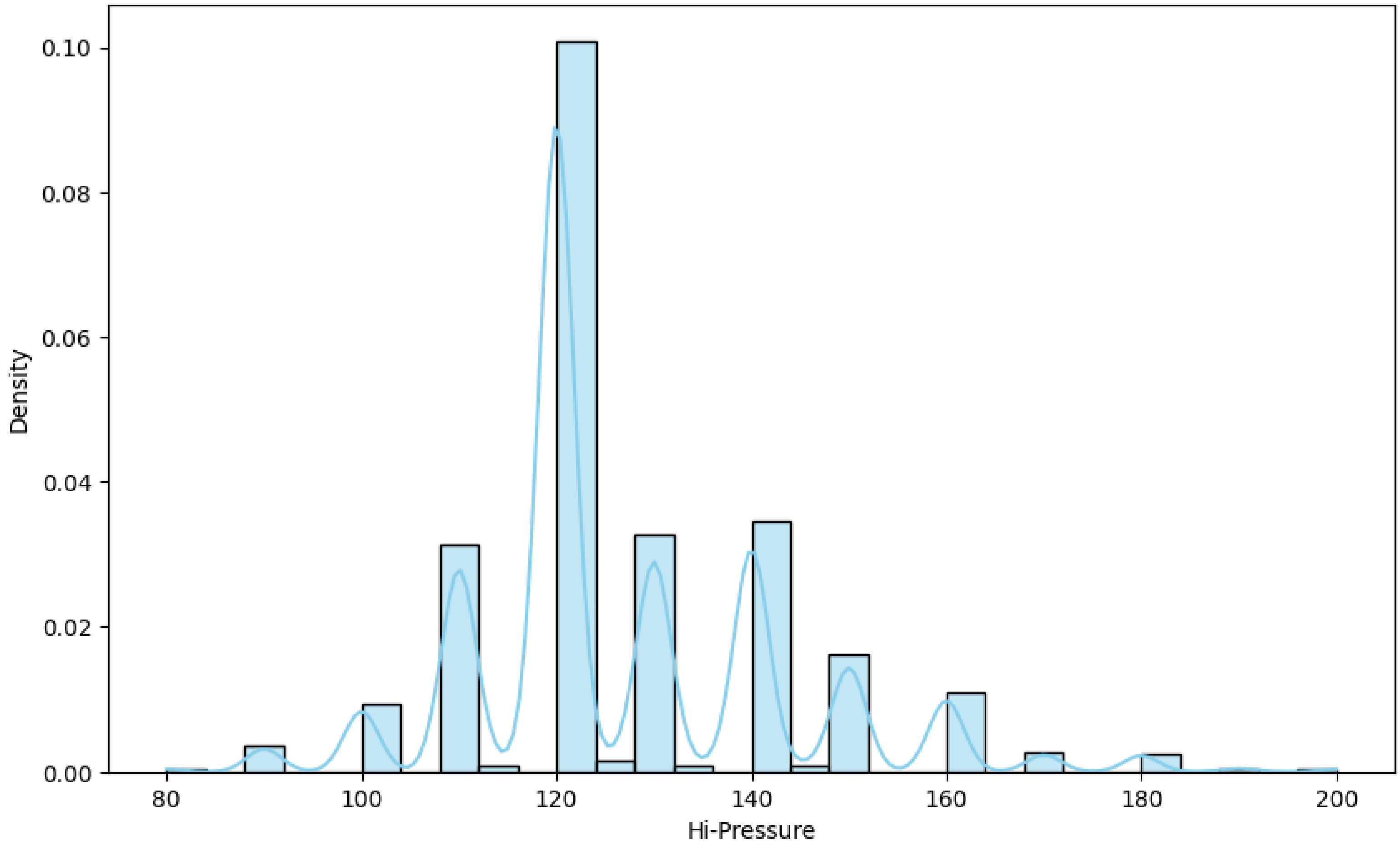


Figure 28- Histogram with KDE (kernel estimate density) - [High Pressure (Hg/mm)]

Fig 28. Shows histogram of high-pressure distribution. Most significant peak is around 120 Hg/mm and normal high-pressure range is between 110Hgmm – 130Hg/mm. This distribution is aligned with the standard distribution of high-pressure according to the Harward health publication (2024)

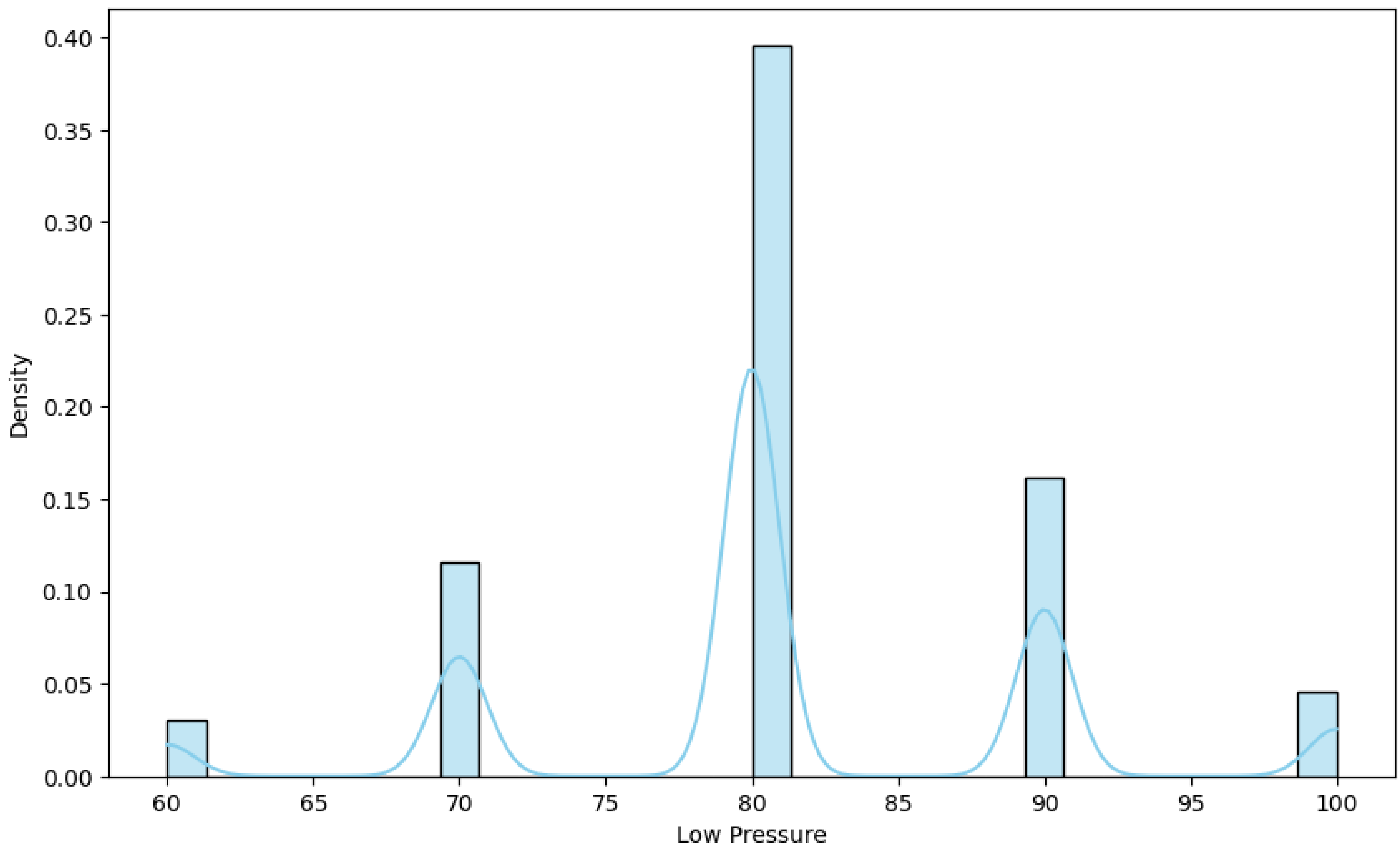


Figure 29 - Histogram with KDE (kernel estimate density) - [Low Pressure (Hg/mm)]

Symantec distribution of low-pressure data shown in figure 29. The peak at 80 can be considered a typical "Low Pressure" value. Primary peak (80) and secondary peaks (70 and 90) acceptable as normal but other peaks considered as high medical attention scenarios. Harward health publication (2024). Low pressure distribution is acceptable throw-out the dataset.

3.3.4 Dataset Balance or Imbalance

A balanced dataset is crucial for reliable and well performed ML model implementation. Specially balance target variables, enable model to train diverse scenarios. This is highly important for delivering accurate results in differentiating learning occurrences and eliminating biases.

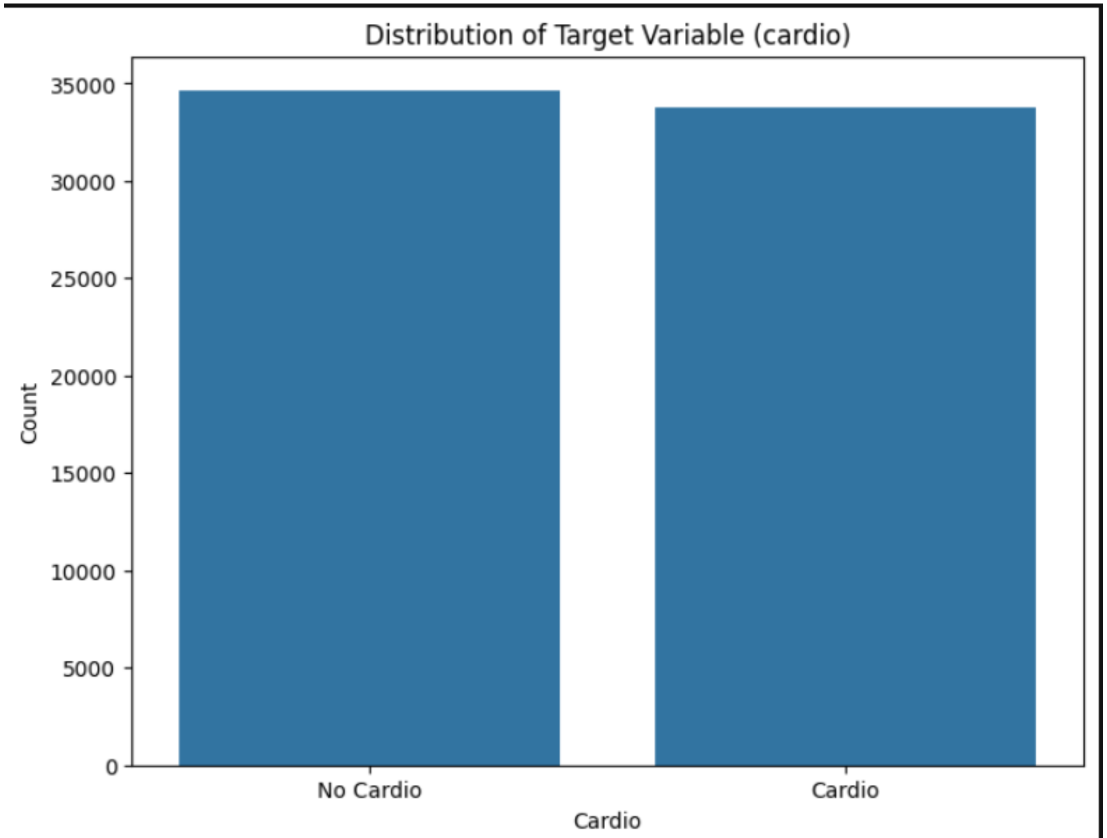


Figure 30- Target variable (Cardio) Distribution

According to fig.30 cardiovascular dataset is well balanced dataset.

3.4 Data Standardization

Data standardization in machine learning is a key functional task for an effective solution. It may increase efficiency, performance and the accuracy of the model. Standardization is the process of transforming different scales of features into common scale without affecting structure or distribution.
In this scenario, *StandardScaler* provided by *sklearn* is used to scaling feature values. *StandardScalere* expressed as,

$z = x - \mu / \sigma$

x - original feature value

μ – mean value of the feature

σ – Standard deviation

z- standardized feature value

3.5 Feature Selection- Co-relation / Mutual-Information / ChiSquare-Test

Hui H. and Cheng W. (2010) discuss the importance of feature selection, process of the feature selection and identification as well as different techniques behind that. Feature selection helps to build a simplified model by avoiding complex and unexplainable relationships and values. Furthermore, helps to achieve shorter training time, variance reduction helps to precision of the estimates and enhancement of generalization by eliminating redundant features and avoid noise.
Understanding co-relation between features with the target variable helps to determine how they are related and identify less related or redundant features that create unnecessary complexities and overlapping.

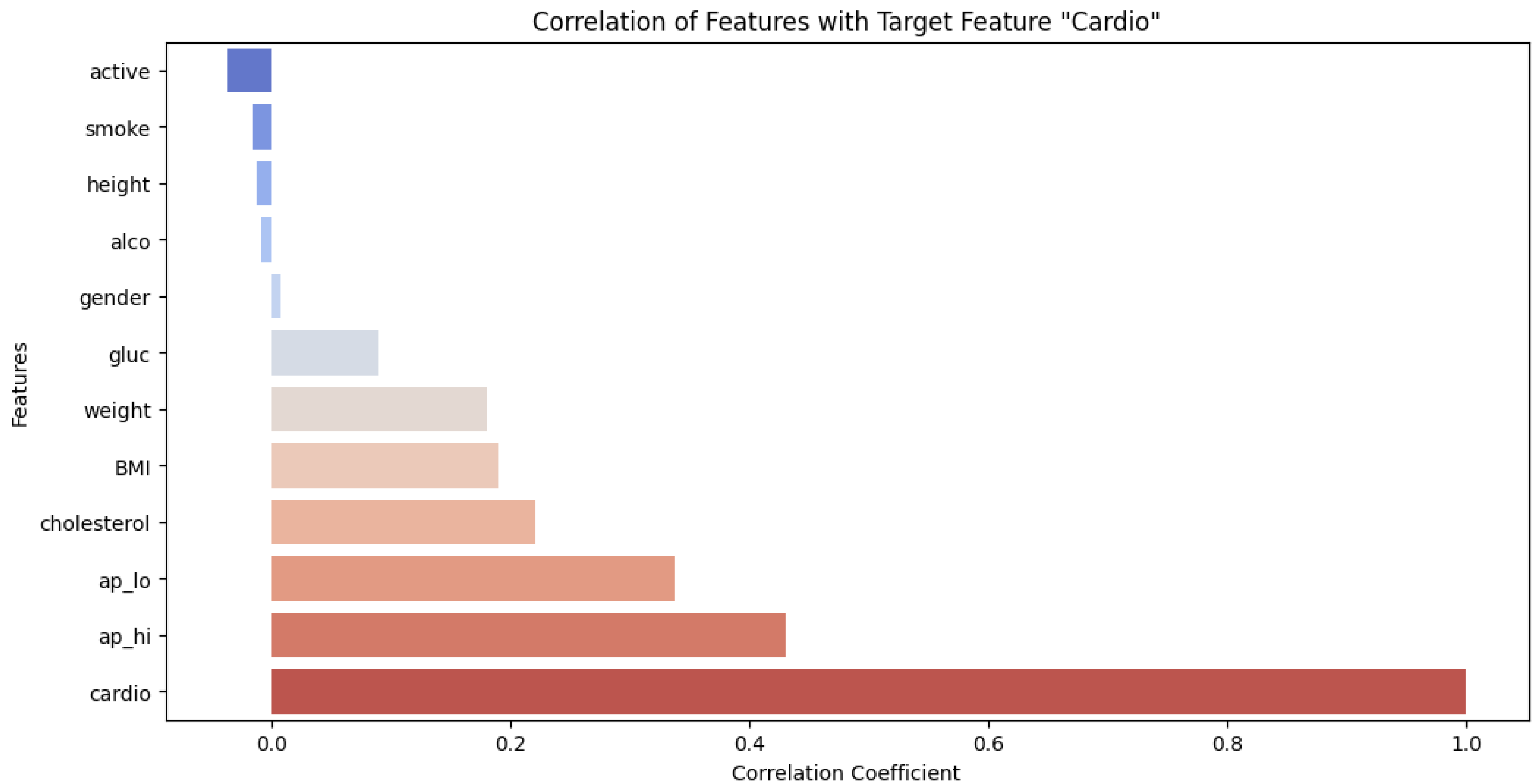


Figure 31 – Correlation of each feature attribute with the target variable 'cardio'

Fig.30 and Fig.31 High blood pressure(ap_hi),Low blood pressure(ap_lo),cholesterol, BMI, weight and blood sugar level(gluc) are highly positive co-relation with the target variable. Furthermore, physical activeness(active),smoke, height and use of alcohol negatively co-related with the target variable.

Syafaah L and Arisnanda D(2020) discussed about risk factor analysis for heart disease prediction which is highly align to this research using chi-square. This analysis is useful for understanding co-relation of categorical variables with the target using, frequency base analysis.

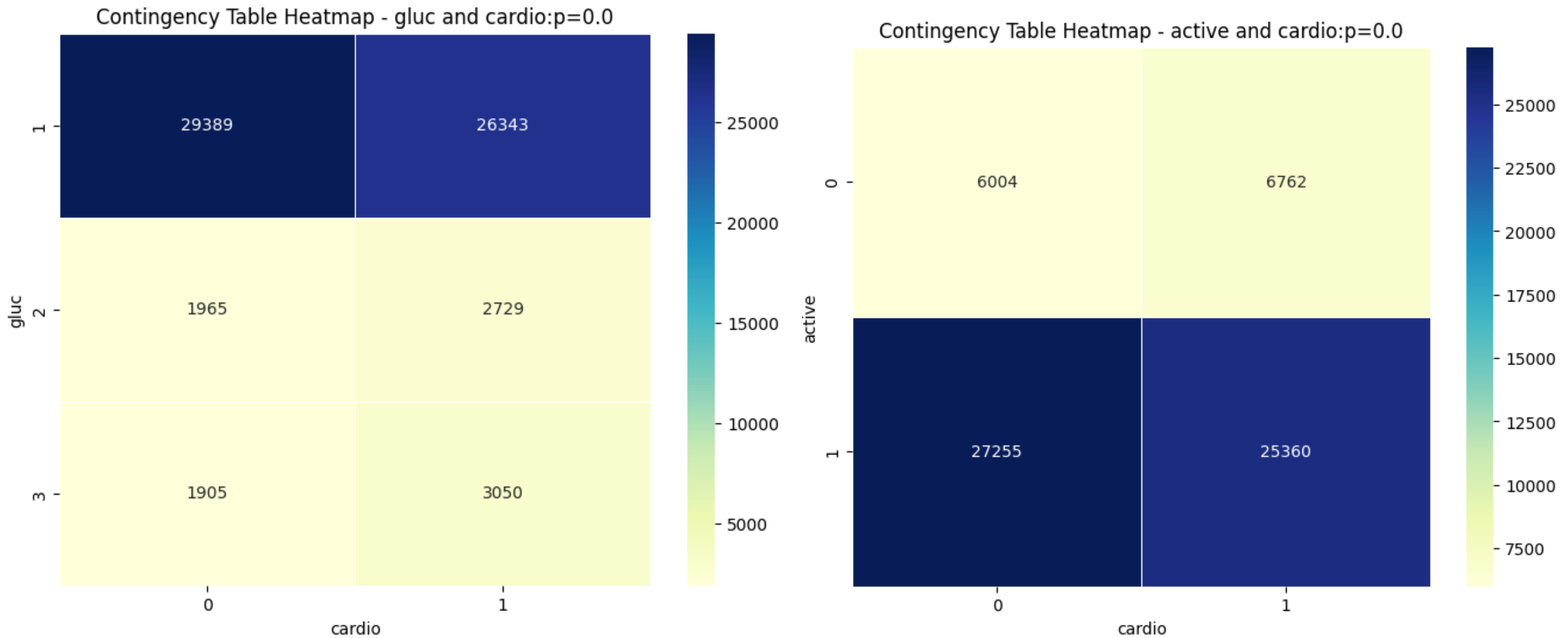


Fig. 32 - illustrates the relationship of two categorical variables (‘gloc’ and ‘active) with the target ‘cardio’ by highlighting the frequency of occurrences demonstrating the strength of the relationships and the co-relation. Same methodology to retain categorical variables to enhance feature understanding. P<=0.05 indicates reject the null hypothesis of no relationship between feature and the target.

Further analysis of features identification decides to go with the ‘Mutual Information’. Muhammad and Jane(2015) discussed importance of employed mutual information on feature selection linear and nonlinear scenarios, especially mutual information works well with both numerical and categorical data. Mutual information is calculated between two variables and measures the reduction in uncertainty for one variable given a known value of the other variable. MI can simply describe

$I(X; Y) = H(X) - H(X \mid Y)$

While 'I' is mutual information of X on Y variable, while H(X) is unpredictable result of occurrence of X and H (X | Y) is conditional entropy measure the remaining uncertainty of X after considering Y

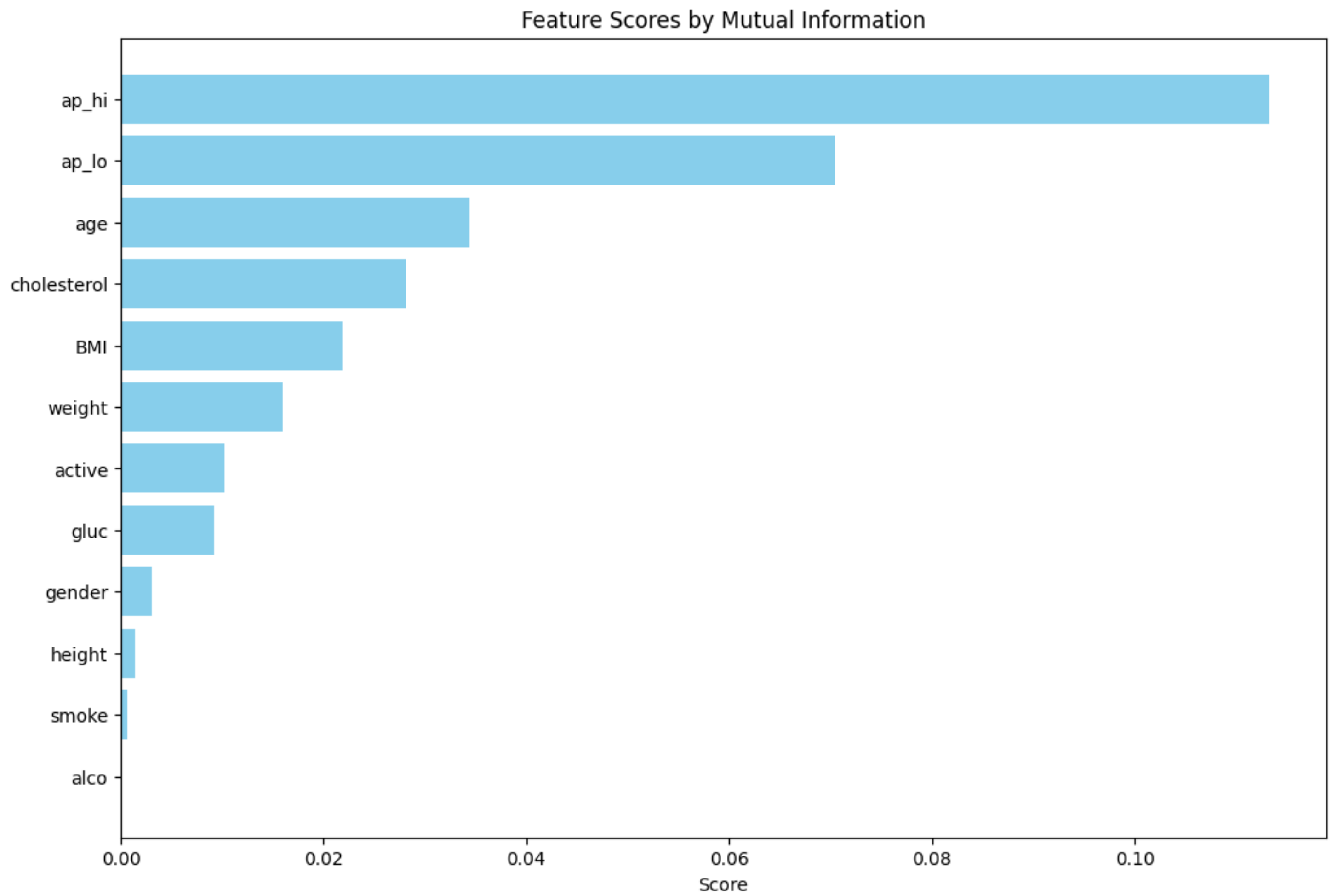


Figure-32 Features scores by Mutual Information.

Fig.32 shows the importance of different features based on their mutual information with the target variable of cardio. It highlights that high blood pressure(ap_hi) and diastolic blood pressure(ap_lo) are the most important features, followed by age and cholesterol levels. Other features like BMI, weight, and activity level also contribute some information, but to a lesser extent. Features like height, gender, smoking, and alcohol consumption have the lowest importance scores, indicating they provide less information about the target variable.

While statistical indicators like co-relation indicators and mutual information scores offer decent information on features attributes, domain knowledge plays a crucial role in the process of feature selection. For this instance, features like blood pressure, age and cholesterol` rank high co-relation as point of statistically on the other hand domain knowledge might prioritize other features like smoking and alcohol consumption base on their real-world impact, even If there is low statistical importance.

3.6 Machine Learning Models / Algorithms Used

After all the data pre-processing, feature engineering and EDA were completed, machine learning models were used to train the cardiovascular dataset. Our study uses machine learning (Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM),K-nearest neighbor(KNN),Random Forest(RF),Naïve Bayes(NB),Light Gradient Boost(LGB) and XG-boost(XGB), and deep learning algorithms with ensemble classifier use of best performing three machine learning models.

3.6.1 Logistic Regression (LR)

Logistic Regression is the widely used machine learning algorithm specifically for binary classification. Logical regression analyzes the relationship between one or more independent variables and classifies data into discrete classes or specific category. Logistic regression uses sigmoid function output probability of 0 or 1 from linear set of input features.

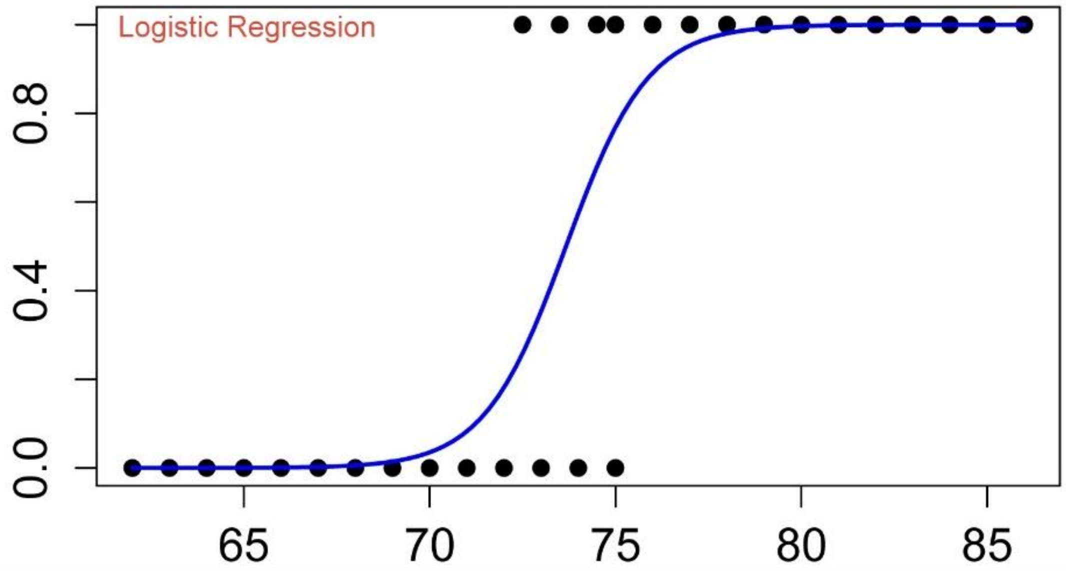


Figure-33 Logistic regression Sigmoid Function behavior with the feature variable.

Fig.33 demonstrates the sigmoid curve, above the 0.5 output is considered as 1 and less than 0.5 considered as 0. The formula of the sigmoid function is,

$$Y = \frac{1}{1 + e^{-Z}}$$

With the replacement of formular of logistic regression to z

$$Y = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

1+e is an Exponential function which converts linear output into logistic representation.

β_0 - intercept

$\beta_1 x$ - coefficient of the linear regression

$\beta_0 + \beta_1 x$ is equivalent to the linear regression $y = mx + c$, that can listed for every feature values as below equation. $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ldots \beta_x x_x$ where x equals to feature values.

3.6.2 Random Forest Classifier (RF)

Random Forest Classifier is the ensemble of multiple decision trees and final decision is based on the voting output of each decision tree. RF can use both classification and regression problem effectively and effectively manner. RF uses bagging bootstrap sampling on the training set which creates random sampling with replacements for effective training. This helps to reduce overfitting and improve accuracy.

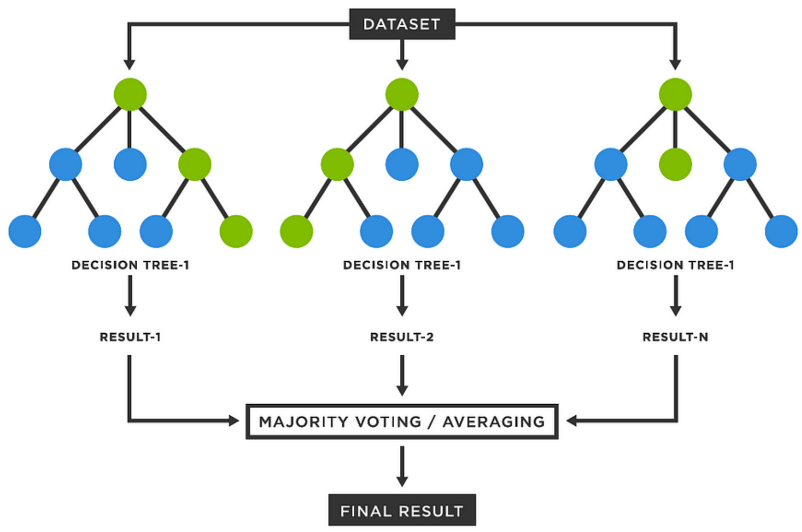


Figure-34. Visualize the Random Forest Diagram (Deniz 2023.)

In RF core algorithm base on decision tree's core concept of Gini impurity. It helps to measure the pure node in a decision tree. In the Random Forest scenario, way the data has been split among each node of the tree by selected feature attribute with minimum Gini Index.

The formular of the Gini Index is given below

$$Gini = 1 - \sum_{j} p_j^2$$

P_j - Data points in the Class (True category or False category)

n - Total data points. (Both categorical classes)

Gini Index = 0: The node is pure, meaning all samples belong to a single class.
Gini Index = 1: The node is highly impure, meaning the samples are equally distributed across all classes.

3.6.3 Support Vector Machine (SVM)

Support Vector Machine is supervising learning algorithms primary on classification. SVM has boundary recognition called hyperplane which separates different classification classes. Identifying the hyperplane with maximum margin among possible hyperplanes helps to better classification on future predictions.

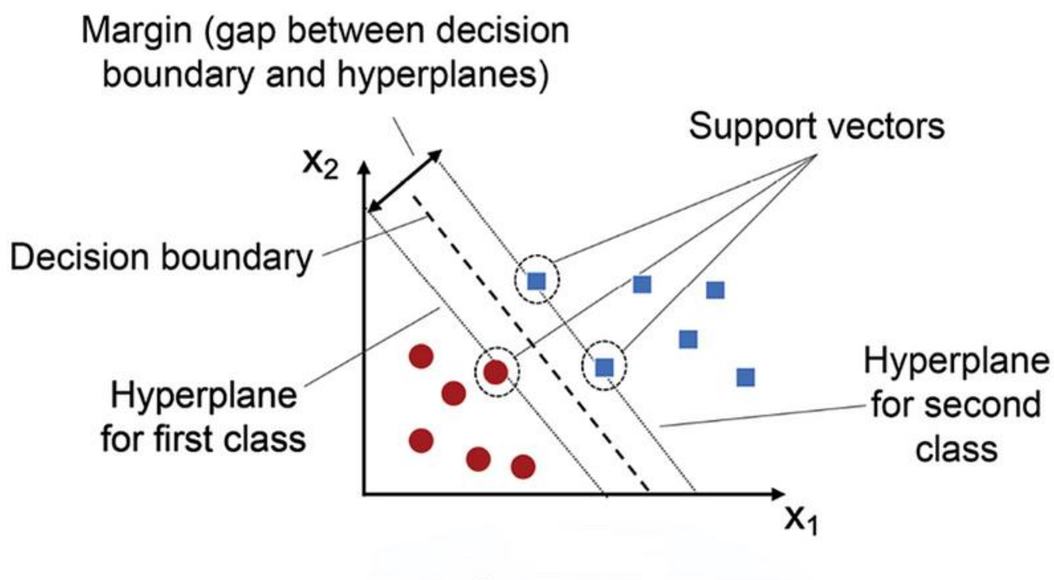


Figure-34. SVM visualization Tasmay P. (2023)

Fig.34 SVM finds a hyperplane that maximizes the separation between classes as above. In two-dimensional space, a hyperplane is simply a line that separates the data points into two classes. In three-dimensional space, a hyperplane is a plane that separates the data points into two classes. Similarly, in N-dimensional space, hyperplane has (N-1)-dimensions. *Tasmay(2023)*

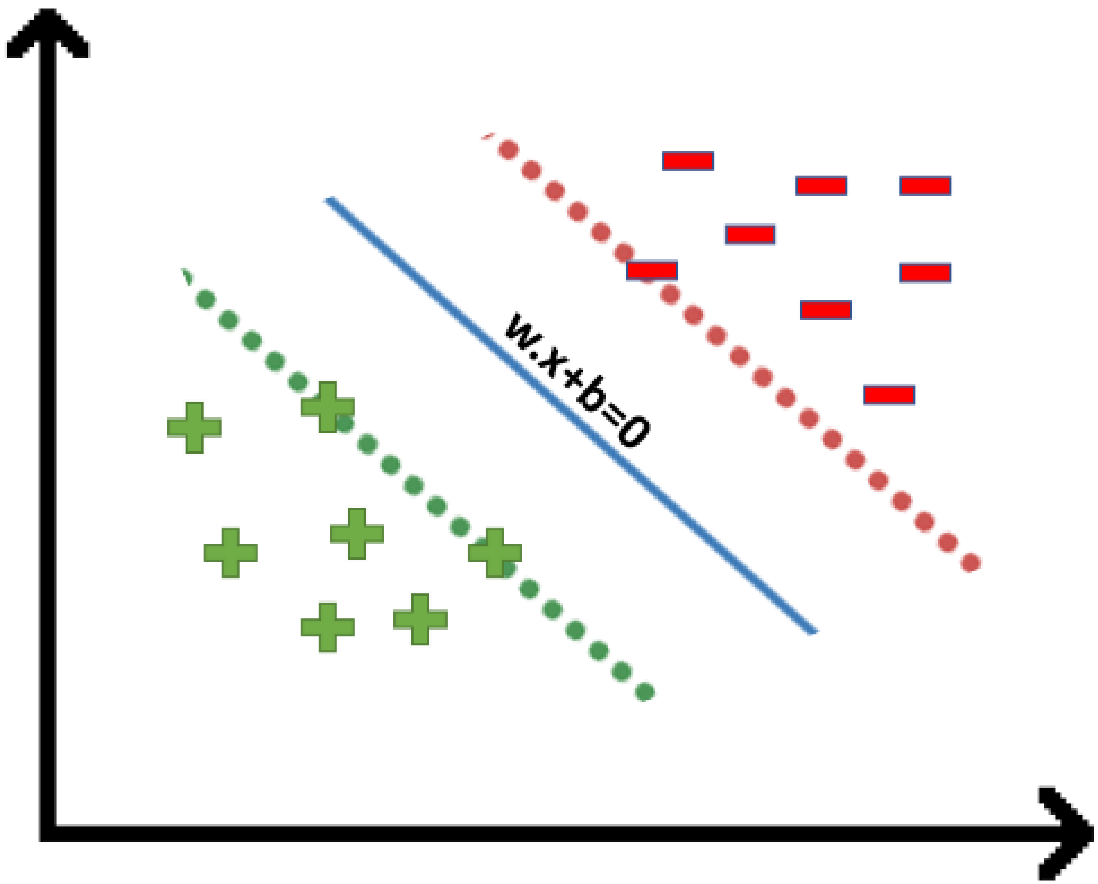


Figure-35 Equation of Hyper Plance. (Anshul S. 2024)

$w * x + b = 0$

w - weight

x - feature vector

b- bias

$$\overrightarrow{X}.\overrightarrow{w} - \mathrm{c} \geq 0$$

$$\text{putting } -\mathrm{c} \text{ as b, we get}$$

$$\overrightarrow{X}.\overrightarrow{w} + \mathrm{b} \geq 0$$

$$\text{hence}$$

$$\mathrm{y} = \begin{cases} +1 & \text{if } \overrightarrow{X}.\overrightarrow{w} + \mathrm{b} \geq 0 \\ -1 & \text{if } \overrightarrow{X}.\overrightarrow{w} + \mathrm{b} < 0 \end{cases}$$

According to Anshul S.(2024)
Yi=+1, the point is on one side of the hyperplane.
Yi=−1, the point is on the other side.

3.6.4 K-nearest Neighbor (KNN)

The K-Nearest Neighbors (KNN) algorithm is a supervised machine learning model suitable for both regression and classification problems .KNN is simplest, but essential classification algorithm brings highest accuracy most of the occasions.

The K-NN algorithm works by finding the K nearest neighbors to a given data point based on a distance metric.

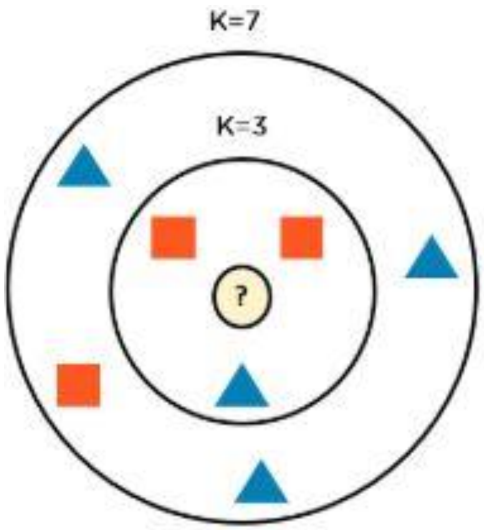


Figure-36 Simplified description of KNN

Fig. 36 describes two instances; First is K=3 and Second is K= 7.
K=3: Draw a circle around the candidate data point to predict, encloses 3 nearest neighbors
K=7: Draw a larger circle that encloses the 7 nearest neighbors. The prediction is based on the majority label or average value among these 7 neighbors. KNN makes prediction based on majority votes from neighbors. Best result K should be odd number. KNN use Distance measures including Euclidean, Manhattan and Minkowski used based on the problem domain. K is a hyperparameter that needs to be evaluated.

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

Figure-37 Euclidean Distance formula (cartesian distance between the two points)

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

Figure-38 Manhattan Distance (absolute difference between the coordinates)

3.6.5 Naïve Bayes (NB)

Naïve bayes is probabilistic classifier based on bayes theory. This is widely used for classification tasks. Bayes theory is based on conditional probability, event occurrence probability that after another event has occurred.
George and Pat(1995) clearly discuss Bayesian theory and concept of naïve bayes and assumption that made.

Probability of B occurring given evidence A has already occurred

Probability of A occurring

$$P(\mathrm{A}|\mathrm{B}) = \frac{P(\mathrm{B}|A) \cdot P(A)}{P(B)}$$

Probability of A occurring given evidence B has already occurred

Probability of B occurring

Figure-39 Nagesh S. (2022) Mathematical formular for buyers theory

Naïve Bayes assumes each feature is independent and all features are equally contributed to the outcome

3.6.6 Extreme Gradient Boost(XGBoost)

XG-Boost is ensemble learning classification use of multiple learning techniques that is under Gradient Boosting decision tree framework. This enables highly efficient and scalable learning opportunities that optimized for speed and performance. Gradient Boosting builds an ensemble of decision trees in a sequential manner. Each new tree corrects the errors of the previous ones, with the goal of minimizing a loss function. This has an effective learning mechanism and regularization technique to prevent over fitting. Tianqi(2016) discusses undelaying concepts behind the extreme gradient boost algorithm.

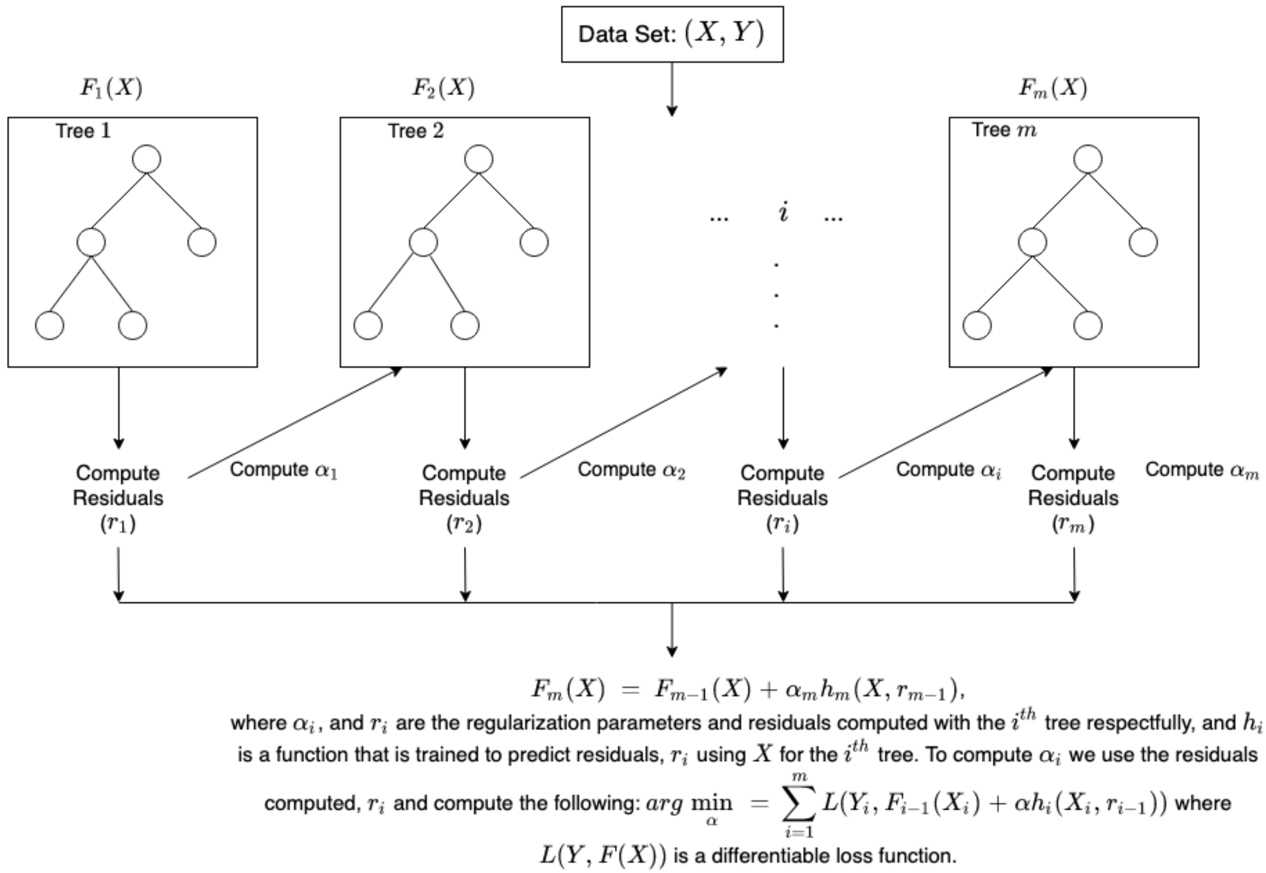


Figure-40 Gradient Boost- Decision tree boosting algorithm. Amazon Web Services, Inc. (2024)

3.6.7 Light Gradient Boost(LGB)

Light Gradient Boost is an ensemble learning framework that brings high levels of accuracy, scalability and efficiency. LGB is structured using decision trees as ensemble manner. Jun and Xin(2019) discussed tree base LGB algorithm and its effective meteorological predictions. Fatimah(2020) demonstrated performance of LGB in sentiment classification. Special research mentioned about LGB capabilities of dealing with uneven datasets and high dimensional data.
LGB's leaf-wise tree growth allows it to build deeper trees, which can capture complex patterns in the data, leading to high accuracy in classification tasks. Furthermore, LGB enables us to tune the model using a wide range of parameters which can more accurately result in classification scenarios.

After pre-processing, allows to find the difference of each leaf node and find the best node with better improvements.

$$\tilde{V}_j(d) = \frac{1}{n}\left(\frac{\left(\sum_{x_i \in A_l} g_i + \frac{1-a}{b}\sum_{x_i \in B_l} g_i\right)^2}{n_l^j(d)} + \frac{\left(\sum_{x_i \in A_r} g_i + \frac{1-a}{b}\sum_{x_i \in B_r} g_i\right)^2}{n_r^j(d)}\right)$$

Guolin and Qi M(2020) discussed Exclusive Feature Bundling (EFB) and Gradient-based One-Side Sampling (GOSS), which effectively manage features especially in large scale scopes and accelerate the training process in efficient manner.

3.6.8 Multi-Layer Dynamic System with voting (MLDS)

Multi-layer dynamic implementation of machine learning is another best practice to develop ensemble model with higher accuracy with better performance. Mohammed Rajith(2021) discusses about multi-layer dynamic system that can improve the current knowledge of each layer by training next layer with previous success predictions. This process of knowledge passing enchases accuracy with the voting classifier which evaluates each model accuracy.

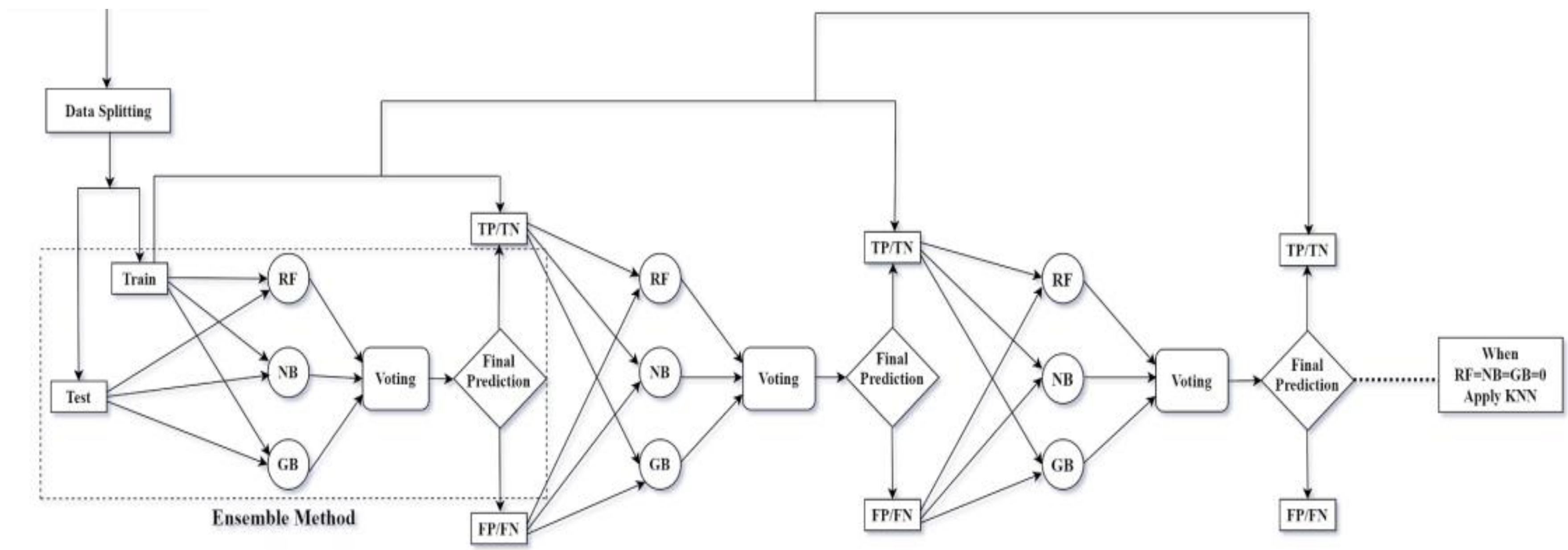
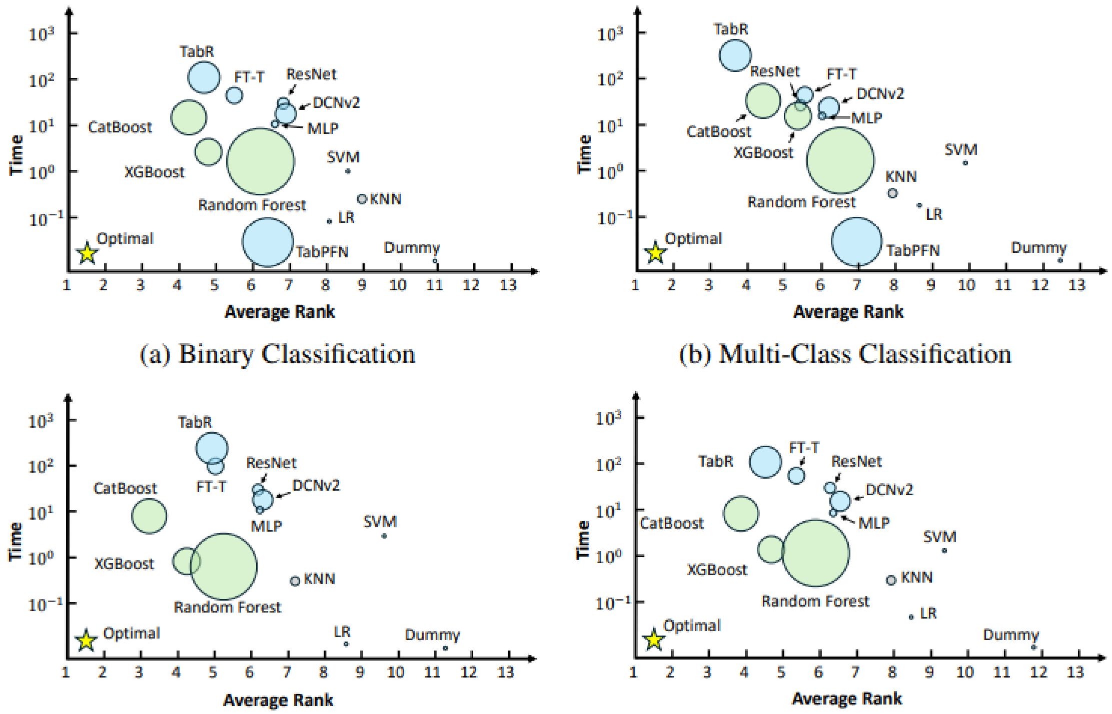


Figure 44 visualizes multi-layer ensemble model. Mohammed N. and Rajib K. (2021)

3.6.9 Deep Learning Classifiers with Tabular Data

Han J. and Si-Y(2020) discussed the deep learning concepts over the tabular data and key factors that influence the deep learning.



(a) Binary Classification (b) Multi-Class Classification

(c) Regression (d) All Tasks

Figure-41 Han J. and Si(2020) Performance-Efficiency-Size comparison of representative tabular methods on our benchmark for(a) binary classification,(b) multi-class classification,(c) regression tasks, and (d) all task types.

Fig-41 visualizes different machine learning models(RF, LR, SVM, KNN) and deep learning models(Tabnet, Rasnet) performance in different benchmarks. Deep learning models have their own ability to capture complex features and behaviors.

3.6.9.1 TabNet - PyTorch

TabNet is a deep neural network which is capable of operating in tabular data environments and well performed on large dataset specially with the complex feature scenarios. Sercan and Pfister (2019) discussed efficiency, performance of TabNet over tabular data and TabNet's sequential attention mechanism that selects features in an effective way. TabNet was created by Google using PyTorch library called 'pytorch-tabnet'.

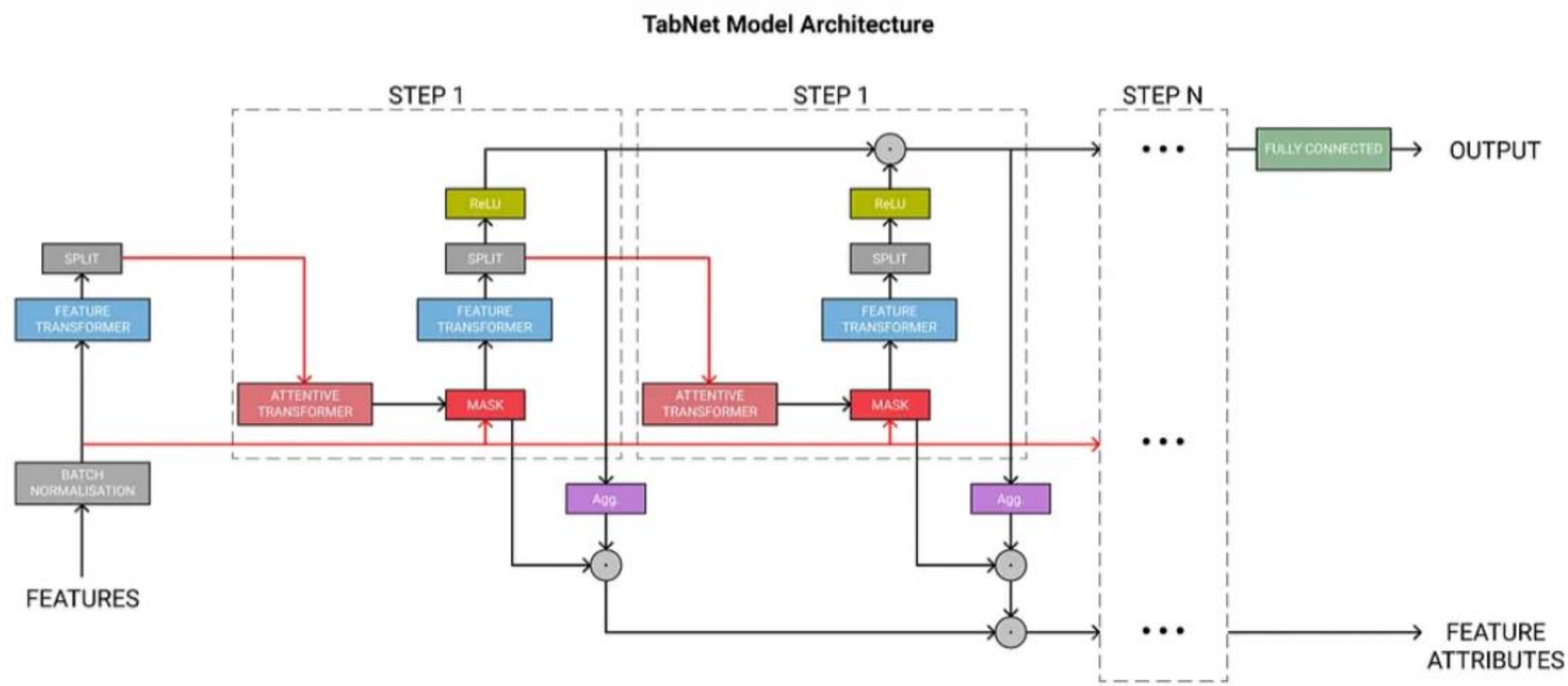


Figure – 42 TabNet Architecture Shafi A(2021)

TabNet, process data in series of steps, each step used feedback and scoring mechanisms to identify the most relevant features. This feature prioritization technique helps to contribute most of the predictions and decisions. Moreover, TabNet has multiple decision steps same as previous, marking aggregate predictions based on each sub-step results.
TabNet is a deep learning model that optimizes tabular data using sequential scoring mechanism to focus most valid features throughout the process.

3.6.9.2 Artificial Neural Network (ANN)

ANN is the computational model that is inspired by the human brain's neural structure. ANN consists of interconnected and interrelated nodes which are called neurons and those are organized as different layers. Information enters from the input layer and passes through the interconnected neuron environment with adjustments until finding the optimum value.

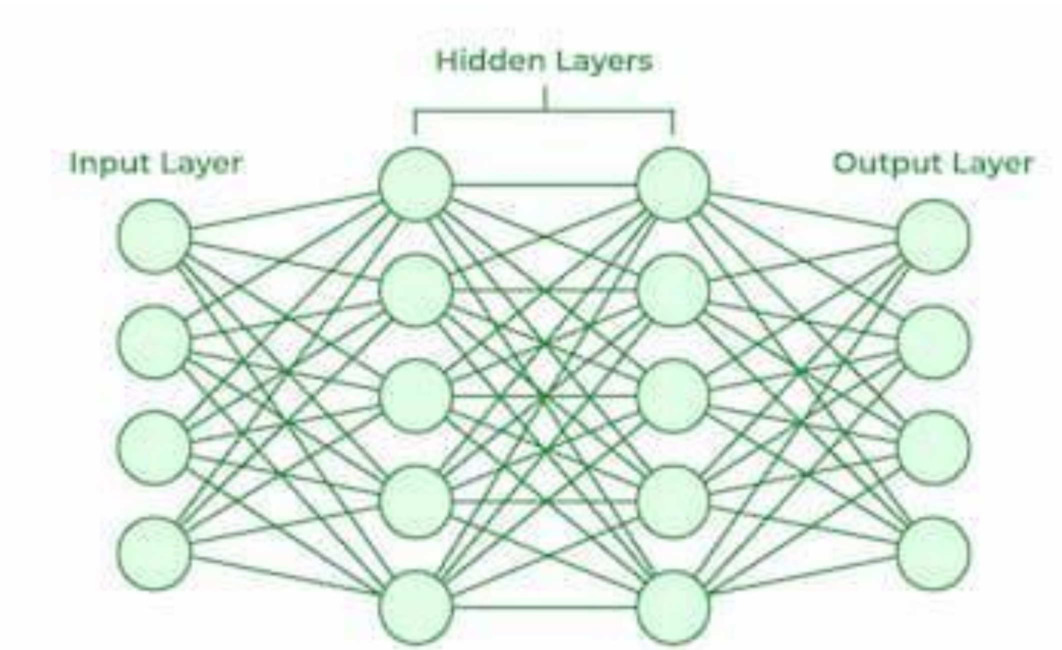


Figure – 43 visualize the ANN architecture

Fig.43 Input layer contains the neurons that equal to number of features. Weight is another core concept indicate that how strong is the connection between two neurons while each hidden layer multiple weight by input adds them and passed to the next layer. This process continues until reaches the output layer to the desired minima.

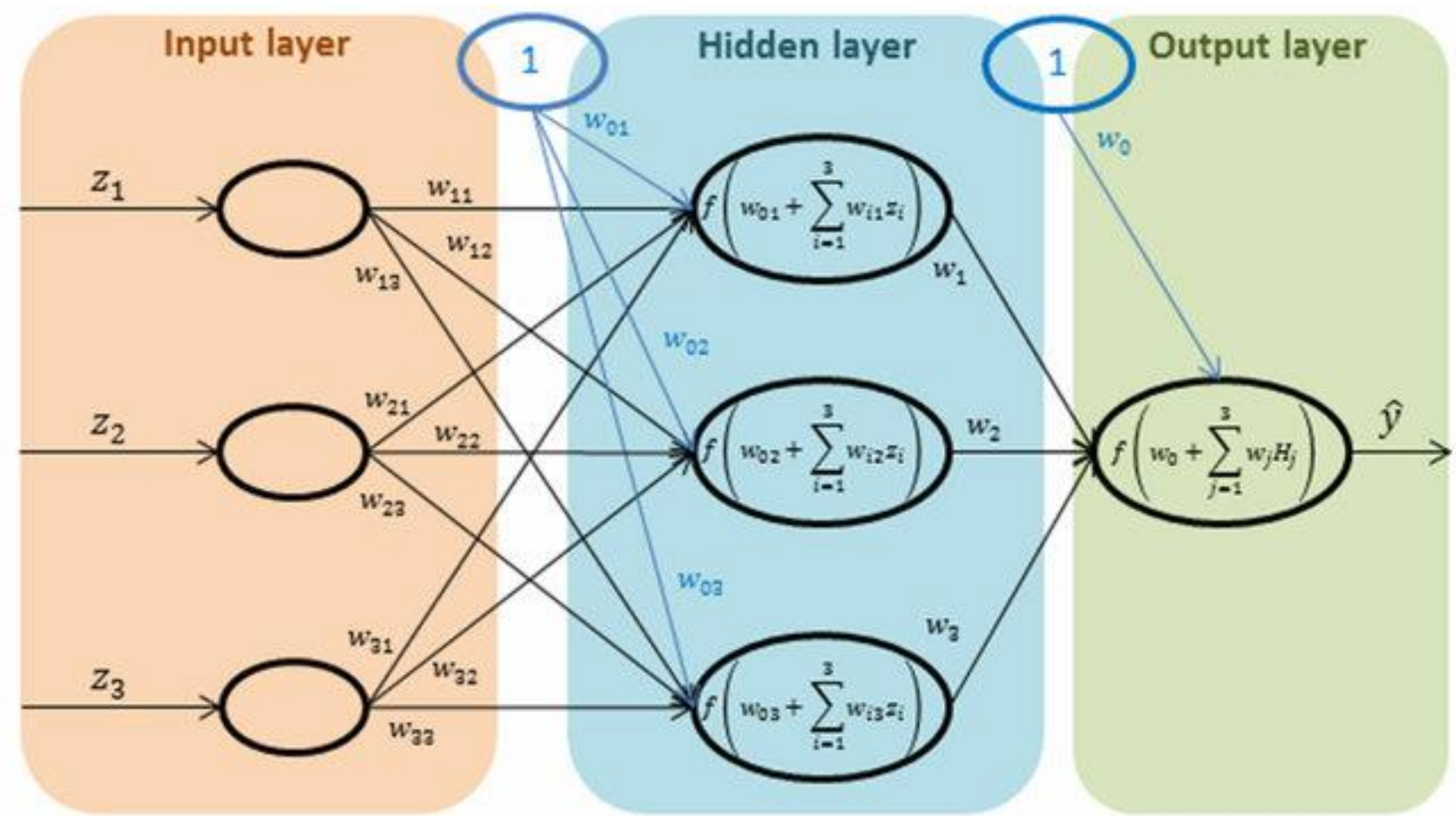


Figure – 43 Example of an artificial neural network (ANN) with three input variables, one hidden layer with three neurons inside and one output neuron (representing target variable). Krzysztof G.(2020)

3.7 Model Evaluation Metrics

Result and the Evaluation method are the key aspect of the latter part of the process. Evaluation matrixes reveal outside to accuracy, performance or the trustworthy of the model or the hypothesis. Nathalie(2020) describes the importance of evaluating and the evaluation techniques including Accuracy Scores, Precision/Recall,F1-score and confusion matrix.

3.7.1 Accuracy Scores

Accuracy score measured by the ratio of correctly predicted instances into total instances or the size of the dataset Shrikant(2023)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

TP=true positive, TN=true negative, FP=false positive, and FN=false negative,

3.7.2 Precision

Precision is a measure, number of positive predictions made correct (true positives), or precision Is the ratio of the predicted positive cases into total positive occurrences.

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision score indicates that model’s predictions of positive has higher level of probability to a correct, Further precision score low indicate the model’s positive predictions accuracy is low.This is very important especially in CVD predictions.

$$precision = \frac{true\ positives}{true\ positives + false\ positives} = \frac{terrorists\ correctly\ identified}{terrorists\ correctly\ identified + individuals\ incorrectly\ labeled\ as\ terroists}$$

Will(2023)

3.7.3 Recall

Recall factor determines the overall performance of the model or the hypothesis. Thus, for all the patients who have heart disease, recall tells how many correct predictions of heart disease.

$$Recall = \frac{True\ Positive(TP)}{True\ Positive(TP) + False\ Negative(FN)}$$

TP- number of positive instances correctly identified
FN- number of positive instances incorrectly identified as negative.
High recall score indicates the successfully identified positive instances, further low recall score indicates missing large number of positive instances. Especially CVD detection, this will be a serious issue.

3.7.4 F1-Score

F1 score is a means of both precision and recall. Steven A.(2022) describes F1 score as harmonic mean of precision and recall.

$$2 * \frac{Precision * Recall}{Precision + Recall}$$

F1-score,specially used in imbalance datasets, the occurrences where one class frequently appears than other.F1-score provide more balanced view. F1 score of near to 1 indicates model is perfect, while F1 score near 0 indicates the model not perfect.

3.7.5 Confusion Matrix

Confusion Matrix is the visualization indicator that describes the performance of classification model. That enables to visualize correct and incorrect predictions of all the classes
Sandeep S. (2022) illustrate the structure of confusion matrix-fig.44

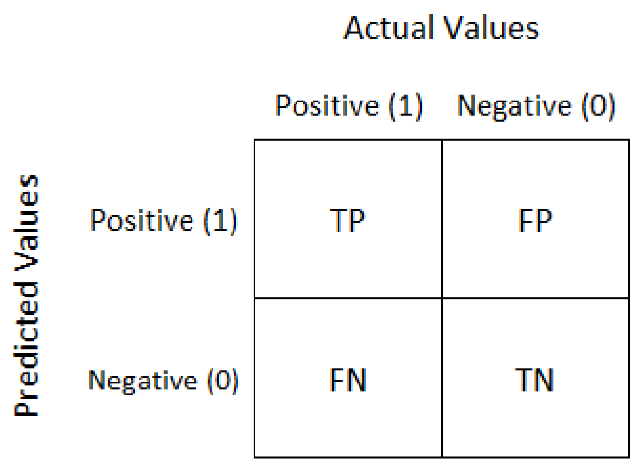


Figure-44 confusion matrix
True Positive(TP)-correct positive predictions.
False Positive(FP)-Incorrect positive predictions
False Negative(FN) -Incorrect negative predictions
True Negative(TN)-Correctly predictions of negative class

3.7.6 Classification Report

Classification report is a detailed overview of the classification model, with class base summarization of calculating metrics such as precision, recall,F1-score and the accuracy score.
According to the Muhammad A.(2023) discuss details of the classification report to understanding insights in depth.

3.8 Model Trainign

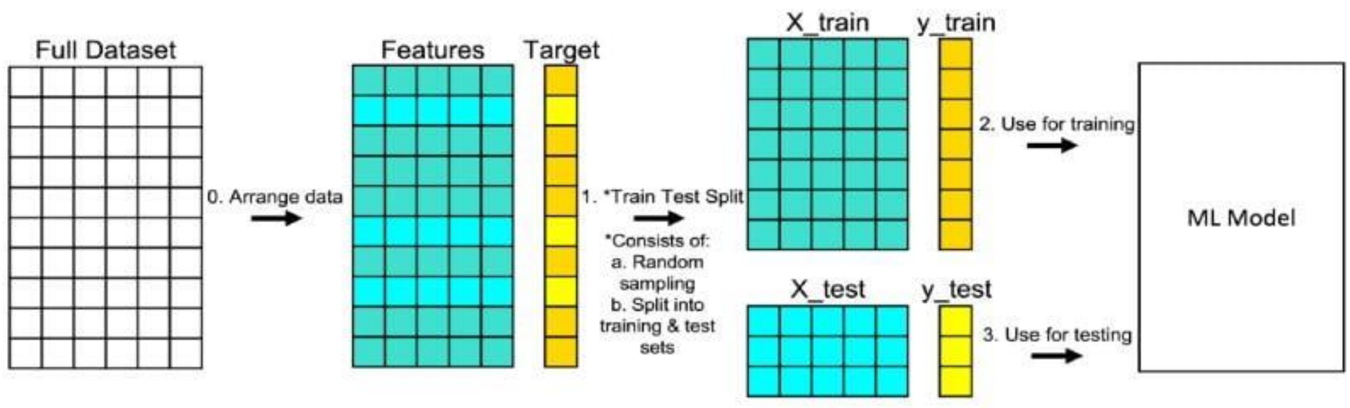


Figure 45- Train and Test Split Data
Source; Build-in Data Science Magazine by Michael Galarnyk 2022.

Reference to figure-45,data set arrange into two components features and target. Features represent all the independent variables or user inputs, and targets represent dependent variables. Split the dataset into two portions randomly. Michael G.(2022) discussed the train and test split and relevant recommend ratios.80% on training. 20% on testing or 70% on training, 30% on testing.

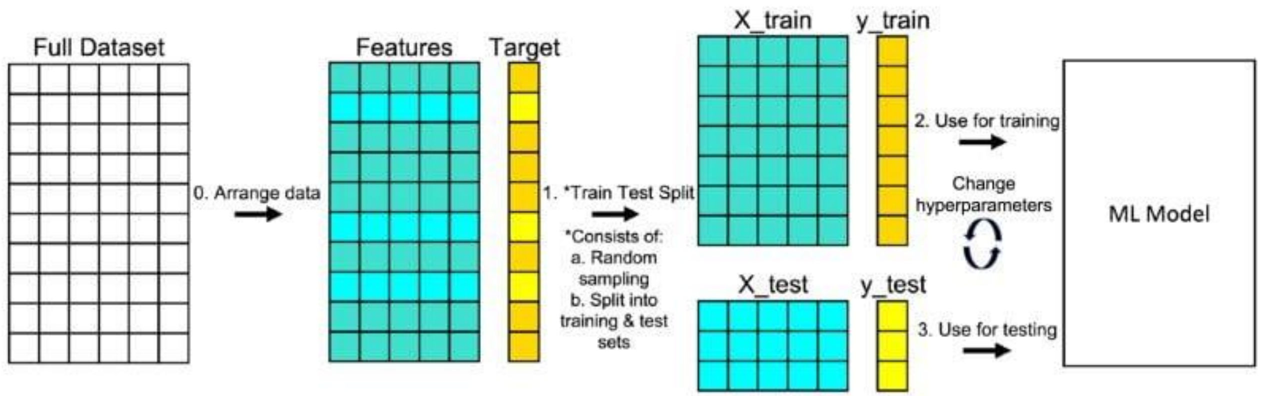


Figure-46. Change hyperparameters" in this image is also known as hyperparameter tuning. | Image: Michael G (2022)

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

sklearn.modelselection.train_test_split()-sklearn library help to split

3.9 Feature Scaling

Feature scaling is standardization and normalization process that requires if the feature data are measured in different scales and different standards. This data preprocessing task is a very crucial step to ensure consistency of measures and enhancement of the model performance. Especially in the gradient descent algorithm the process of find global minima, smooth update rate is required. With the different scalers process being complex. Further distance-based algorithms effected with different scales because they are using distances between data points to determine their similarity.

```
scaler = StandardScaler()
X_train_scalled=scaler.fit_transform(X_train)
X_test_scalled=scaler.transform(X_test)
```

Fig 46a. Standard scaler to scale the features.

4. Results

The first part of this chapter discusses the model building, key outcomes and evaluation results of each machine learning classifier. In the second part of this chapter, dedicated to comparison of outcomes and evaluation results. The final part of this chapter discusses the deployment of the best model, application development and integration.

4.1 Model Building, Key Outcomes and Evaluation Results

Model Building with hyper parameter tunning eligible to search best model configuration by tunning the model parameters.Nguyen (2019) discussed advantages in Bayesian search. Specially discussed on efficiency on selecting regions on parameter space and reduced resources specially CPU and memory. Bayesian use past evaluation result as information gain keep information base decision on each iteration.

The combination that yields the best performance is selected, and the final model is built with these optimized hyperparameters. The resulting model object is then ready for deployment, where it can make predictions on new data.

Evaluation result and confusion matrix derived from the single function(fig.47) that input predictions and actual results as input. Results are stored in dictionaries under the name of each classifier.

```
Train_Score={}
Test_Score={}
Accuracy_Score={}
Percision_Score={}
Recall_Score={}
F_Score={}
```

```
def checkAccuracy(y_pred,y_test):
  print('Performance Metrics')
  acc_score=accuracy_score(y_test,y_pred)
  pre_Score=precision_score(y_test,y_pred,average='weighted')
  rec_score=recall_score(y_test,y_pred,average='weighted')
  f_score=f1_score(y_test,y_pred,average='weighted')
  model_score_test = round(best_model.score(X_test_scalled, y_test) * 100, 2)
  model_score_train = round(best_model.score(X_train_scalled, y_train) * 100, 2)
  # add in to the dictionary
  m_table=[["Model Train Score",model_score_train],["Model Test Score",model_score_test],["Accuracy Score",round(acc_score*100,2)],
    ["Percision Score",round(pre_Score*100,2)],["Recall Score",round(rec_score*100,2)],["F_Score          ",round(f_score*100,2)]]
  print(tabulate(m_table, headers=["Metric", "Score (%)"], tablefmt="pretty"))
  print("\nClassification Report:")
  print(classification_report(y_test, y_pred))
  dict_model_score_train['LR']=model_score_train
  dict_model_score_test['LR']=model_score_test
  dict_acc_score["LR"]=round(acc_score*100,2)
  dict_percision_score['LR']=round(pre_Score*100,2)
  dict_recall_score['LR']=round(rec_score*100,2)
  dict_f1_score['LR']=round(f_score*100,2)
 #confusion matrix
  cm=confusion_matrix(y_test,y_pred)
  display=ConfusionMatrixDisplay(confusion_matrix=cm,display_labels=['Risk','Safe'])
  display.plot(cmap=plt.cm.Blues)
  plt.title('Confusion Matrix:Heart-Care Analysis- Logistic Regresssion')
```

Figure 47- Score Dictionary and Accuracy calculation function
Check accuracy function output accuracy and score matrixes

4.1.1 Logistic Regression

```
param_spaces = [
    {'C': Real(1e-6, 1e+6, prior='log-uniform'), 'penalty': Categorical(['l2', 'none']), 'solver': Categorical(['newton-cg', 'lbfgs', 'sag'])},
    {'C': Real(1e-6, 1e+6, prior='log-uniform'), 'penalty': Categorical(['l1', 'l2']), 'solver': Categorical(['liblinear'])},
    {'C': Real(1e-6, 1e+6, prior='log-uniform'), 'penalty': Categorical(['elasticnet']), 'solver': Categorical(['saga']), 'l1_ratio': Real(0, 1)},
    {'C': Real(1e-6, 1e+6, prior='log-uniform'), 'penalty': Categorical(['l2']), 'solver': Categorical(['saga'])}
]
```

```
opt = BayesSearchCV(
    estimator=model,
    search_spaces=param_spaces,
    n_iter=32,
    cv=5,
    random_state=42,
    n_jobs=-1,  # Use all available CPU cores
    optimizer_kwargs={'base_estimator': 'GP'}  # Specify Gaussian Process as the base estimator
)
```

Figure 48- Model building and hyper parameter tunning -Logistic Regression

Performance Metrics

Metric	Score (%)
Model Train Score	72.76
Model Test Score	72.33
Accuracy Score	72.33
Percision Score	72.51
Recall Score	72.33
F_Score	72.21

Figure 49- Performance Metrics – Logistic regression

Logistic regression manages to achieve 72.76% accuracy on training and 72.33% score in accuracy or testing. The difference is only 0.43%, indicates that model is not overfitting and good stability over new data. Precision score of 72.51%

indicates that model capable of predicting positive predictions (predict someone has cardiovascular disease, if she or he has) is 72.51% of the time.

A recall score of 72.33% indicates the ability to identify positive cases especially in the scenario like cardiovascular disease prediction. The F1 score of 72.21% suggests that the model has a balanced performance between precision and recall, means model not bias towards false positive or false negative

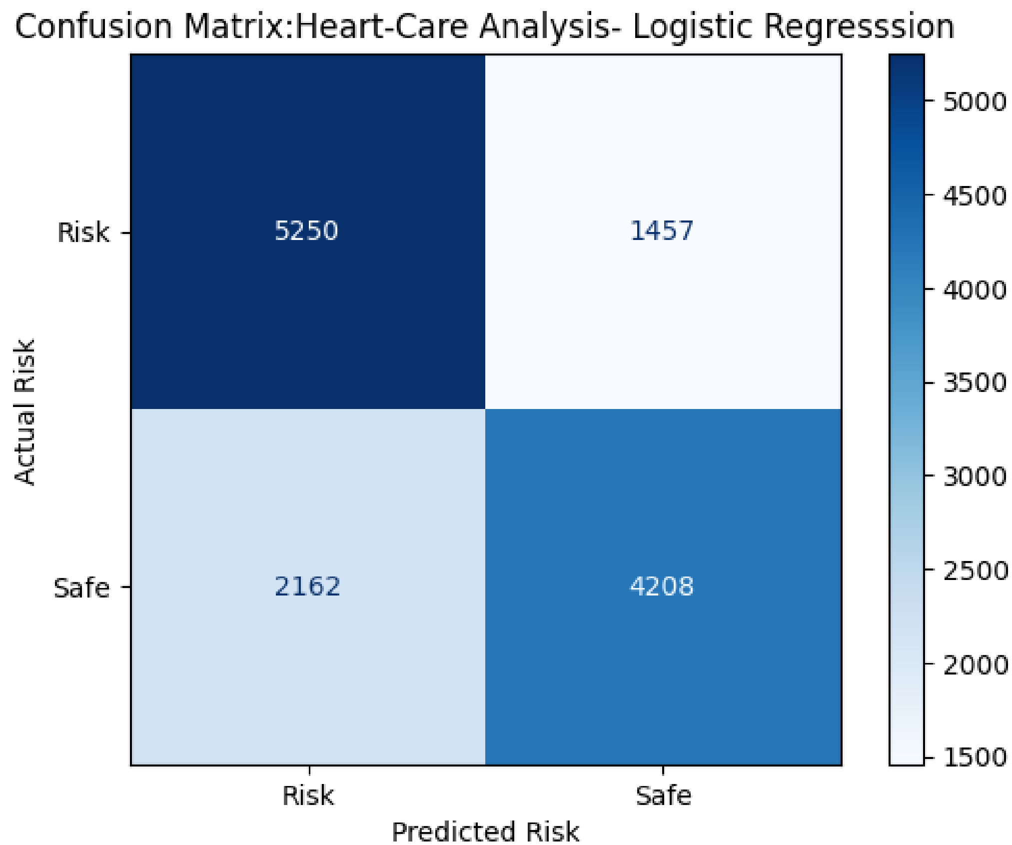


Figure – 50. Confusion matrix – Logistic Regression

According to confusion matrix LR model correctly classified 9,458 occurrences (5250 true positives and 4208 true negatives). Furthermore 2162 patients who have cardiovascular disease are not identified as risk and 1457 patients identified as safe but in real world, they are on risk.

4.1.2 K-Nearest Neighbor

```
modelKNN = KNeighborsClassifier()

parameter_List={'n_neighbors': Integer(1, 30),
'weights': Categorical(['uniform', 'distance']),
'metric': Categorical(['euclidean', 'manhattan', 'minkowski'])}
```

```
optimizer=BayesSearchCV(modelKNN,parameter_List,n_iter=30,cv=5,random_state=42,n_jobs=-1)
```

Figure 51- Model building and hyper parameter tunning -KNN

```
Performance Metrics
+------------------+-----------+
|      Metric      | Score (%) |
+------------------+-----------+
| Model Train Score |   74.09   |
| Model Test Score |   71.67   |
|  Accuracy Score  |   71.67   |
|  Percision Score |   71.86   |
|   Recall Score   |   71.67   |
|     F_Score      |   71.54   |
+------------------+-----------+

Classification Report:
              precision    recall  f1-score   support

           0       0.70      0.78      0.74      6707
           1       0.74      0.65      0.69      6370

    accuracy                           0.72     13077
   macro avg       0.72      0.72      0.71     13077
weighted avg       0.72      0.72      0.72     13077
```

Figure 52- Performance Matrix and Classification Report -KNN

KNN's training accuracy(74.09%) is slightly higher than the testing accuracy(71.67%), indicating overfitting. R*egularization* and *K-fold cross validation* are needed to improve generalization. Classification report shows balanced class distribution, but slightly better on class0 (78% recall, 74% F1-score),however lower recall on class1 indicates model missed considerable true positives.

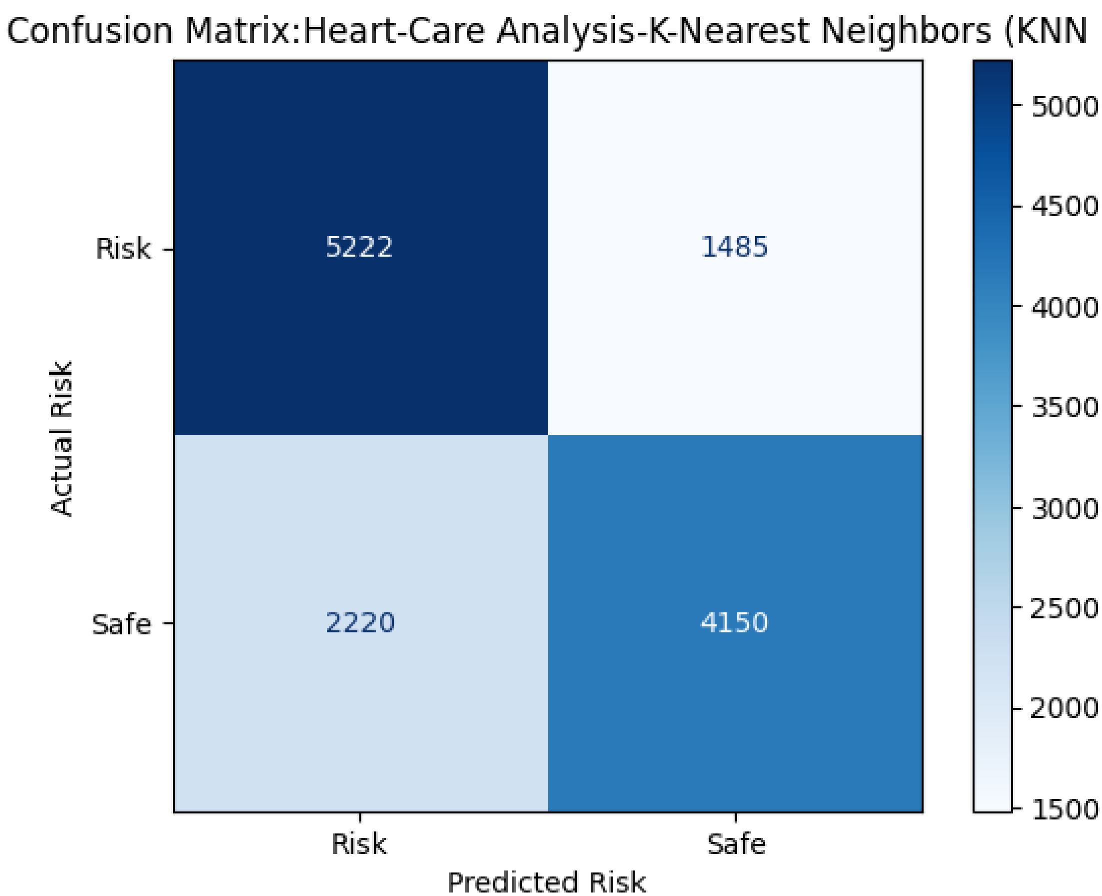


Figure-53-Confusion Matrix–KNN

Fig-53 shows the false negatives(1485) suggest the model might miss some "Risk" cases, which is critical in heart-care context. Furthermore,2220 occurrences KNN identified ‘Safe’ instances as ‘Risk’ creates unnecessary stress over patients.

4.1.3 Random Forest Classifier [RF]

```
param_space_RF = {
    'n_estimators': Integer(10, 200),
    'max_depth': Integer(1, 30),
    'min_samples_split': Integer(2, 20),
    'min_samples_leaf': Integer(1, 20),
    'max_features': Categorical(['sqrt', 'log2']),
    'bootstrap': Categorical([True, False])
}
```

```
optimizer_RF = BayesSearchCV(
    estimator=model_RF,
    search_spaces=param_space_RF,
    n_iter=32,
    cv=5,
    random_state=42,
    n_jobs=-1
)
```

Figure-54-Model building and hyper parameter tunning-RF

```
Performance Metrics
+-------------------+-----------+
|      Metric       | Score (%) |
+-------------------+-----------+
| Model Train Score |   74.86   |
| Model Test Score  |   73.29   |
|  Accuracy Score   |   73.29   |
|  Percision Score  |   73.4    |
|   Recall Score    |   73.29   |
|      F_Score      |   73.21   |
+-------------------+-----------+

Classification Report:
              precision    recall  f1-score   support

           0       0.72      0.78      0.75      6707
           1       0.75      0.68      0.71      6370

    accuracy                           0.73     13077
   macro avg       0.73      0.73      0.73     13077
weighted avg       0.73      0.73      0.73     13077
```

Figure-55 Performance Matrix and Classification Report-RF

RF achieved a training score of 74.86% and test score of 73.29% indicates that model fairly predicts well at unseen data. Accuracy scores 73% which is acceptable. Precision score for non-risk category(class 0) 72% and risk(75%), which is strong performance on risk identification. However, the recall for risk(68%), suggesting that the model misses 32% of actual risk cases, critical in a heart disease context.F1-scores for ‘Risk’ and ‘Non-Risk ‘are 71% and 75% respectively means RF performs better in detecting non-risk patients.

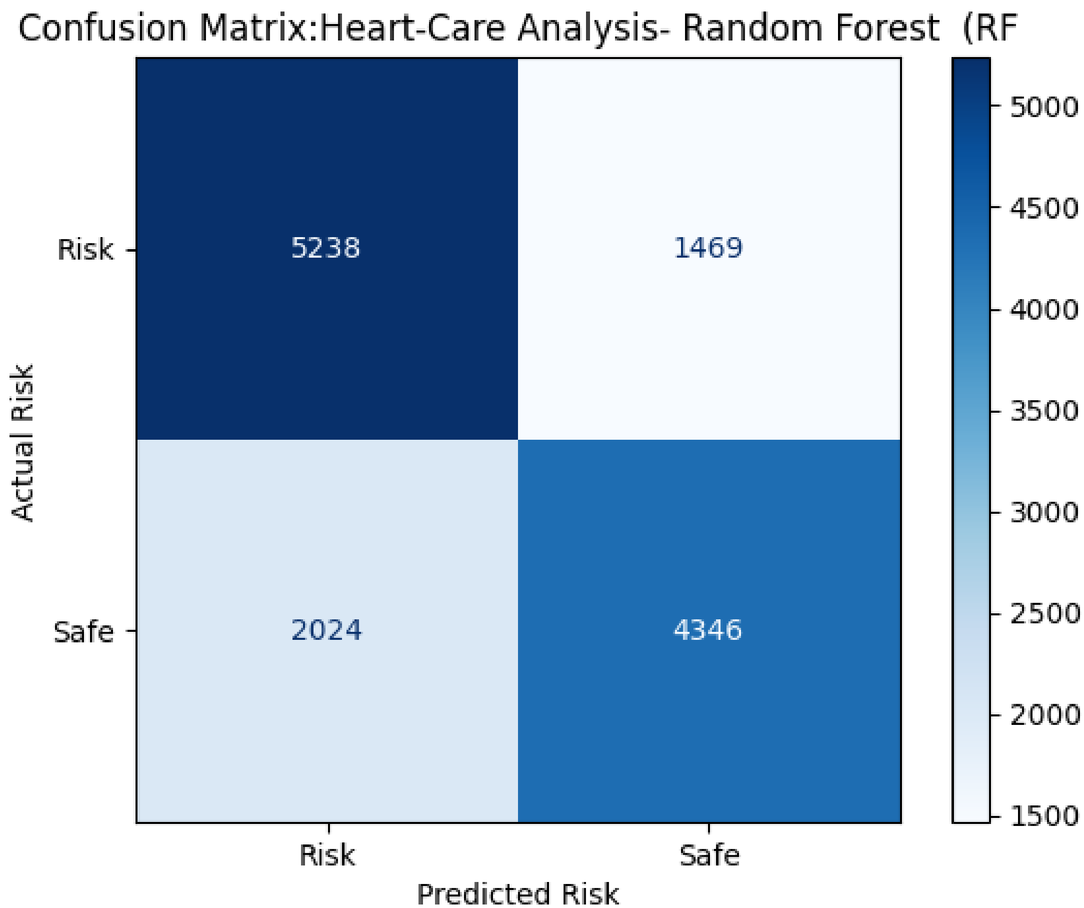


Figure 56- Confusion Matrix – RF

The true positives and true negatives indicate that the model has good ability to correctly identify both risk and safe cases, critical for effective medical decision-making. Furthermore, RF misclassified 1469 actual 'risk' instances as 'safe', make concerns in CVD context and 2024 safe instance as risk.

4.1.4 Support Vector Machine

```
param_list_svc = {
    'C': Real(1e-6, 1e+2, prior='log-uniform'),
    'gamma': Real(1e-6, 1e+1, prior='log-uniform'),
    'kernel': Categorical(['linear', 'poly', 'rbf', 'sigmoid']),
     'degree': Integer(2, 5),
    'coef0': Real(0, 10),
}
```

```
optimizer_SVC= BayesSearchCV(
    estimator=model_svc,
    search_spaces=param_list_svc,
    n_iter=8,
    cv=3,
    random_state=42,
    n_jobs=-1
)
```

Figure 57- Model building and hyper parameter tunning -SVM

```
Performance Metrics
+-------------------+-----------+
|      Metric       | Score (%) |
+-------------------+-----------+
| Model Train Score |   73.42   |
| Model Test Score  |   73.0    |
|  Accuracy Score   |   73.0    |
|  Percision Score  |   73.3    |
|   Recall Score    |   73.0    |
|      F_Score      |   72.84   |
+-------------------+-----------+

Classification Report:
              precision    recall  f1-score   support

           0       0.71      0.80      0.75      6707
           1       0.76      0.66      0.70      6370

    accuracy                           0.73     13077
   macro avg       0.73      0.73      0.73     13077
weighted avg       0.73      0.73      0.73     13077
```

Figure 58- Performance Matrix and Classification Report -SVM

Figure 58 shows training score of SVM(73.42%) with less difference of testing score(73%) indicate minimum overfitting. All scores are closely aligned except recall score of class 1('risk') indicates SVM missing some true positive, that crucial for health decisions. Otherwise, performs well in identifying "non-risk" cases with recall for class0(80%) and good precision(76%) on class1.

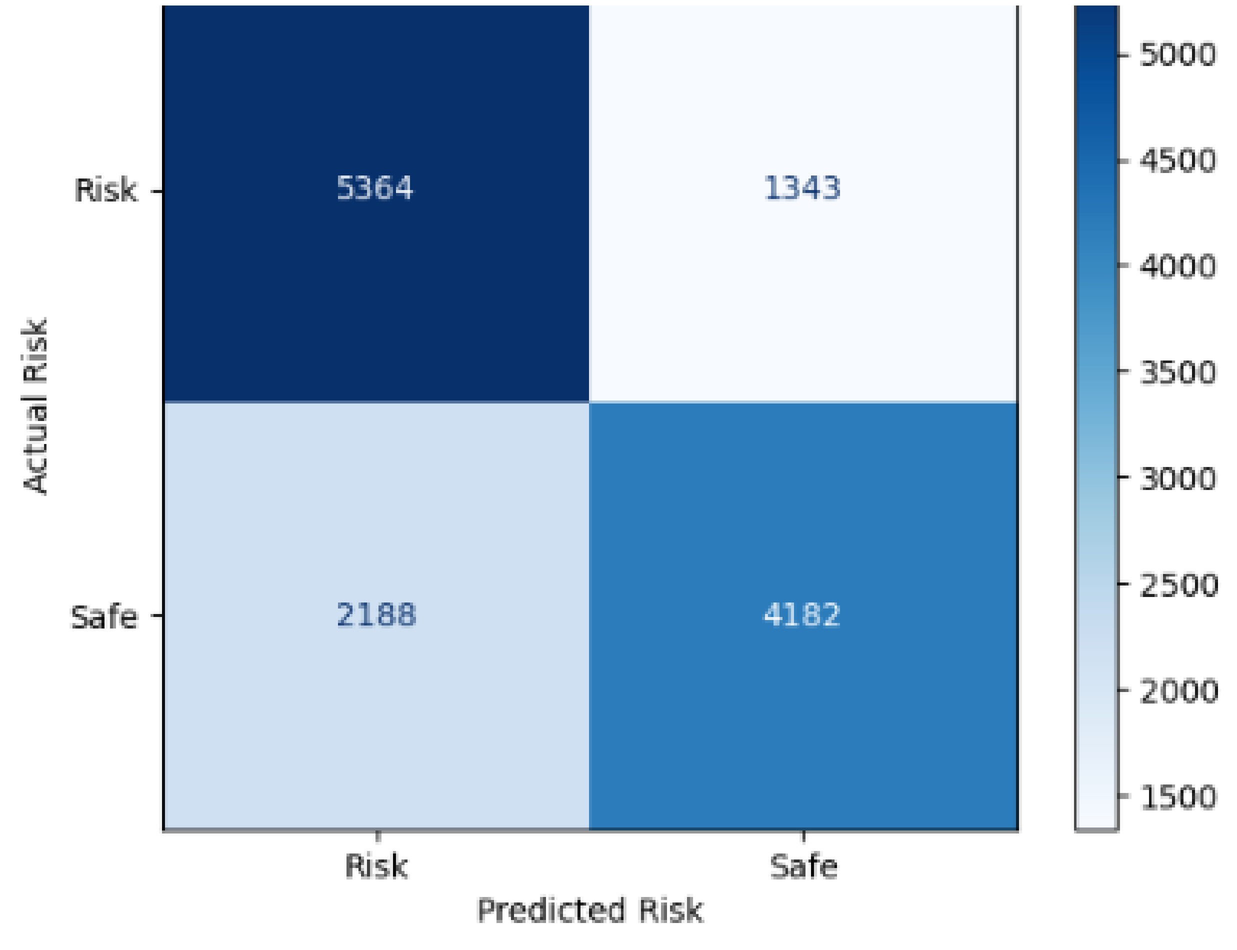


Figure 59- Confusion Matrix – SVM

SVM correctly identified 5364 true positive 4182 true negative incidents indicates that model is performing efficiently, however there is notable concerns 1343 "risk" cases as "safe," cases which is harmful to the model's credibility.

4.1.5 Naïve Bais

```
param_list_NB = {
    'var_smoothing': Real(1e-12, 1e-9, prior='log-uniform')
}
```

```
optimzer_NB = BayesSearchCV(
        estimator=model_NB,
        search_spaces=param_list_NB,
        n_iter=32,
        cv=5,
        random_state=42,
        n_jobs=-1
)
```

Figure 60- Model building and hyper parameter tunning -NB

Performance Metrics

Metric	Score (%)
Model Train Score	71.04
Model Test Score	71.03
Accuracy Score	71.03
Percision Score	71.5
Recall Score	71.03
F_Score	70.77

Classification Report:

	precision	recall	f1-score	support
0	0.69	0.80	0.74	6707
1	0.74	0.62	0.68	6370
accuracy			0.71	13077
macro avg	0.72	0.71	0.71	13077
weighted avg	0.72	0.71	0.71	13077

Figure 61- Performance Matrix and Classification Report -NB

Fig-61 shows the equivalent train and test illustrates no overfitting or underfitting. The accuracy of 71.03% reflects correct classification instance, while the precision of 71.5% indicates good identification of positive cases while 71.03% and 62% of class(risk) recall talks about missing true positives is notable.F1-Score(70.77%) indicates performance consistency.

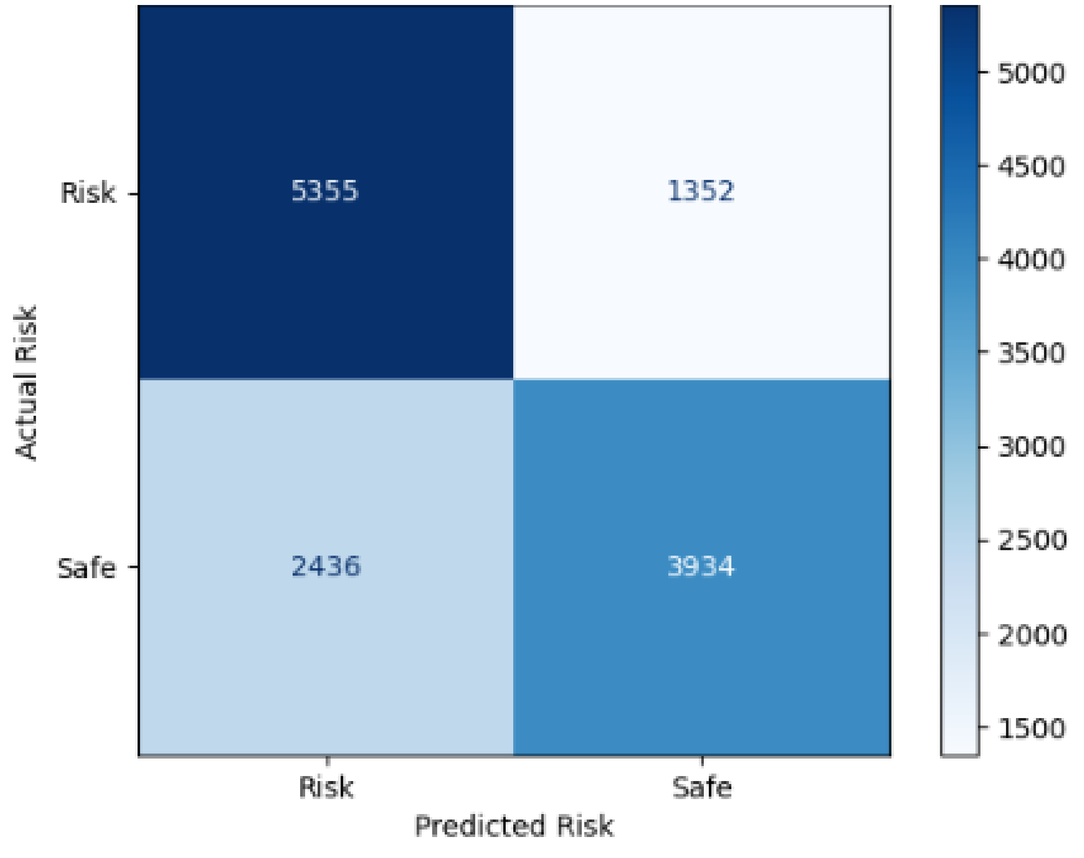


Figure 62- Performance Matrix and Classification Report -NB

The model correctly identified 5,355 "Risk" cases and 3,934 "Safe" cases. However, it misclassified 1,352 "Risk" cases as "Safe" and 2436 ‘safe cases as ‘risk’

4.1.6 Decision Tree

```
param_list_DT = {
    'max_depth': Integer(1, 30),
    'min_samples_split': Integer(2, 20),
    'min_samples_leaf': Integer(1, 20),
    'criterion': Categorical(['gini', 'entropy']),
}
```

```
optimizer_DT = BayesSearchCV(
    estimator=model_DT,
    search_spaces=param_list_DT,
    n_iter=32,
    cv=5,
    random_state=42,
    n_jobs=-1
)
```

Figure 63-Model building and hyper parameter tunning-DT

```
Performance Metrics
+-------------------+-----------+
|      Metric       | Score (%) |
+-------------------+-----------+
| Model Train Score |   73.27   |
| Model Test Score  |   72.68   |
|  Accuracy Score   |   72.68   |
|  Percision Score  |   72.8    |
|   Recall Score    |   72.68   |
|      F_Score      |   72.6    |
+-------------------+-----------+

Classification Report:
              precision    recall  f1-score   support

           0       0.72      0.78      0.74      6707
           1       0.74      0.67      0.71      6370

    accuracy                           0.73     13077
   macro avg       0.73      0.73      0.73     13077
weighted avg       0.73      0.73      0.73     13077
```

Figure 64- Performance Matrix and Classification Report -DT

DT also scored 73.2% in training and 72.7% in testing with balance generalization. Furthermore, achieves good balance in precision, recall and f-score. However, recall score on ‘risk’ is 67% indicates, model missing 33% of actual risk occurrences.

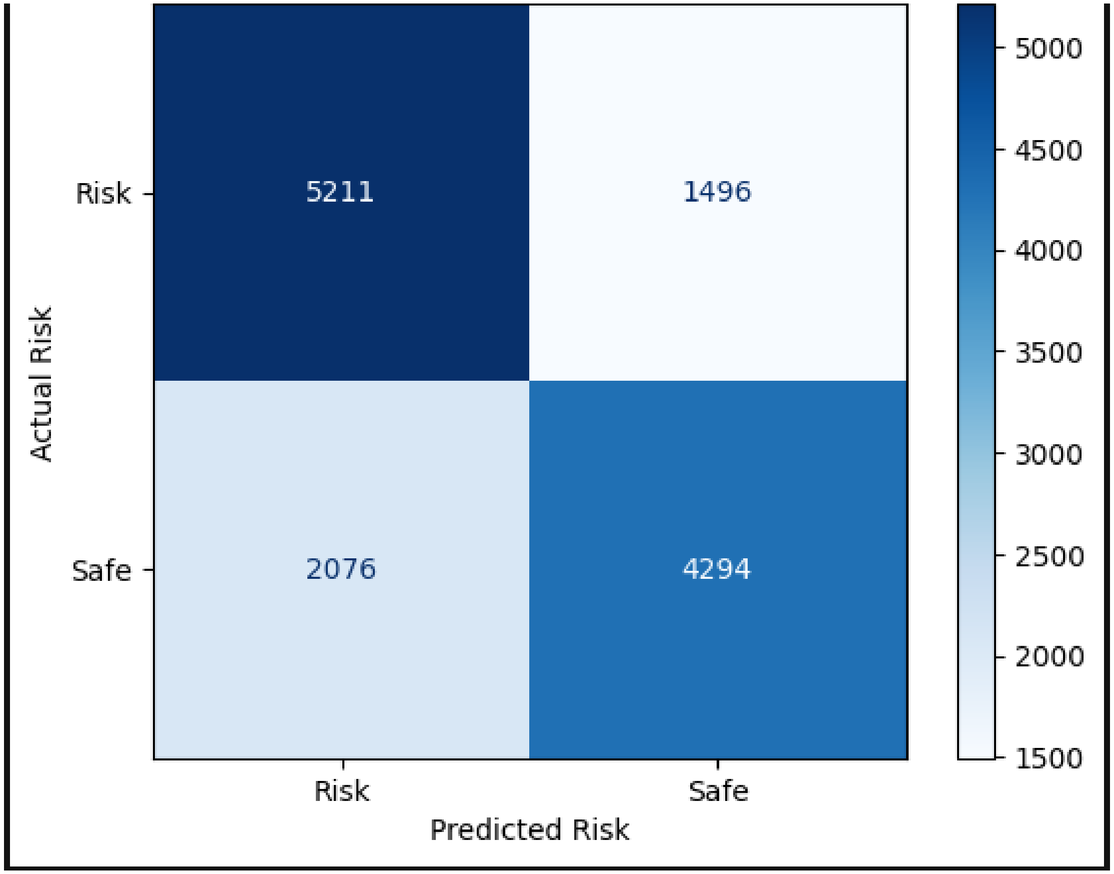


Figure 64- Confusion Matrix -DT

Fig.64 1496 actual ‘risk’ patients detect as ‘safe’ which is significant amount in cardiovascular prediction scenario. Furthermore, 2076 occurrence model’s incorrect predictions on ‘non risk’ patients as ‘risk’

4.1.7 Extreme Gradient Boost

```
param_space = {
    'n_estimators': Integer(50, 1000),
    'learning_rate': Real(0.01, 0.3, prior='log-uniform'),
    'max_depth': Integer(3, 10),
    'min_child_weight': Integer(1, 10),
    'subsample': Real(0.6, 1.0),
    'colsample_bytree': Real(0.6, 1.0)
}
xgb_clf = XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state=42)
```

```
bayes_search = BayesSearchCV(
    estimator=xgb_clf,
    search_spaces=param_space,
    n_iter=32,
    cv=3,
    scoring='accuracy',
    random_state=42,
    n_jobs=-1,
    verbose=1
)
```

Figure 65-Model building and hyperparameter tuning-XG-Boost.

Performance Metrics

Metric	Score (%)
Model Train Score	74.61
Model Test Score	73.2
Accuracy Score	73.2
Percision Score	73.29
Recall Score	73.2
F_Score	73.14

Classification Report:

	precision	recall	f1-score	support
0	0.72	0.78	0.75	6707
1	0.74	0.69	0.71	6370
accuracy			0.73	13077
macro avg	0.73	0.73	0.73	13077
weighted avg	0.73	0.73	0.73	13077

Figure 66-Performance Matrix and Classification Report -XG-Boost

Fig-67 shows close train and test scores indicating the overall model is generalizing well. Accuracy scores 73.2% show that high performance of the model with the Precision(73.29%),Recall (73.2%) and F1-Score (73.14%) good overall benchmark specially handling false positives and false negative. The classification report of XG-Boost highlights that the model has a slightly higher recall for class0 (78%) compared to class1 (69%), indicating a potential bias towards class 0.

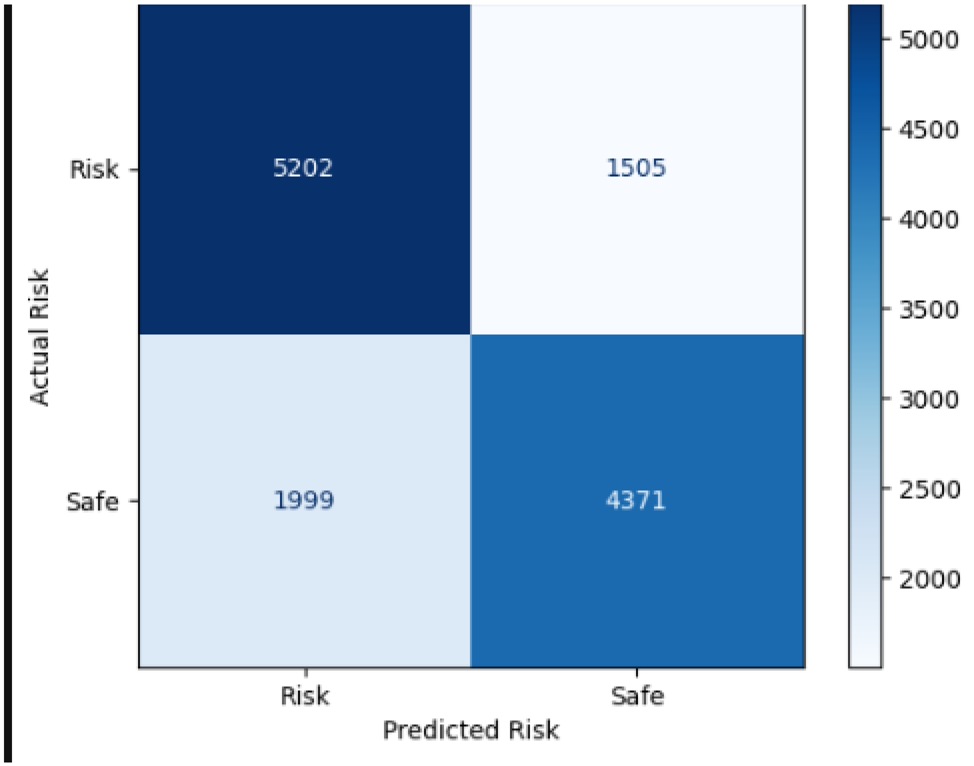


Figure 67- Confusion Matrix-XG-Boost

confusion matrix reveals that successful predictions on ‘risk’ and ‘safe’ are 5202 and 4371 respectively. It shows 1505 false negatives, where actual "risk" cases were misclassified as "safe," which can cause serious misclassification on CVD prediction. Additionally, the model produces 1505 false positives, incorrectly labeling "Safe" cases as "Risk," causing unnecessary burden over the patients.

4.1.8 Light Gradient Boost

```
space_lgb = {
            'learning_rate': hp.quniform('learning_rate', 0, 0.05, 0.0001),
            'n_estimators': hp.choice('n_estimators', range(100, 1000)),
            'max_depth':  hp.choice('max_depth', np.arange(2, 12, dtype=int)),
            'num_leaves': hp.choice('num_leaves', 2*np.arange(2, 2**11, dtype=int)),
            'min_child_weight': hp.quniform('min_child_weight', 1, 9, 0.025),
            'colsample_bytree': hp.quniform('colsample_bytree', 0.5, 1, 0.005),
            'objective': 'binary',
            'boosting_type': 'gbdt',
            }

best = fmin(fn=hyperopt_lgb_score, space=space_lgb, algo=tpe.suggest, max_evals=10)
```

```
{'boosting_type': 'gbdt',
 'colsample_bytree': 0.605,
 'learning_rate': 0.00030000000000000003,
 'max_depth': 3,
 'min_child_weight': 2.45,
 'n_estimators': 716,
 'num_leaves': 2832,
 'objective': 'binary'}
```

Figure 68- Model building and hyper parameter tunning-LGB

```
Performance Metrics
+-------------------+-----------+
|      Metric       | Score (%) |
+-------------------+-----------+
| Model Train Score |   83.73   |
| Model Test Score  |   71.83   |
|  Accuracy Score   |   71.83   |
|  Percision Score  |   71.87   |
|   Recall Score    |   71.83   |
|      F_Score      |   71.78   |
+-------------------+-----------+

Classification Report:
              precision    recall  f1-score   support

           0       0.71      0.76      0.73      6707
           1       0.73      0.68      0.70      6370

    accuracy                           0.72     13077
   macro avg       0.72      0.72      0.72     13077
weighted avg       0.72      0.72      0.72     13077
```

Figure 69- Performance Matrix and Classification Report -LGB

Training and testing accuracy has significant difference, indicating that model is overfitting. Regularization required before applying with unseen data. Recall scores on class0(76%) and class0(68%) show LGB more effective on detecting ‘safe’ over ‘risk’. Weighted averages(72%) visualize better consistence, but improvements need.

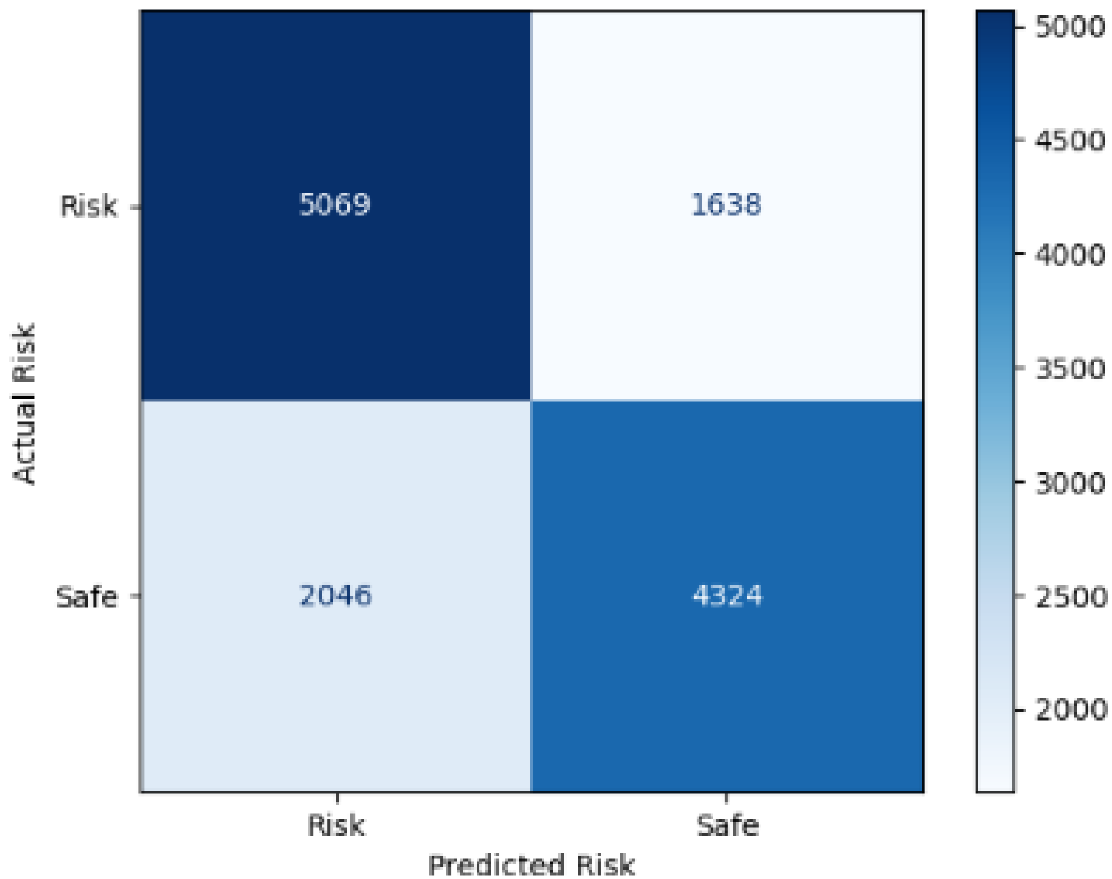


Figure 70- Confusion Matrix -LGB

Figure-70 shows most instances are correctly classified, but focus is needed on misclassifying "safe" cases as "risk."

4.1.9 Multi-layer voting classifier

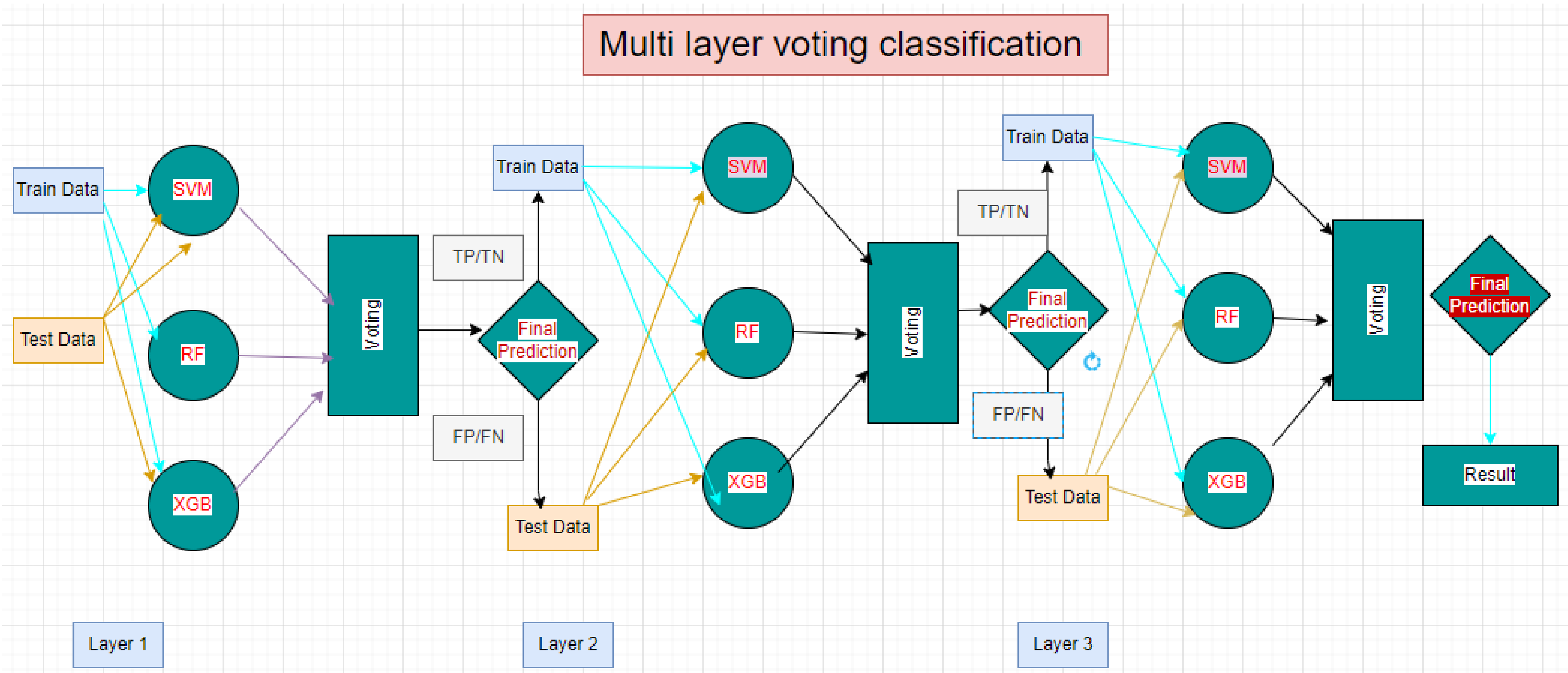


Figure 71- Multi-layer voting classifier using best three accurate models

Fig.71 illustrates multilayer classifier with three most accurate models-SVM, RF and XG-Boost. Each layer trained with the successful predictions of prior layer and tested on the failures of the prior layer.

Performance Metrics

Metric	Score (%)
Model Train Score	74.76
Model Test Score	73.11
Accuracy Score	73.11
Percision Score	73.21
Recall Score	73.11
F_Score	73.04

Classification Report:

	precision	recall	f1-score	support
0	0.72	0.78	0.75	6707
1	0.74	0.68	0.71	6370
accuracy			0.73	13077
macro avg	0.73	0.73	0.73	13077
weighted avg	0.73	0.73	0.73	13077

Figure 72- Performance Matrix and Classification Report -Ensemble classifier

Fig-72 shows significant overfitting, with considerable recall difference between 'risk' and 'non-risk' means model is less effective at determination 'risk'. Higher F1-score for class 0 (0.75) compared to class 1 (0.71).

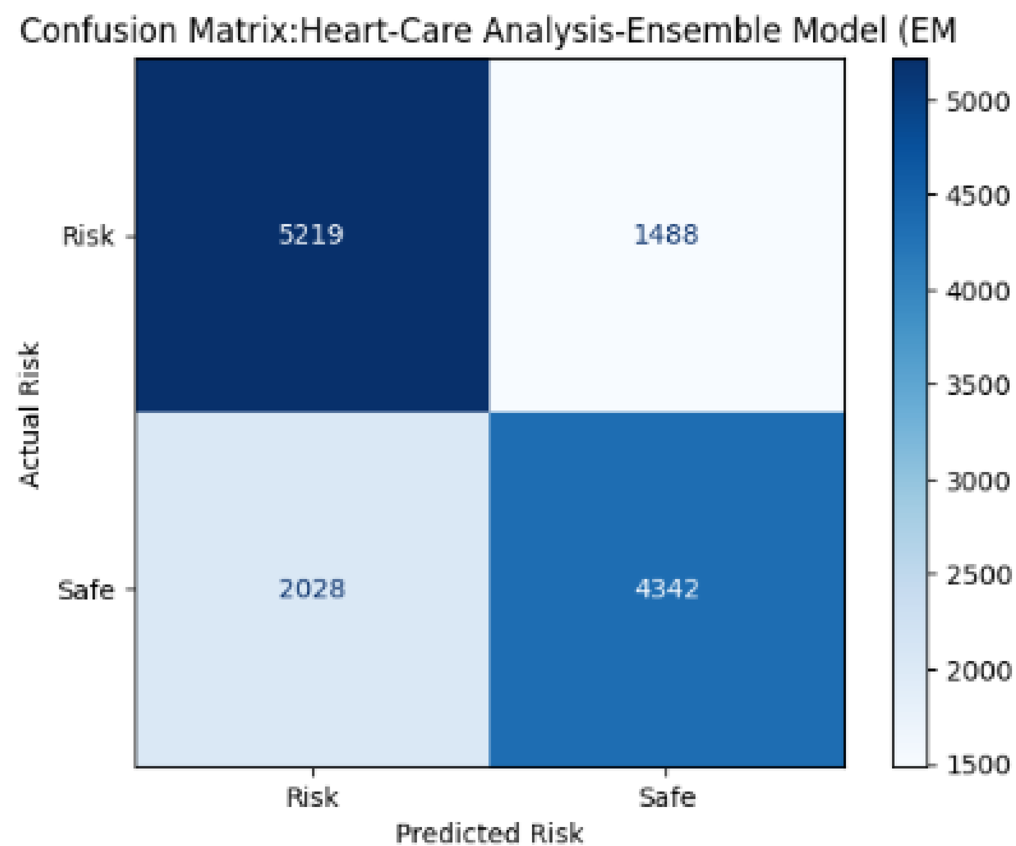


Figure 73- Confusion Matrix- EM

The model demonstrates a reasonable ability to identify both “Risk” and “Safe” cases, with higher accuracy in detecting “Risk” cases (5219- true positives) compared to “Safe” cases (4342-true negatives). Missing risk of 1488 is notable amount in the context of cardiovascular environment. Furthermore, the false negatives (2028) occurrences are critical concern, as these represent models’ weakness om detecting actual “Risk” cases.

4.1.10 TabNet Deep Learning Model

```
best_clf = TabNetClassifier(
    n_d=best_params['n_d'], n_a=best_params['n_a'], n_steps=best_params['n_steps'],
    gamma=best_params['gamma'], lambda_sparse=best_params['lambda_sparse'],
    optimizer_fn=torch.optim.Adam, optimizer_params=dict(lr=best_params['lr']),
    mask_type=best_params['mask_type'], n_shared=best_params['n_shared'],
    n_independent=best_params['n_independent'], verbose=0, seed=42
)
```

```
Performance Metrics
+-----------------------+-----------+
|         Metric        | Score (%) |
+-----------------------+-----------+
| Model  ROC Score Train |   79.13   |
| Model ROC  Score Test |   78.73   |
|     Accuracy Score    |   72.74   |
|    Percision Score    |   72.81   |
|      Recall Score     |   72.74   |
|        F_Score        |   72.67   |
+-----------------------+-----------+

Classification Report:
              precision    recall  f1-score   support

           0       0.72      0.77      0.74      6707
           1       0.74      0.68      0.71      6370

    accuracy                           0.73     13077
   macro avg       0.73      0.73      0.73     13077
weighted avg       0.73      0.73      0.73     13077
```

Figure 74 - Performance Matrix and Classification Report - TabNet Deep Learning Model

According to Figure 74, TabNet achieved an AUC ROC score of 80.12% on the training data and 70.54% on the testing data, demonstrating a good ability to differentiate between 'risk' and 'safe' categories, even on unseen data. The slight difference between the training score (79.54%) and the testing score (80.12%) suggests that the model generalizes well without overfitting

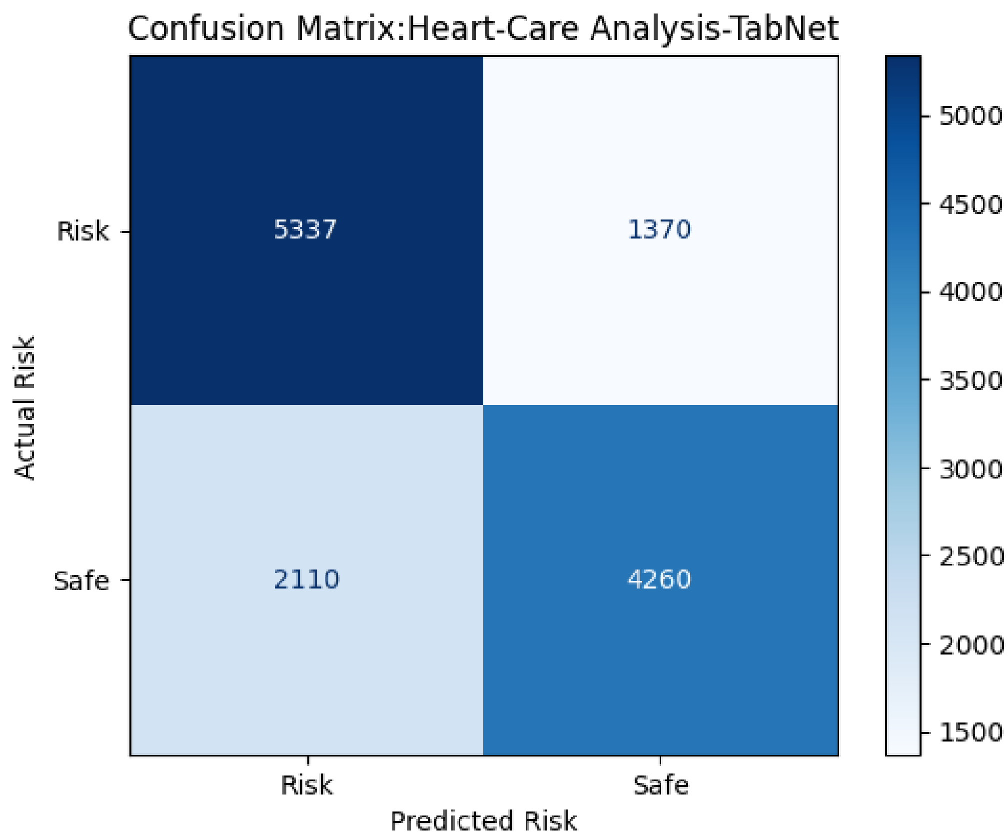


Figure75- Confusion Matrix-TabNet

The model has a strong ability to identify ‘Risk’ incidents and ‘Safe’ incidents correctly. However, 1370 incidents have been reported failed prediction of ‘Safe’ against actual ‘Risk' incidents. On other hand 2110 incidents wrongly predicted ‘Risk’ over ‘Safe’.

4.1.11 Artificial Neural Network

```
def build_model(hp):
    model=Sequential()
    model.add(Dense(units=hp.Int('units',min_value=32,max_value=512,step=32),activation=hp.Choice('activation',['relu','tanh','sigmoid']),
                    input_dim=X_train_scalled.shape[1]))
    model.add(Dropout(hp.Float('dropout_rate',min_value=0.1,max_value=0.5,step=0.1)))
    model.add(Dense(units=hp.Int('units2',min_value=32,max_value=256,step=32),activation=hp.Choice('activation2',['relu','tanh','sigmoid'])))
    model.add(Dropout(hp.Float('dropout_rate2',min_value=0.1,max_value=0.5,step=0.1)))
    model.add(Dense(1,activation='sigmoid'))
    model.compile(optimizer=Adam(hp.Choice('learning_rate',[0.001,0.01,0.1])),loss='binary_crossentropy',metrics=['accuracy'])
    return model
```

```
tuner=kt.RandomSearch(build_model,objective='val_accuracy',max_trials=20,executions_per_trial=1,directory='project',project_name='heart_disease_soft_layer1')
```

Figure-76

```python
configure_layers=[

    (1, 16),
    (1, 32),
    (1, 64),
    (1, 128),
    (2, 16),
    (2, 32),
    (2, 64),
    (2, 128),
    (3, 16),
    (3, 32),
    (3, 64),
    (3, 128),
    (4, 16),
    (4, 32),
    (4, 64),
    (4, 128),
]
def build_model(num_layers, num_neurons,l2_reg=0.01):
    model = Sequential()
    model.add(Dense(12, activation=activation, kernel_initializer=initializer,input_shape=(12,)))
   #model.add(activation)
    for i in range(num_layers):
        model.add(Dense(num_neurons,activation='relu',kernel_initializer=initializer,kernel_regularizer=l2(l2_reg)))
        #model.add(activation)
    model.add(Dense(1, activation='sigmoid', kernel_initializer=initializer))
    return model
```

Figure-77

Fig.76 and Fig.77 illustrate the Hyperparameter space of the ANN.

```python
for num_layers, num_neurons in configure_layers:

  model = build_model(num_layers, num_neurons)
  optim = Adam(learning_rate=learning_rate)
  model.compile(optimizer=optim, loss='binary_crossentropy', metrics=['accuracy'])
  history = model.fit(X_train, y_train, batch_size=batch_size, epochs=epochs, validation_split=0.2, verbose=1)


  val_accuracy = max(history.history['val_accuracy'])
  final_results_dict[(num_layers, num_neurons)] = val_accuracy
  print(f'Validation accuracy with {num_layers} layers and {num_neurons} neurons per layer: {val_accuracy}')
  if val_accuracy > best_accuracy:
    best_accuracy = val_accuracy
    model.save('best_model_optimized.h5')

best_config = max(final_results_dict.items(), key=lambda x: x[1])
best_layers, best_neurons = best_config[0]
best_accuracy = best_config[1]

print(f'Best configuration: {best_layers} layers, {best_neurons} neurons per layer with validation accuracy: {best_accuracy}')


num_layers_list = [key[0] for key in final_results_dict.keys()]
num_neurons_list = [key[1] for key in final_results_dict.keys()]
val_accuracy_list = [value for value in final_results_dict.values()]


plt.figure(figsize=(10, 6))
scatter = plt.scatter(num_layers_list, num_neurons_list, c=val_accuracy_list, cmap='viridis', s=100)
plt.colorbar(scatter, label='Validation Accuracy')
```

Figure 79

Fig.79 illustrate the Model Building code snippet using best parameters -ANN

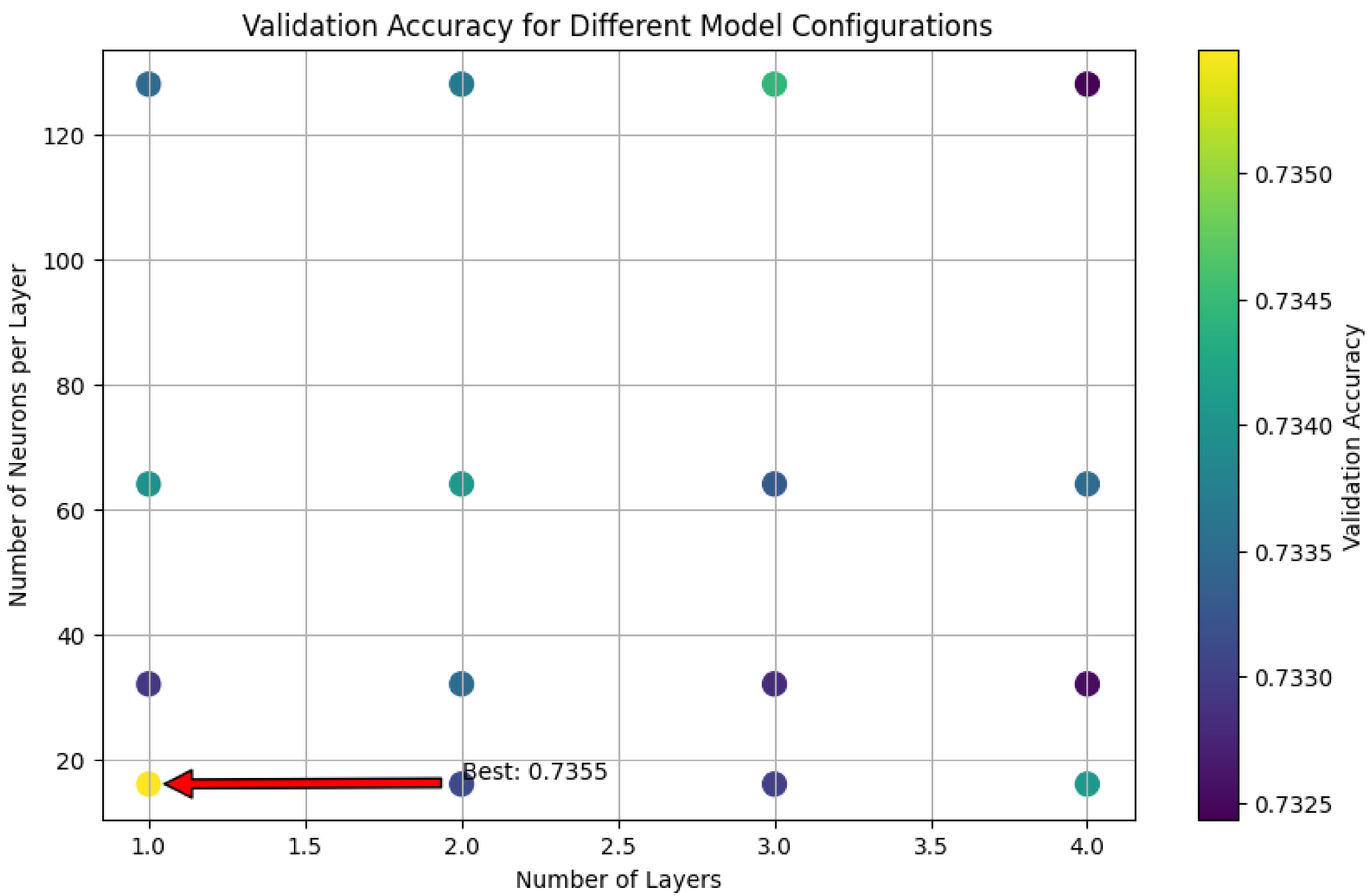

Validation Accuracy for Different Model Configurations
Number of Neurons per Layer
Number of Layers
Validation Accuracy
Best: 0.7355
120
100
80
60
40
20
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.7350
0.7345
0.7340
0.7335
0.7330
0.7325

Figure 80

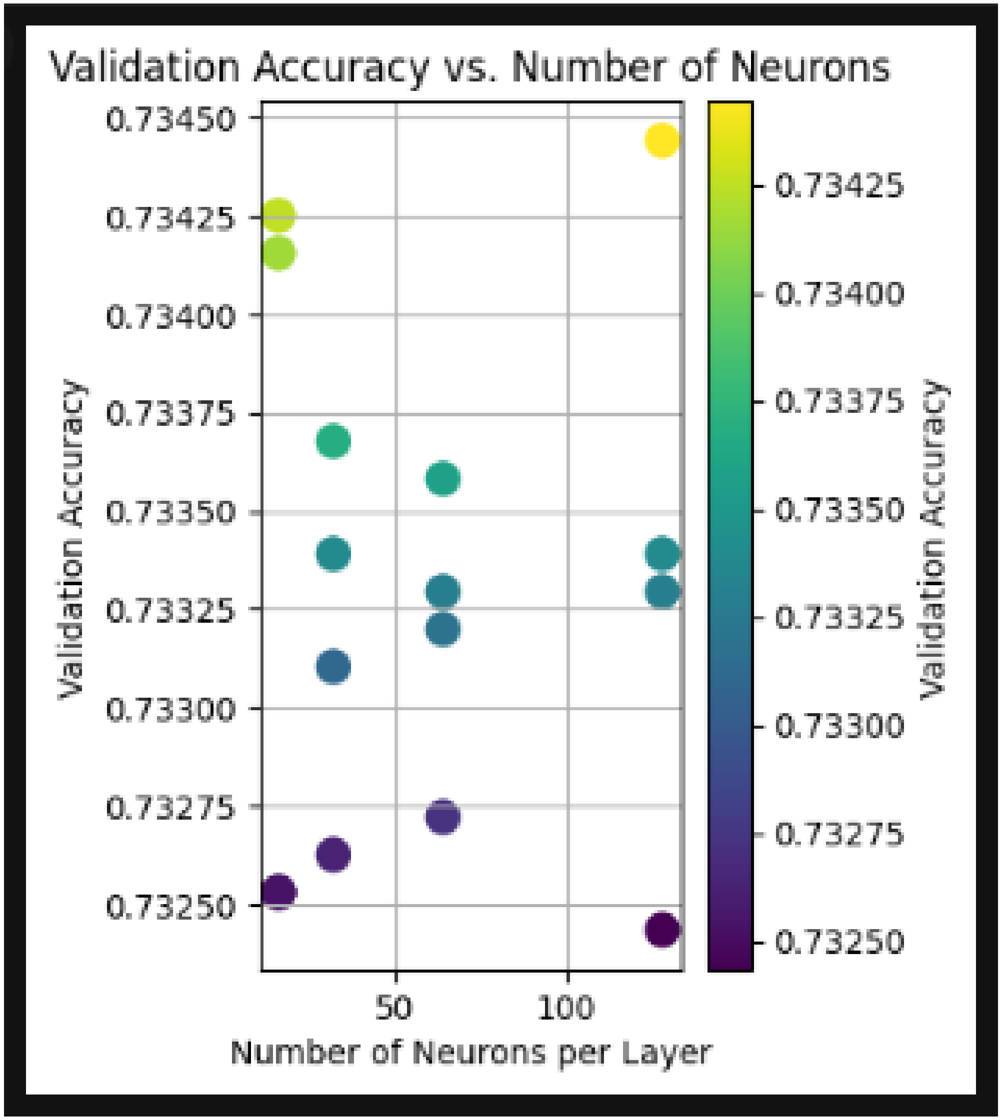

Validation Accuracy vs. Number of Neurons
0.73450
0.73425
0.73400
0.73375
0.73350
0.73325
0.73300
0.73275
0.73250
Validation Accuracy
50
100
Number of Neurons per Layer
0.73425
0.73400
0.73375
0.73350
0.73325
0.73300
0.73275
0.73250
Validation Accuracy

Figure 81

Figures 80 and 81 illustrate the optimal accuracy achieved by varying two key parameters: the number of layers (Figure 80) and the number of neurons per layer (Figure 81).

	precision	recall	f1-score	support
No Risk	0.72	0.78	0.75	6707
Risk	0.74	0.68	0.71	6370
accuracy			0.73	13077
macro avg	0.73	0.73	0.73	13077
weighted avg	0.73	0.73	0.73	13077

Figure-82 Performance Matrics-ANN

ANN achieved 73% overall accuracy, however 'no-risk' class has 78% success rate on positive predictions, while risk predictions Is quite lags on 68% means ANN failed 32% actual risk incidents predict as 'no risk'

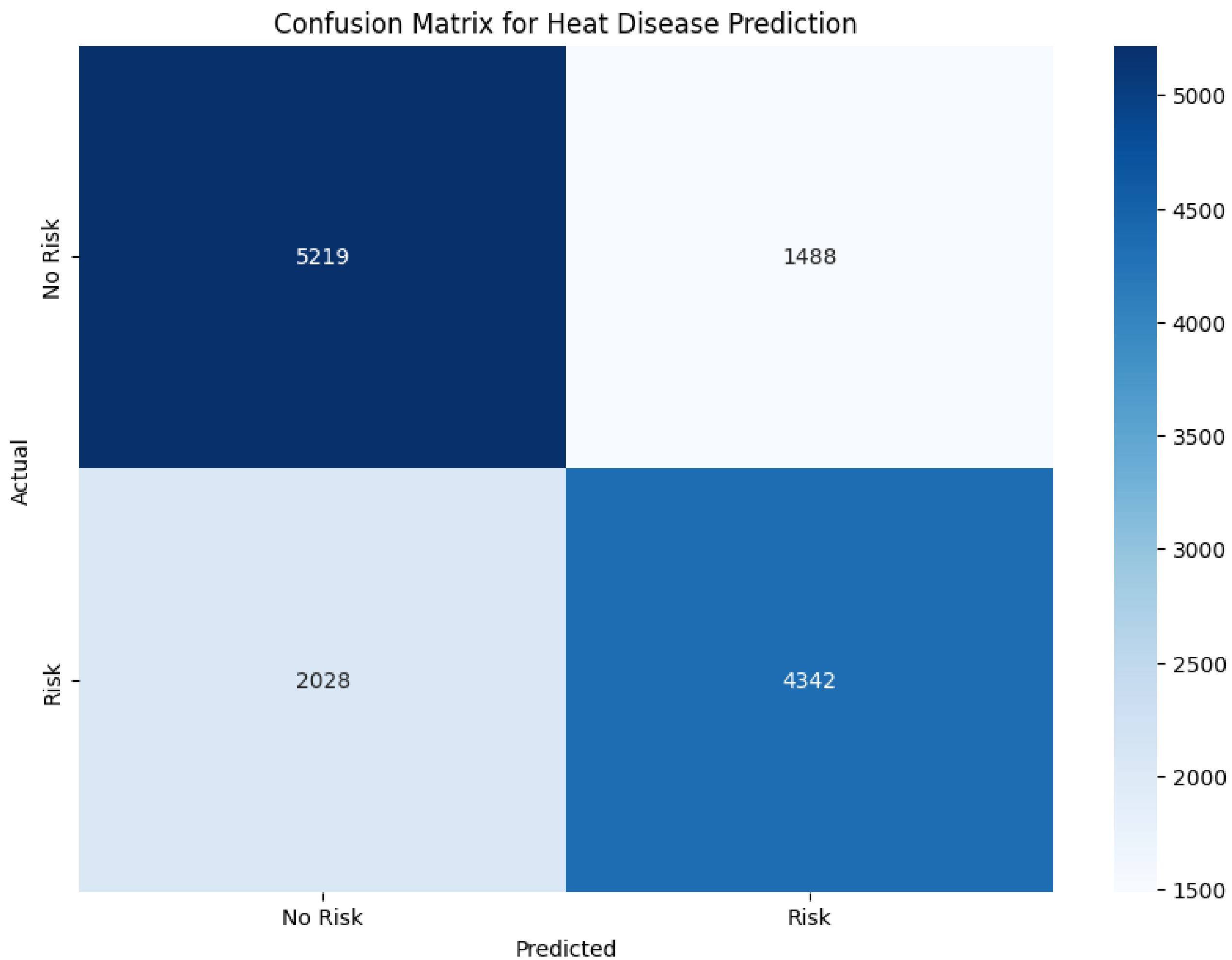


Figure-73 Confusion Matrix-ANN

ANN correctly predicts 5219 of True positive cases and 4324 of True negative cases. ANN identified 2028 false negative incidents and 1488 false positive incidents which is 26% of overall test data.

4.2 Model Comparison

To ensure consistency throughout the training and testing process, the dataset split into training and testing using the same random state. To identify highly accurate and best performing model among 8 machine learning models (RF,KNN,DT,SVM,LR,XG,LGB),2-deep learning classifier(ANN/TabNet) and ensemble classifier. Model comparison based on train score, test score, accuracy, precision and recall, furthermore detail classification report and confusion matrix use for enchasing comparisons.

Fig.84 illustrates train score, test score, accuracy score, precision score, recall score and f-score as line chart. Further in fig.85 visualize the bar graph related to test score, train score and accuracy score for each classifier.

Considering the training and testing scores clearly see that LGB, TabNet, XG and RF has highest training scores but except TabNet other classifiers have significant difference between train and test result indicates that overfitting scenario. Furthermore, with the balanced precision and recall, better F1 scores can be seen in TabNet, SVM and XG classifiers.

4.2.1 Model Selection

The process of model selection Malcolm R(1999) discussed importance consider not just only the performance metrics like accuracy, precision, recall or F1 score. Papers discussed about model's ability of on perform well with seen and unseen data in equal manner. LGB and TabNet have the best training scores, however small variance of TabNet train and test scores indicate stability across different data. TabNet maintains higher accuracy with the balanced F1-score which is remarkable results on both false positives and false negatives.

Further, Antons(2022) discussed TabNet's capabilities of learning through deep neural networks and simulation decision marking on tree base model, this hybrid approach deliver more effective than traditional models.

TabNet's higher f1Score indicates handling precision and recall in imbalanced datasets very well. This makes TabNet the most reliable and robust choice among the classifiers considered.

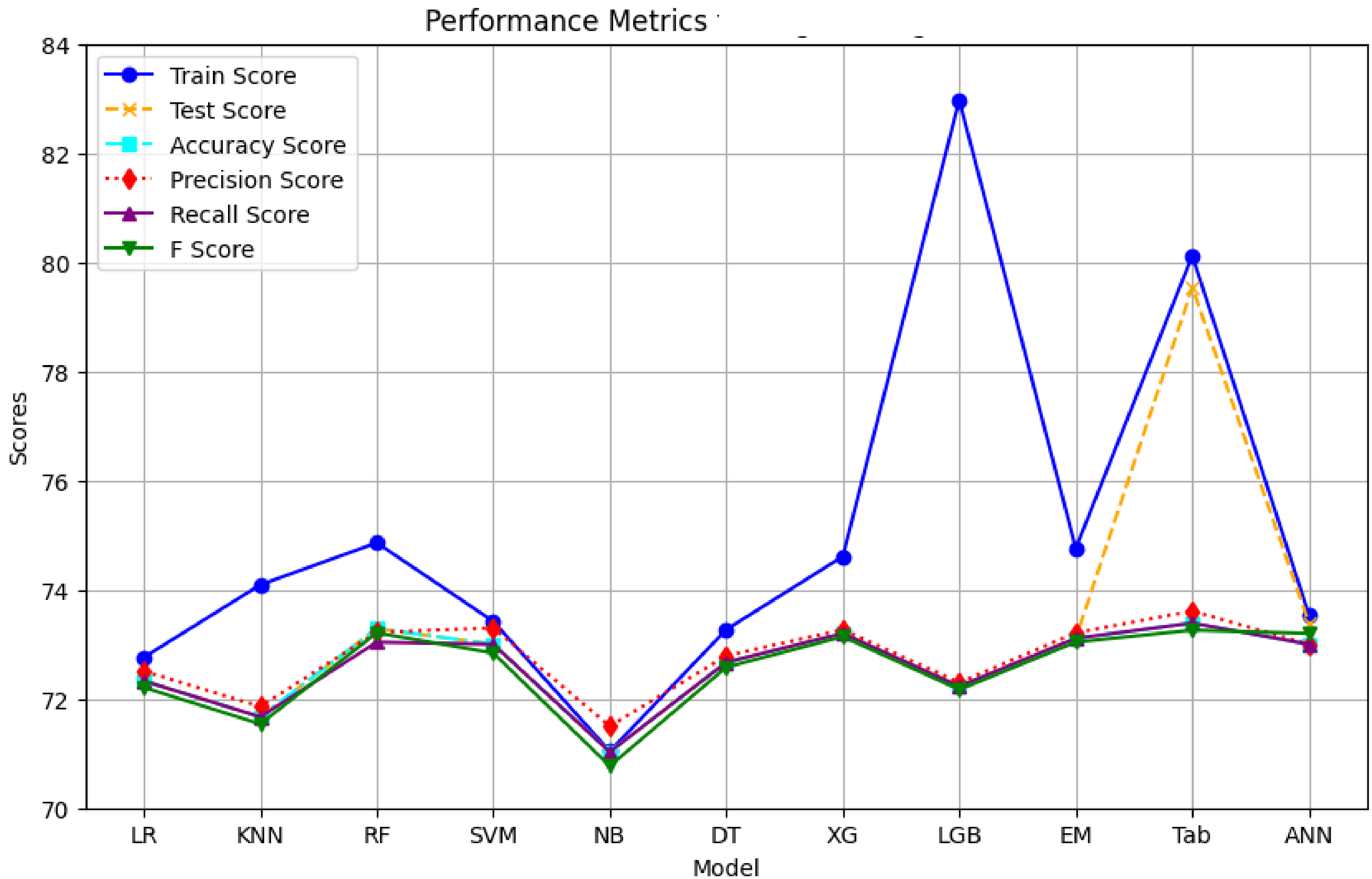


Figure 84- Model Comparison line chart

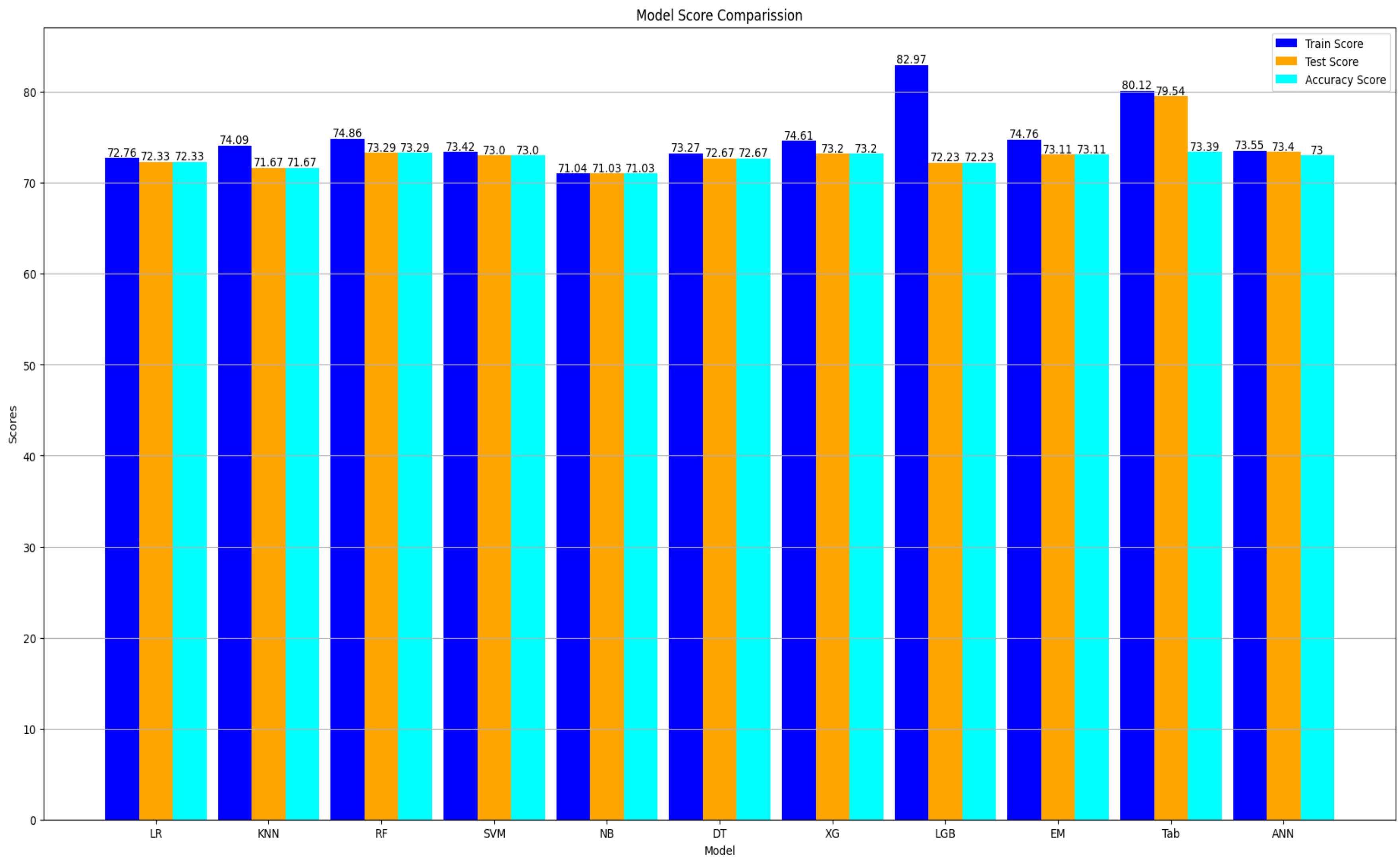


Figure-85 ***Model Comparison- Bar Chart (Train Score, Test Score and Accuracy Score)***

4.3 Model Deployment, Integration and Application Development

Deploying the successful model to the real world is a straightforward process.

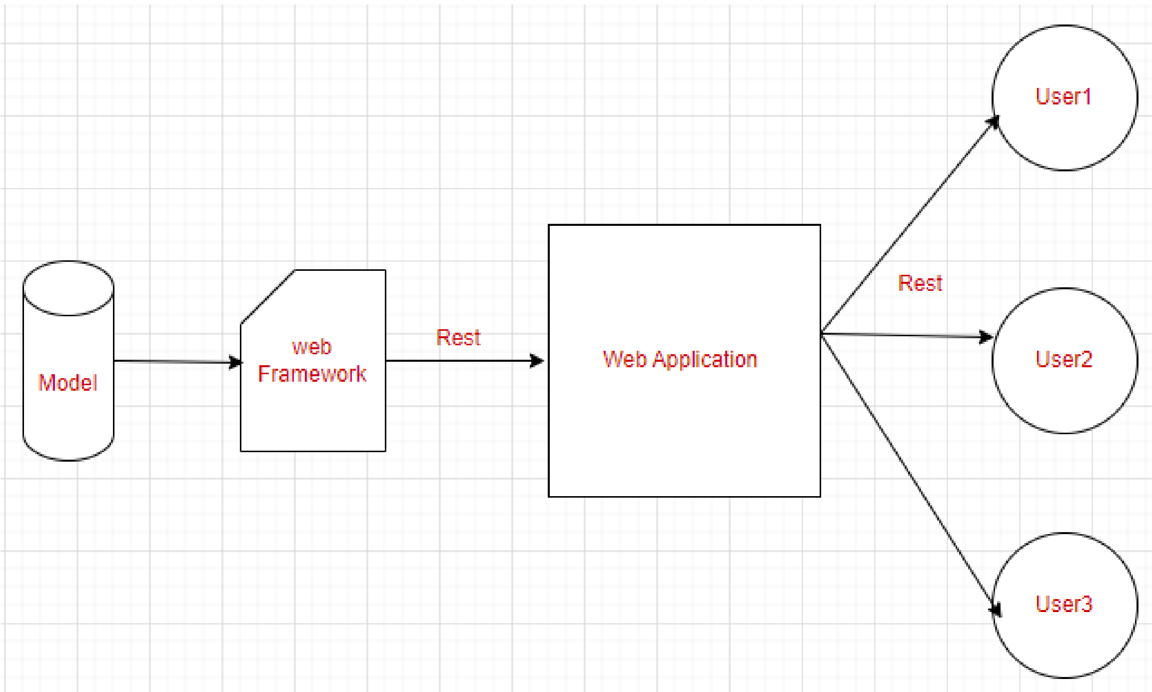


Figure 86 - Heart Care Application Architecture

According to Fig.86 TabNet classifier deployed using FastAPI web framework. Junqiao C. (2023) discussed the reliability and efficiency of FastAPI over other frameworks such as Django and Flask. Implementation using FastAPI is simple and straightforward. Web Application develop using reactJS with tailwind UI which is widely used front end java script framework

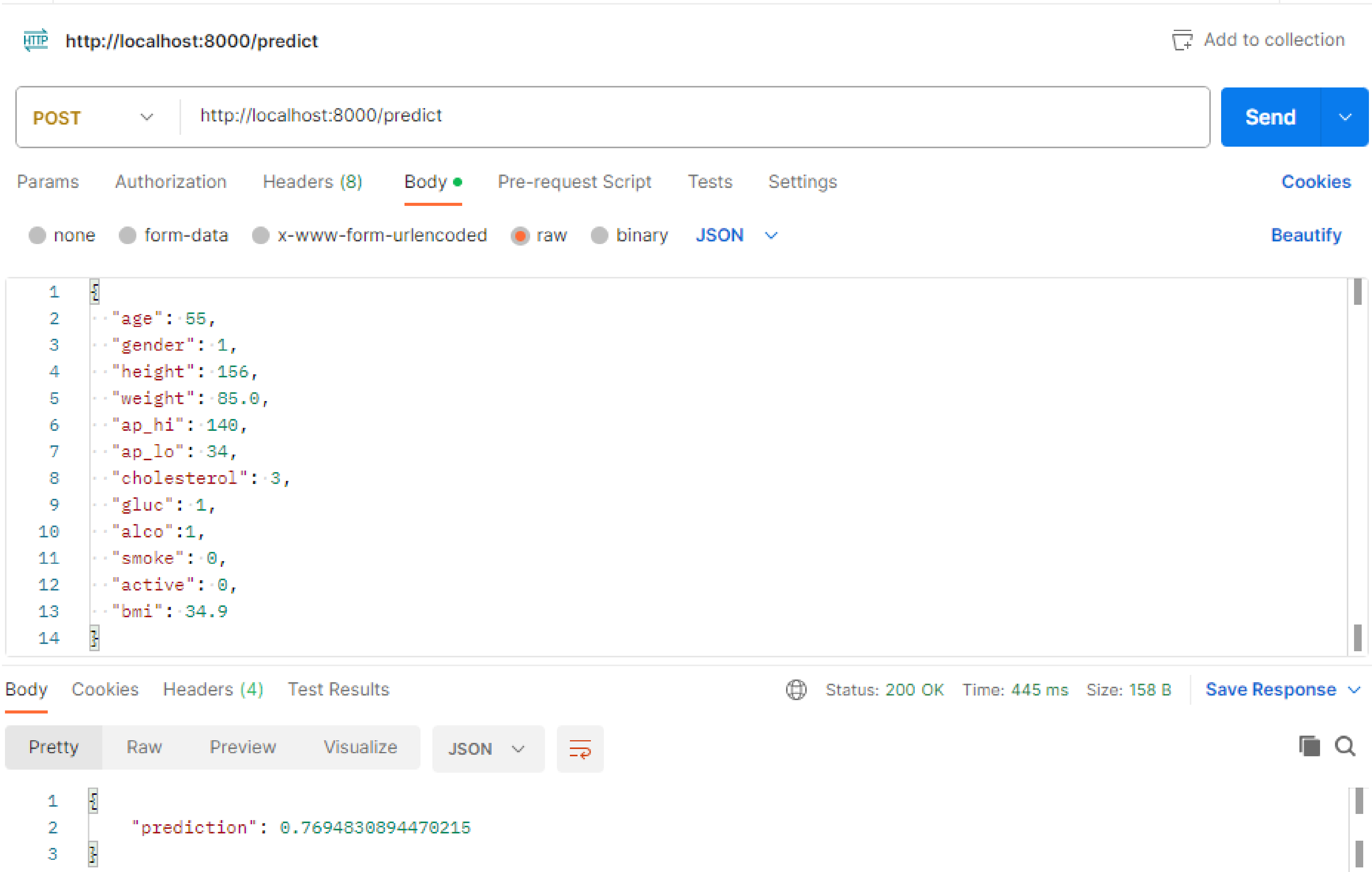

http://localhost:8000/predict
Add to collection
POST
http://localhost:8000/predict
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
{
"age": 55,
"gender": 1,
"height": 156,
"weight": 85.0,
"ap_hi": 140,
"ap_lo": 34,
"cholesterol": 3,
"gluc": 1,
"alco":1,
"smoke": 0,
"active": 0,
"bmi": 34.9
}
Body
Cookies
Headers (4)
Test Results
Status: 200 OK
Time: 445 ms
Size: 158 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"prediction": 0.7694830894470215
}

Figure 87

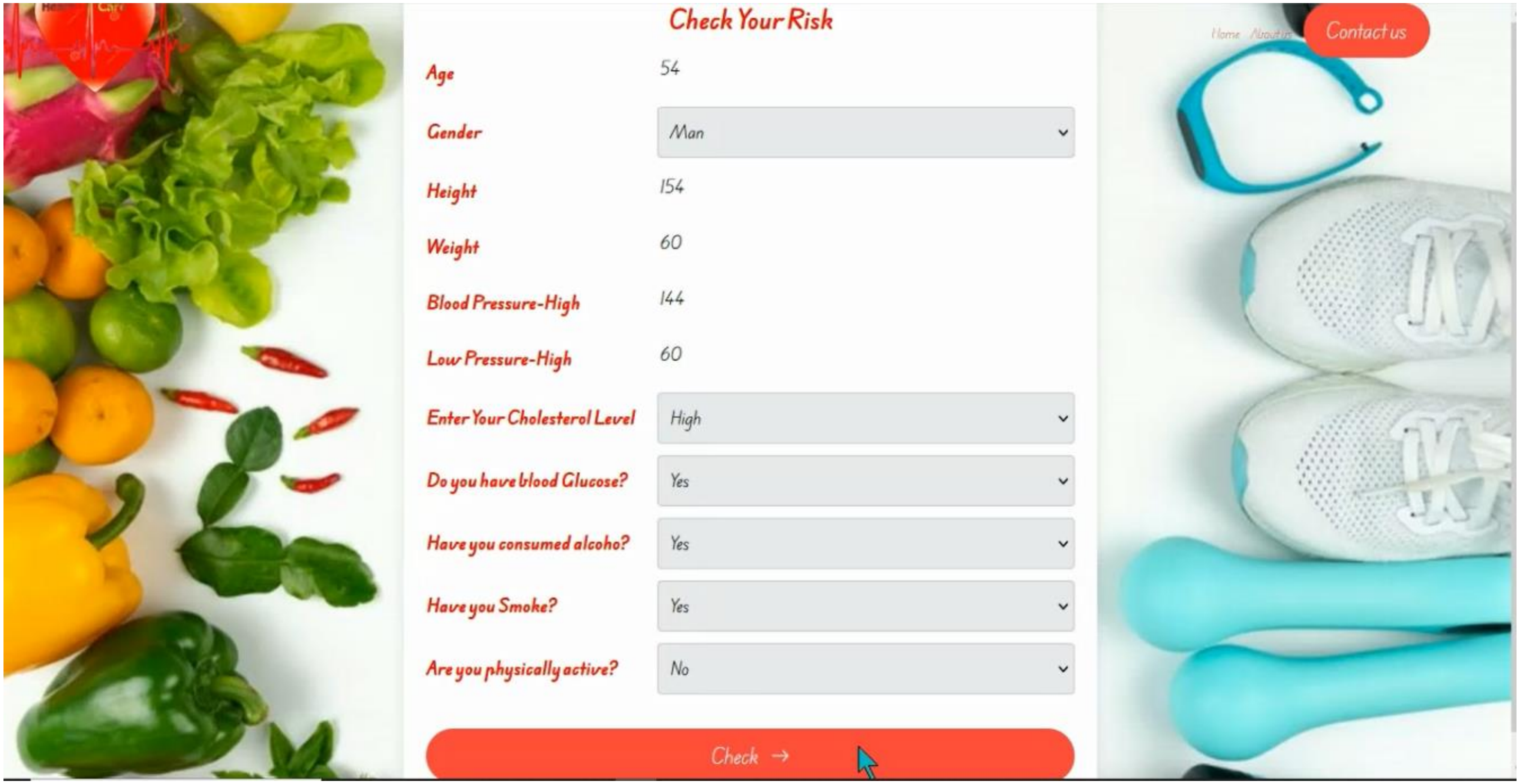

Contact us
Check Your Risk
Age
54
Gender
Man
Height
154
Weight
60
Blood Pressure-High
144
Low Pressure-High
60
Enter Your Cholesterol Level
High
Do you have blood Glucose?
Yes
Have you consumed alcoho?
Yes
Have you Smoke?
Yes
Are you physically active?
No
Check →

Figure 88

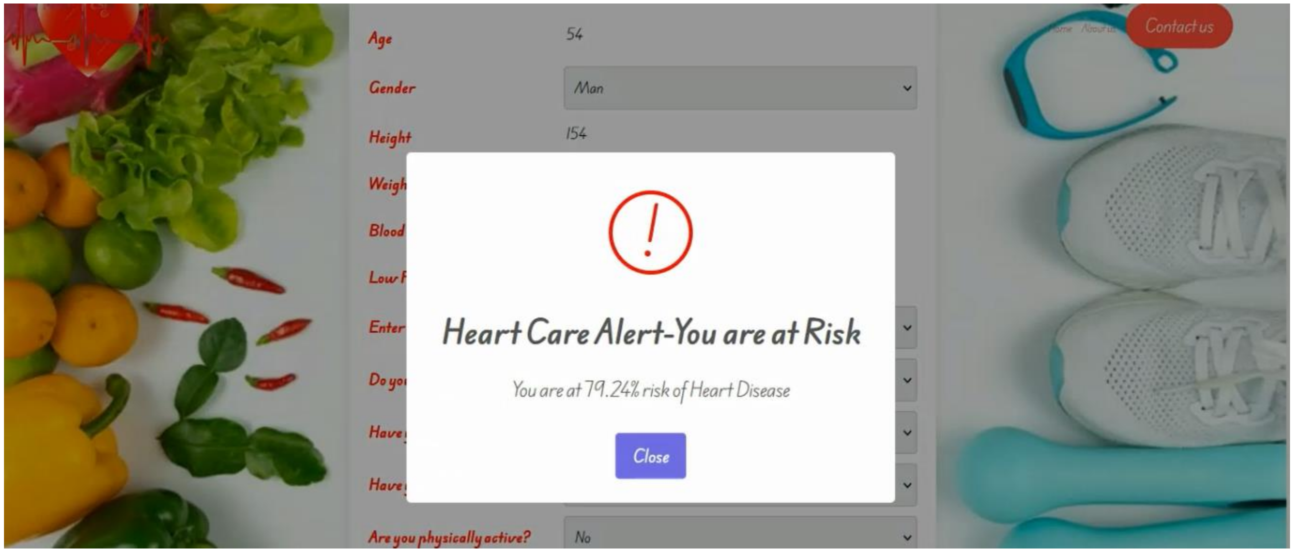
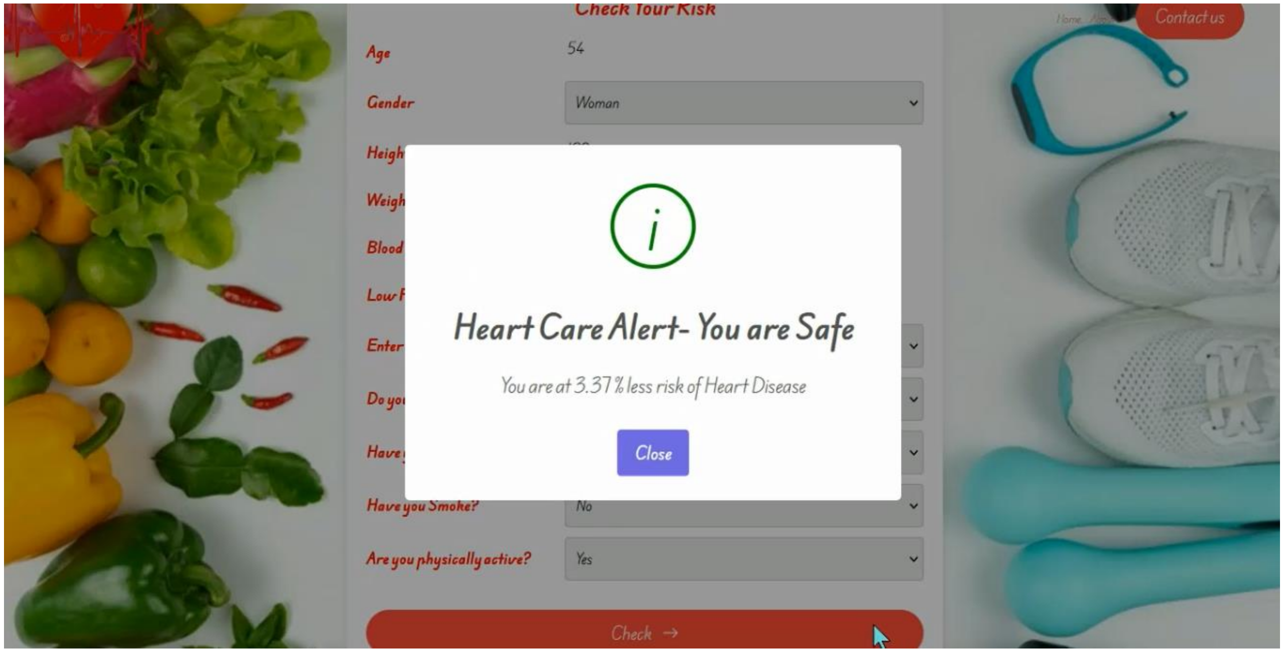

Contact us
Age
54
Gender
Man
Height
154
Heart Care Alert-You are at Risk
You are at 79.24% risk of Heart Disease
Close
Are you physically active?
No

Figure 89


Contact us
Age
54
Gender
Woman
Heart Care Alert- You are Safe
You are at 3.37 % less risk of Heart Disease
Close
Have you Smoke?
No
Are you physically active?
Yes
Check

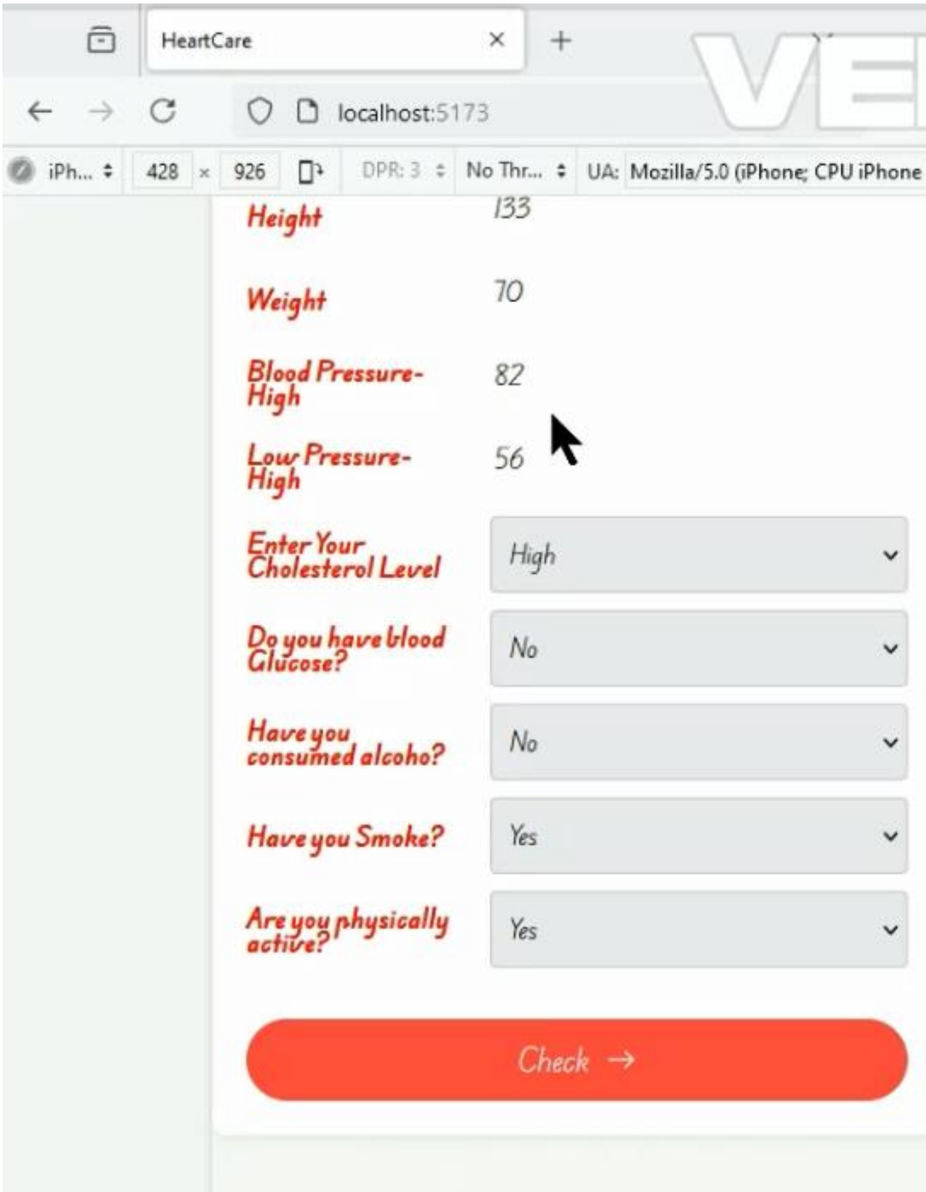

HeartCare
localhost:5173
iPh...
428
926
DPR: 3
No Thr...
UA: Mozilla/5.0 (iPhone; CPU iPhone
Height
133
Weight
70
Blood Pressure-High
82
Low Pressure-High
56
Enter Your Cholesterol Level
High
Do you have blood Glucose?
No
Have you consumed alcoho?
No
Have you Smoke?
Yes
Are you physically active?
Yes
Check →

Figure 91

Figure 87, postman screen shows the rest request and response as model input and output
Figures 89, 90, and 91 illustrate screenshots from the Heart Care cardiovascular disease prediction application, showcasing its interface and functionality. Figure 90 shows the responsive application interface with capable for different screen sizes.

5. Learning Outcomes and Discussion

This section discusses learning outcomes gained from this study, quality of the work and its contribution to address the existing problem.

5.1 Findings and Learning

CVD is a leading cause of death globally, create significant impacts to the global healthcare, economy and the society. Past few years number of research, studies and products come up in digital context to full fill the vacuum of CVD prediction. Largely focused on pre-programmed software and minority employing machine learning application, often limited to traditional classifiers and limited training, testing spaces.

Therefore, using machine learning and deep learning concepts and techniques extensively, with wide range of classifiers under broad training and testing dataset, given opportunity to develop highly accurate, reliable and predictive output.

This study highlights the importance identifying best parameters and process of effective hyper parameter tunning can make significant contribution to the result, furthermore, importance of sampling techniques and using different measuring techniques, methods and scores to verify each classifier behavior as well as the performance.

Research contributes to study, save and load different models, especially in traditional pickles, TensorFlow, karas classifiers as well as py-touch models. Furthermore, learning expanded to different web frameworks that exposure model as a web service. In addition to that different java scripts frameworks including react and tailwind ability explore and experience.

5.2 Novelty

Several studies and research on the area of cardiovascular disease prediction, however most of the research and studies are based on limited dataset. As example Senthilkumar(2019) and Devansh(2020) studied limited to 300 data. This small dataset can generate bias predictions and accuracy limitations. In this study focus on wide and large datasets with 11 most crucial features that directly enforce to the cardio health.

This study addresses the research gap of developing effective cardiovascular disease predictions model using deep learning. Mohammed and Rajib(2021) suggest deep learning classifier for effective accuracy and performance.

The purpose of this study is to build a high-performance, accurate deep learning base application that is accessible to the public. Unlike most of the existing applications that use pre-defined software algorithms, that may vary depending on the situation. This study focuses on developing a well adaptable solution for cardiovascular disease predictions.

5.3 Reflection

In reflection of this study, I allocate considerable time to researching the related works, find appropriate dataset and gain deep understanding of existing and ongoing works. I have tested different approaches, a variety of feature engineering techniques and models. For example, I tried different models without any parameter tunning achieved less accuracy, same time I tried different ensemble models without using any regularization techniques achieve considerable accuracy differences on training and testing.

On the other hand, I experience considerable delays on the train, test and validate. Especially for large datasets on SVM, XG-boost, ANN and TabNet classifiers. I changed my training environment to Paid subscription of google colab to use enchase GPU and Memory to success my tasks. Based on my experience, deep learning classifiers required over two hours for tuning and training.

5.4 Limitations and Boundaries

Considerable limitations of this project are infrastructure and the cost. Especially in the process of training and Hyper Parameter tunning perform to the large datasets GPUs and high memory requirements challenge the further developments and experiments. Furthermore, application deployment cost on cloud significantly limits potential enhancements and review public responses.

6. Conclusion and Future works

Cardiovascular disease is considered one of the leading causes of death around the world. Early diagnostic can help to prevent or reduce risk on progression of the disease. My proposed TabNet, deep learning classifier achieved 80% of training accuracy,79% accuracy on testing and 73.4% on overall accuracy. Furthermore, in this study evaluate another eight machine learning classifiers including RF, DT, LR, SVM, XGB, LGB, KNN, NB, Ensemble voting classifier(SVM/LR/XGB) and ANN.

DBSCAN is used for identifying and removing outliers.

Furthermore, the study involves developing cardiovascular disease predictions application using TabNet classifier as core-classifier. Which enables effective and accurate predictions.

In the future, I recommend continuing to study additional deep learning classifiers to enhance predictive analysis. Further suggest continuing study on splitting dataset in various portions such as (70:30),(60:40),(75:25),(85:15) to determine different scenarios.

The next step will be deploying Heart-Care applications on a cloud platform such as AWS or Azure, and open for public use. As further enhancement I proposed personalizing user profile that easily access his past feature details and provide patients to access integrated environment for receive expert advice.

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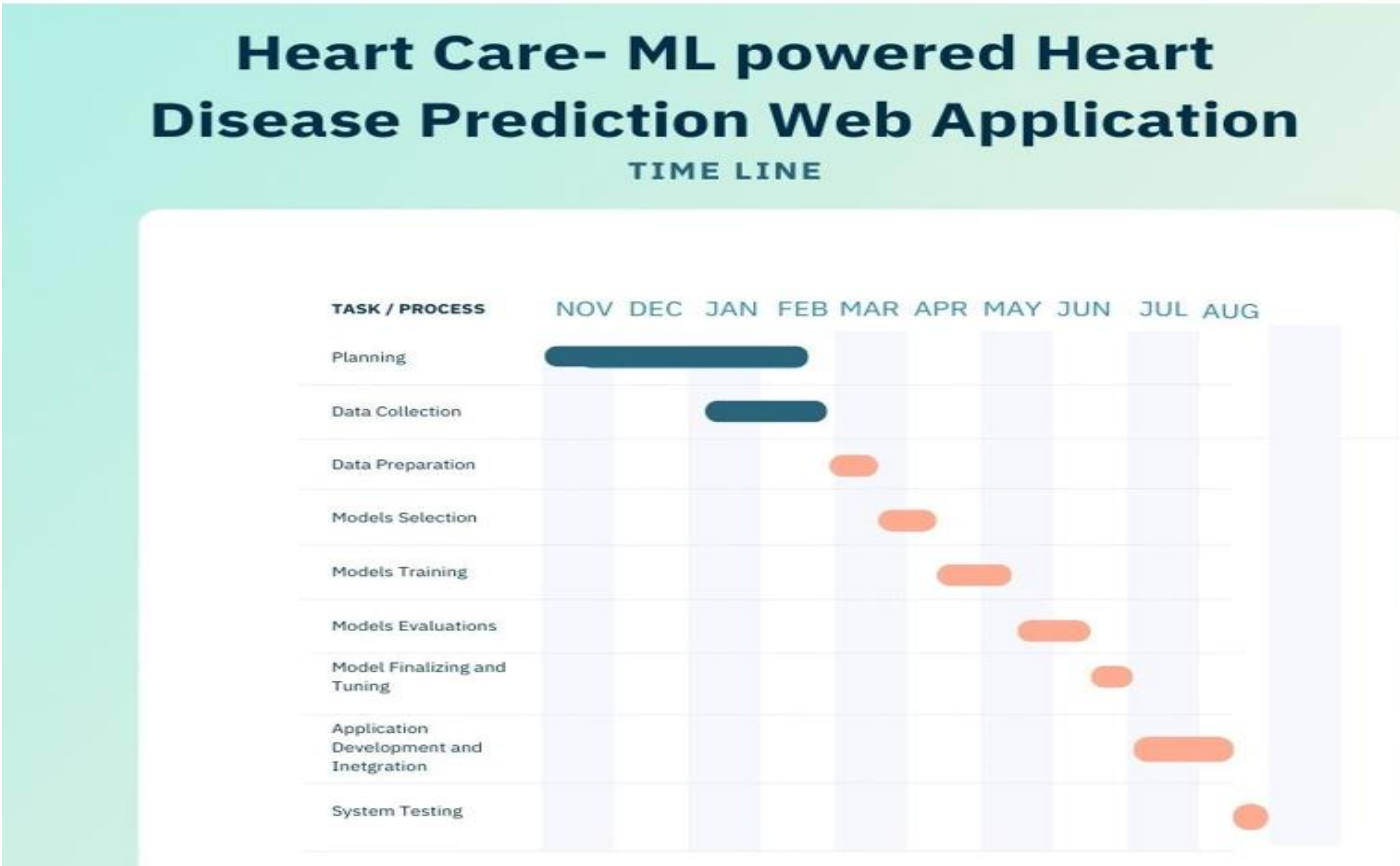
8. Appendix

8.1 Appendix A – Gantt Chart



8.2 Infrastructure

Processor	T4 GPU
GPU Name	Tesla T4
Version	535.104.05
CUDA Version:	12.2
System RAM	51 GB
GPU RAM	15 GB
Disk	202 GB
Model Api URL	http://localhost:8000/predict
Application URL	http://localhost:5173/HeartCare

8.3 Supervisory Meeting

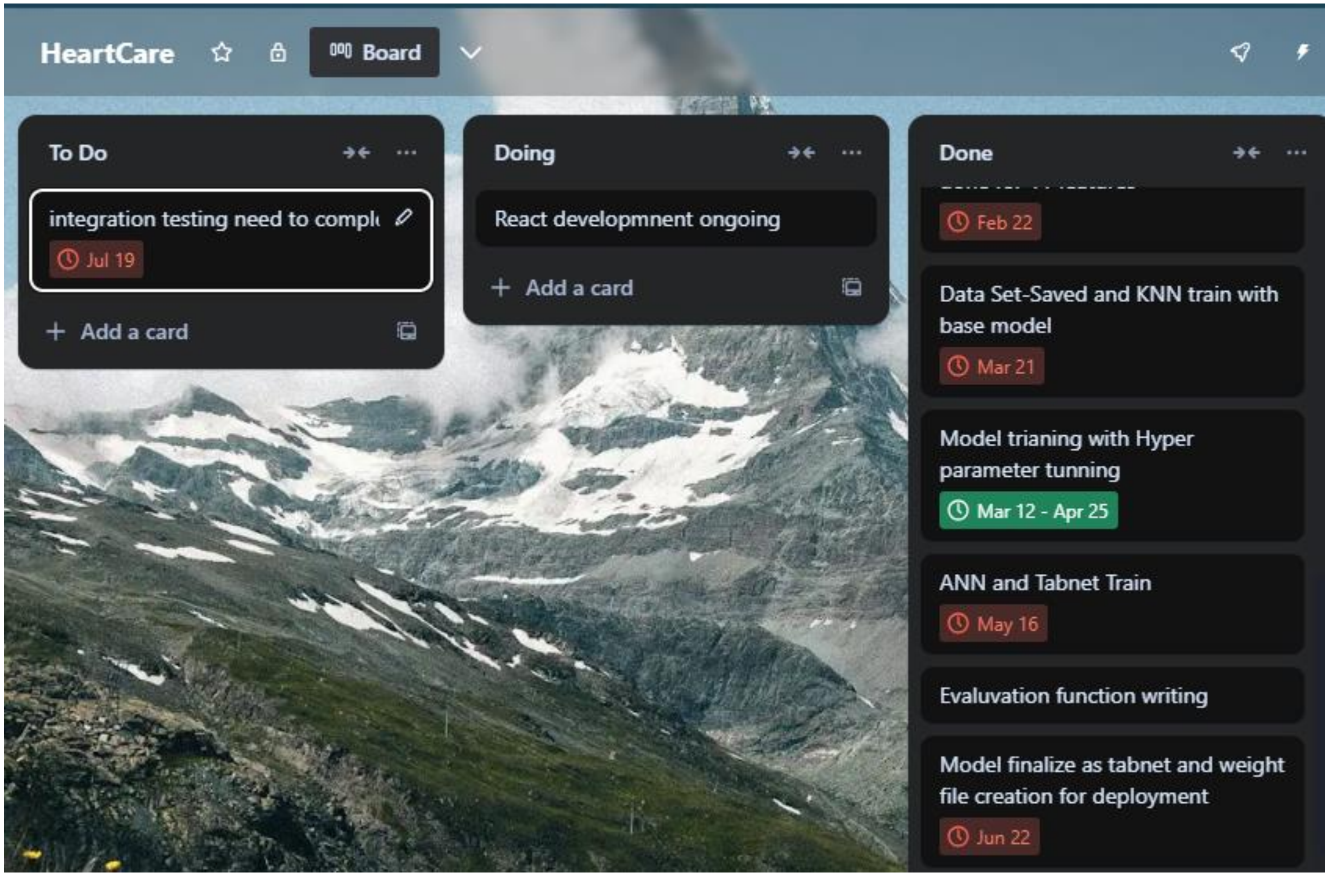
Supervisor	Mahmoud Elbattah
Date	04/07/2024
Meeting Agenda	Focus on Project Topic and initiatives
Notes	Brief description about project plan and existing development
Suggestions and Feedback	More focus on existing literature
Date	19/07/2024
Meeting Agenda	Interim discussion about existing development, Ethical form submit and future plan
Suggestions	Pay attention on deep learning
Date	01/08/2024
Teams Instance Message	Share existing works
Suggestions	Instruction received for report writing.
Date	14/082024

Teams Instance Message	Send Half Complete Report
Suggestion	Ask to complete full report.

Email Communication and Teams Communication

Date	Description
2024/5/27	Initial project discusses and meeting request
2024/6/16	Project progress and meeting request
2024/7/04	Discussed about ethical aspects and related dataset
2024/7/12	Project progress notification
2024/7/16	Project progress
2024/8/01	Dissertation submit for review
2024/8/24	Project Draft send for review
2024/8/27	Information about ethical consideration and file format

Trello boards and google Calander used for effective project management and time measurement tool.



8.4 Libraries

Pandas	Powerful python library for data manipulation and analyze using structured data handling such as Data Frame, Series
MatplotLib	Powerful python library for create static, visualize by interactive plots, charts and graphs
Sklearn	Sklearn library contain tools for datamining, data analysis and building models
Pytorch	Pytorch python library for deep learning, provides framework for building, training and testing deep learning models. TabNet is Pytorch based deep learning model
Seaborn	Seaborn provides high level attractive and statical graphics
Keras	Keras running top of deep learning frameworks like TensorFlow. Keras module designed to wrap more user-friendly modules and libraries
Tensoflow	TensorFlow is widely used machine learning framework, that contain libraries, tools and other related resources to build deep learning model.
DBSCAN	Density based clustering and noise identification library
React	JavaScript library
Sweetalert2	Java script library for customization of alert and messages

8.5 Personal Reflection

This dissertation was a great opportunity to incorporate my passion for Data Science to solve burning real world problem, specifically throughout this project I gained comprehensive experience on the entire life cycle of the Data Science solution, and achieved extreme level of self-satisfaction by completion of the successful project that can make meaningful impact to the people’s lives.

I managed to get positive experience in every process of the CVD prediction project, in literature review I manage to get a deep understanding of the problem domain and identified crucial ethical and legal limitations, rules that need to be followed in sensitive domain like health. Furthermore, from literature review I understand that there are considerable gaps in the use of deep learning classifier, in CVD prediction. Moreover, literature review guides me to strong future direction use of effective methodological approaches with better technical guidance.

In technical perspective I learn clustering techniques like DBSCAN(density based spatial and clustering of applications with noise), to efficiently handle noises in the dataset. Furthermore, I learn about different machine learning techniques , deep learning classification models and the multi-layer ensemble techniques that combinedly contribute for successful predictions.

I was fortunate to deep drive latest deep learning classifiers like Google's TabNet, Artificial Neural Network classifiers and those parameterize tunning, which widely used impact on tabular predictions. On the other hand, I learn evaluation of classification scenarios in different context and deeply look at the different evaluation reports which tell model's performance in different point of views. Example. Which model is capable of predicting 'risk' factors and what is good for 'non risk' classifications.

This project helped me to develop stronger analytical, problem-solving skills with better understanding of technical and conceptual scenarios, especially in statistical concepts behind these classifiers.

I am continuing to work on this project to open Heart-Care application to public use as first version and continue further development with highly diversified, up-to-date dataset to gain better performance and trustworthiness.